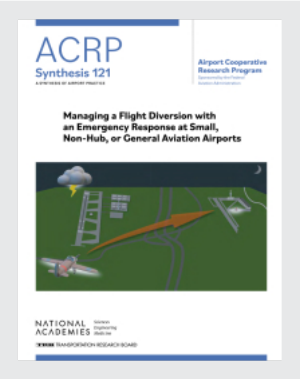
NATIONAL ACADEMIES Sciences Engineering Medicine

NATIONAL ACADEMIES PRESS Washington, DC

This PDF is available at https://nap.nationalacademies.org/26900



Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports (2023)

BUY THIS BOOK

FIND RELATED TITLES

DETAILS

168 pages | 8.5 x 11 | PAPERBACK

ISBN 978-0-309-68773-7 | DOI 10.17226/26900

CONTRIBUTORS

Rose Agnew, Christina Coverdell; Airport Cooperative Research Program; Transportation Research Board; National Academies of Sciences, Engineering, and Medicine

SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2023. *Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports*. Washington, DC: The National Academies Press. https://doi.org/10.17226/26900.

Visit the National Academies Press at nap.edu and login or register to get:

– Access to free PDF downloads of thousands of publications
– 10% off the price of print publications
– Email or social media notifications of new titles related to your interests
– Special offers and discounts

All downloadable National Academies titles are free to be used for personal and/or non-commercial academic use. Users may also freely post links to our titles on this website; non-commercial academic users are encouraged to link to the version on this website rather than distribute a downloaded PDF to ensure that all users are accessing the latest authoritative version of the work. All other uses require written permission. (Request Permission)

This PDF is protected by copyright and owned by the National Academy of Sciences; unless otherwise indicated, the National Academy of Sciences retains copyright to all materials in this PDF with all rights reserved.

AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP SYNTHESIS 121

Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

A Synthesis of Airport Practice

Rose Agnew
Christina Coverdell
Aviation Innovation, LLC
St. Paul, MN

Subscriber Categories
Aviation • Security and Emergencies

Research sponsored by the Federal Aviation Administration

NATIONAL ACADEMIES *Sciences Engineering Medicine*

TRB TRANSPORTATION RESEARCH BOARD

2023

Copyright National Academy of Sciences. All rights reserved.

AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

ACRP SYNTHESIS 121

Project 11-03, Topic S04-26
ISSN 1935-9187
ISBN 978-0-309-68773-7
Library of Congress Control Number 2023930056

© 2023 by the National Academy of Sciences. National Academies of Sciences, Engineering, and Medicine and the graphical logo are trademarks of the National Academy of Sciences. All rights reserved.

COPYRIGHT INFORMATION

Authors herein are responsible for the authenticity of their materials and for obtaining written permissions from publishers or persons who own the copyright to any previously published or copyrighted material used herein.

Cooperative Research Programs (CRP) grants permission to reproduce material in this publication for classroom and not-for-profit purposes. Permission is given with the understanding that none of the material will be used to imply TRB, AASHTO, APTA, FAA, FHWA, FTA, GHSA, or NHTSA endorsement of a particular product, method, or practice. It is expected that those reproducing the material in this document for educational and not-for-profit uses will give appropriate acknowledgment of the source of any reprinted or reproduced material. For other uses of the material, request permission from CRP.

Cover art credit: Cover art designed by Aviation Innovation, LLC.

NOTICE

The report was reviewed by the technical panel and accepted for publication according to procedures established and overseen by the Transportation Research Board and approved by the National Academies of Sciences, Engineering, and Medicine.

The opinions and conclusions expressed or implied in this report are those of the researchers who performed the research and are not necessarily those of the Transportation Research Board; the National Academies of Sciences, Engineering, and Medicine; or the program sponsors.

The Transportation Research Board does not develop, issue, or publish standards or specifications. The Transportation Research Board manages applied research projects which provide the scientific foundation that may be used by Transportation Research Board sponsors, industry associations, or other organizations as the basis for revised practices, procedures, or specifications.

The Transportation Research Board; the National Academies of Sciences, Engineering, and Medicine; and the sponsors of the Airport Cooperative Research Program do not endorse products or manufacturers. Trade or manufacturers' names or logos appear herein solely because they are considered essential to the object of the report.

Published reports of the

AIRPORT COOPERATIVE RESEARCH PROGRAM

are available from

Transportation Research Board
Business Office
500 Fifth Street, NW
Washington, DC 20001

and can be ordered through the Internet by going to

https://www.mytrb.org/MyTRB/Store/default.aspx

Printed in the United States of America

Copyright National Academy of Sciences. All rights reserved.

NATIONAL ACADEMIES *Sciences Engineering Medicine*

The **National Academy of Sciences** was established in 1863 by an Act of Congress, signed by President Lincoln, as a private, non-governmental institution to advise the nation on issues related to science and technology. Members are elected by their peers for outstanding contributions to research. Dr. Marcia McNutt is president.

The **National Academy of Engineering** was established in 1964 under the charter of the National Academy of Sciences to bring the practices of engineering to advising the nation. Members are elected by their peers for extraordinary contributions to engineering. Dr. John L. Anderson is president.

The **National Academy of Medicine** (formerly the Institute of Medicine) was established in 1970 under the charter of the National Academy of Sciences to advise the nation on medical and health issues. Members are elected by their peers for distinguished contributions to medicine and health. Dr. Victor J. Dzau is president.

The three Academies work together as the **National Academies of Sciences, Engineering, and Medicine** to provide independent, objective analysis and advice to the nation and conduct other activities to solve complex problems and inform public policy decisions. The National Academies also encourage education and research, recognize outstanding contributions to knowledge, and increase public understanding in matters of science, engineering, and medicine.

Learn more about the National Academies of Sciences, Engineering, and Medicine at **www.nationalacademies.org**.

The **Transportation Research Board** is one of seven major programs of the National Academies of Sciences, Engineering, and Medicine. The mission of the Transportation Research Board is to provide leadership in transportation improvements and innovation through trusted, timely, impartial, and evidence-based information exchange, research, and advice regarding all modes of transportation. The Board's varied activities annually engage about 8,000 engineers, scientists, and other transportation researchers and practitioners from the public and private sectors and academia, all of whom contribute their expertise in the public interest. The program is supported by state transportation departments, federal agencies including the component administrations of the U.S. Department of Transportation, and other organizations and individuals interested in the development of transportation.

Learn more about the Transportation Research Board at **www.TRB.org**.

Copyright National Academy of Sciences. All rights reserved.

COOPERATIVE RESEARCH PROGRAMS

CRP STAFF FOR ACRP SYNTHESIS 121

Christopher J. Hedges, *Director, Cooperative Research Programs*
Waseem Dekelbab, *Deputy Director, Cooperative Research Programs*
Marci A. Greenberger, *Manager, Airport Cooperative Research Program*
Jordan Christensen, *Senior Program Officer*
Demisha Williams, *Senior Program Assistant*
Natalie Barnes, *Director of Publications*
Heather DiAngelis, *Associate Director of Publications*

ACRP PROJECT 11-03 PANEL

Joshua D. Abramson, *Barge Design Solutions, Inc., Nashville, TN* (Chair)
Gloria G. Bender, *TransSolutions, LLC, Fort Worth, TX*
David A. Byers, *Quadrex Aviation LLC, Melbourne, FL*
Traci Clark, *Allegheny County (PA) Airport Authority, West Mifflin, PA*
Brenda L. Enos, *TRC, Boston, MA*
Cameron Thompson, *Birmingham Airport Authority, Birmingham, AL*
Keila Walker-Denis, *Orlando International Airport, Orlando, FL*
Scott Wintner, *San Jose Norman Y. Mineta International Airport, San Jose, CA*
Patrick Magnotta, *FAA Liaison*
Liying Gu, *Airports Council International–North America Liaison*
Christine L. Gerencher, *TRB Liaison*

TOPIC S04-26 PANEL

Kathleen Henderson, *Airports Council International–North America, Ottawa, ON*
Max Zhaoyu Li, *University of Michigan, Ann Arbor, Ann Arbor, MI*
David M. McCurdy, *Tarrant County (TX) Administrator's Office, Fort Worth, TX*
Michael Tobin, *Charlotte Douglas International Airport, Charlotte, NC*
Kellyann Valentine, *Port Authority of New York and New Jersey, New York, NY*
Kelli White, *Emergency Response Partners LLC, Burlingame, CA*
Kent Duffy, *FAA Liaison*
Paul James Eubanks, *Airports Council International–North America Liaison*

ABOUT THE ACRP SYNTHESIS PROGRAM

Airport administrators, engineers, and researchers often face problems for which information already exists, either in documented form or as undocumented experience and practice. This information may be fragmented, scattered, and unevaluated. As a consequence, full knowledge of what has been learned about a problem may not be brought to bear on its solution. Costly research findings may go unused, valuable experience may be overlooked, and due consideration may not be given to recommended practices for solving or alleviating the problem.

There is information on nearly every subject of concern to the airport industry. Much of it derives from research or from the work of practitioners faced with problems in their day-to-day work. To provide a systematic means for assembling and evaluating such useful information and to make it available to the entire airport community, the Airport Cooperative Research Program authorized the Transportation Research Board to undertake a continuing project. This project, ACRP Project 11-03, "Synthesis of Information Related to Airport Practices," searches out and synthesizes useful knowledge from all available sources and prepares concise, documented reports on specific topics. Reports from this endeavor constitute an ACRP report series, Synthesis of Airport Practice.

This synthesis series reports on current knowledge and practice, in a compact format, without the detailed directions usually found in handbooks or design manuals. Each report in the series provides a compendium of the best knowledge available on those measures found to be the most successful in resolving specific problems.

FOREWORD

By Jordan Christensen
Staff Officer
Transportation Research Board

Unexpected flight diversions may impact airport operations from routine to emergency incidents. This synthesis compiles practices that small, non-hub, and general aviation airports use when planning for and responding to flight diversions that involve a routine gas and go, an incident, or an emergency. The objectives of this synthesis are (1) to discover and describe current practices for handling diverted flights that involve an emergency response and (2) to provide information on practices to small hub, non-hub, and general aviation airport that may receive a diverted passenger flight and not have appropriate staff, resources, or equipment available. Information used in this study was attained through a literature review, surveys, and interviews with 25 diverse airports, which produced seven case examples found in Chapter 6. For immediate use, Appendix C contains example diversion checklists.

Rose Agnew, Aviation Innovation, LLC, St. Paul, Minnesota, synthesized the information and wrote the report, assisted by Christina Coverdell. The members of the topic panel are acknowledged on page iv. This synthesis is an immediately useful document that records the practices that were acceptable within the limitations of the knowledge available at the time of its preparation. As progress in research and practice continues, new knowledge will be added to that now at hand.

Copyright National Academy of Sciences. All rights reserved.

AUTHOR ACKNOWLEDGMENTS

Many interviews took place to collect the data found in this synthesis. Thank you to the following people who are listed and who assisted in this endeavor:

Alex Martichuski, FlightAware
Alvin Lorenzo, Sioux Gateway/Brig Gen Bud Day Field
Amber Christenson, Delta Air Lines
Blaine Peterson, Flying Cloud Airport and Airlake Airport
Bob McNally, Lincoln Airport
Chris Waugh, Halifax Stanfield International Airport
Darren Anderson, Hector International Airport
Garnett Schafer, Rick Husband Amarillo International Airport
Ian Turner, La Crosse Regional Airport
Joe Harris, Minneapolis-St. Paul International Airport
John Hawkins, Iqaluit International Airport
Jonathon Perout, Sioux Falls Regional Airport
Joseph Hennessy, Minneapolis-St. Paul International Airport
Kelly Gerads, Minneapolis-St. Paul International Airport
Kirk White, St. John's International Airport
Kurt Claussen, Rochester International Airport
Mark Cihlar, Central Wisconsin Airport
Mark Libby, Passur Aerospace Inc.
Matt Aubry, Eppley Field (Omaha, NE)
Michael Baker, Ocala International Airport
Michael Lawrence, Premier Jet Center
Michal P. Junik, Dane County Regional-Truax Field
Mike Wilson, St. Paul Downtown Airport and Crystal Airport
Paul Borgstrom, Delta Air Lines
Paul Wipf, Sioux Falls Regional Airport
Philip Tiedeman, Lake Elmo Airport and Anoka County-Blaine Airport
Ramon Sanchez, Palm Springs International Airport
Ray Laroche, Punta Gorda Airport
Richard King, Sioux Falls Regional Airport
Rick Audette, Grand Forks International Airport
Robert Dockry, Minneapolis-St. Paul International Airport
Ryan Welch, Duluth International Airport
Scott Andre Remillard, Saab
Timothy Reid, Portland International Jetport
Todd Gibbs, The Eastern Iowa Airport
Tom Smith, Dallas-Fort Worth International Airport

Copyright National Academy of Sciences. All rights reserved.

CONTENTS

1 Summary

6 **Chapter 1** Introduction
6 Flight Diversions
7 How Flight Diversions Impact Smaller Airports
9 Scope of Work
9 Synthesis Overview

11 **Chapter 2** Defining the Stakeholders
11 Internal Airport Stakeholders
13 External Airport Stakeholders
13 Roles of Mutual Aid and Network Partners During Diversions

16 **Chapter 3** Methodology
16 Literature Review
16 Survey
16 Interviews
19 Case Examples

20 **Chapter 4** Literature Review
20 Introduction to Flight Diversions
23 Before a Diversion: Planning, Training, and Coordinating
30 During a Diversion: Managing and Mitigating
34 After a Diversion: Restoring Operations
35 Lessons Learned and Good Practices

37 **Chapter 5** Survey and Interview Responses
37 Survey Responses
53 Interview Responses
60 Challenges

63 **Chapter 6** Case Examples
63 Small Hub Airports
69 Non-Hub Airports
74 General Aviation Airports

79 **Chapter 7** Benefits of Collaborative Planning

80 **Chapter 8** Diverted Flight Challenges
80 Uncertainty Related to Increased Diversions Versus Smarter Planning
82 Small Staffing Size of Smaller Airports
82 Planning- and Training- and Exercise-Related Challenges
83 Diversion Costs
83 Coordination Issues
83 Compounding Issues

Copyright National Academy of Sciences. All rights reserved.

84 **Chapter 9** Conclusion and Summary of Findings
84 Conclusion and Summary
84 Gaps in Knowledge
85 Further Areas of Study
87 Glossary
91 Acronyms
93 References
95 Bibliography
97 **Appendix A** Survey Questionnaire and Interview Guide
113 **Appendix B** Survey Responses
127 **Appendix C** Sample Diversion Plans, Checklists, and Training Exercises

Note: Photographs, figures, and tables in this report may have been converted from color to grayscale for printing. The electronic version of the report (posted on the web at nap.nationalacademies.org) retains the color versions.

Copyright National Academy of Sciences. All rights reserved.

SUMMARY

Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

This synthesis explores how smaller airports manage various levels of unexpected aircraft diversions from routine to emergency. It investigates which types of diversions these airports can manage with their own capabilities. It also indicates who these airports call on for additional assistance. It discusses the resiliency of smaller airports, or rather, their ability to recover quickly from flight diversions to maintain regularly scheduled flight schedules and operations.

Capabilities are different for each airport because of a myriad of factors explored in this synthesis. Factors associated with "capability" include an airport's

- Size (e.g., annual enplanements),
- Structure (e.g., governance and contracting structures),
- Physical infrastructure (e.g., runway type and length, terminal size, ramp space and number of gates, and impacts to it like construction),
- Number of personnel and expertise (e.g., training and qualifications),
- Resources (e.g., supplies like fuel, equipment like airstairs, and technology for communicating critical information),
- Preparedness (e.g., plans, training, and collaborative networks),
- Notification factors (e.g., tracking, timing, and airport accessibility),
- Communication factors (e.g., common nomenclature and established protocols), and
- Hours of operation.

The objectives of this synthesis are to describe current practices that small hub (SH), nonhub (NH), and general aviation (GA) airports use to manage the following types of aircraft diversions: (1) routine gas and go technical stops; (2) diversions involving an incident, especially those that exceed an airport's capability to manage a situation; and (3) diversions requiring an emergency response.

The information for this synthesis initially came from an extensive literature review and a survey that was sent to approximately 300 SH, NH, and GA airports in the United States (71 survey responses). Interviews were also conducted with leaders at eight SH, eight NH, and nine GA airports (including those GA airports categorized as nonprimary commercial service [CS] and reliever [R] airports); pilots at one airline and representatives from the industry group Airlines for America (A4A); a fixed-base operator (FBO); four technology partners; three regional irregular operations (IROPS) networks; a reliever airport group; and the Federal Aviation Administration (FAA).

Flight diversions are typically caused by factors outside the aircraft (e.g., weather), with the physical aircraft (e.g., low fuel levels, minor mechanical issues, and structural damage),

Copyright National Academy of Sciences. All rights reserved.

and inside the aircraft (e.g., disruptive passengers and medical issues). Research demonstrates that most diverted flights to smaller airports are "gas and go" technical stops in which no passenger deplaning occurs. These types of diversions are considered routine operations for most smaller airports and are easy for them to manage. Smaller airports look forward to these diversions as they make money from them by collecting landing fees and FBO services.

On rare occasions, smaller airports receive flight diversions with incident-level issues like minor aircraft mechanical issues or disruptive passengers on board. Sometimes they receive a diverted flight with an emergency-level issue like engine failure or a medical emergency. But, when one of these types of diverted flights lands at an airport, the authority under which first responders operate is the primary factor that determines whether smaller airports can manage the situation without exceeding their capabilities.

For many SH and NH airports, first responders such as aircraft rescue and firefighting (ARFF), law enforcement, and medical responders are considered under an airport's authority, either as in-house staff or third-party contracted staff. For many GA airports, first responders are considered mutual aid and are contracted through interlocal or joint-use agreements or are managed through another department under municipal control. But these generalizations do not always hold true. It is difficult to make broad statements about SH, NH, and GA airports because of their various governance and management structures, contractual service arrangements, and operating characteristics–all of which are unique for each airport across various sizes.

Research does show that SH and NH airports can effectively manage most incident- and emergency-level flight diversions. This is especially true for joint-use civil-military airports that often have a higher ARFF index level. SH and NH airports are challenged most when they receive non-station international flight diversions or multiple flight diversions at once, particularly if they have limited gate or parking space for aircraft or terminal space for deplaned passengers.

GA airports tend to exceed their capabilities more quickly than SH and NH airports because of their limited hours of operation, very small staff size, limited or no federal agency presence, and fewer, if any, in-house emergency response resources. This does not hold true for GA airports that are considered nonprimary commercial service (CS) airports as they service larger aircraft and have ARFF presence. However, since GA airports tend to have fewer incoming and outgoing daily flights, emergencies can often be mitigated without impacting normal flight operations. But passengers may experience extended delays if there are fewer alternative flights available to take them to their final destinations.

Unless they are part of a regional diversion network, smaller airports do not receive early notification of flight diversions; they hear from air traffic control (ATC) or an airline pilot about a diversion just before or just as they witness it landing on their runway. For smaller airports located in remote locations, incident- or emergency-related diversions can be challenging to manage without adequate warning when they rely on first responders from mutual aid located miles away. However, with newer flight-tracking technology websites, smaller airport staff can stay abreast of changing conditions whether they are on or off the airport.

Once a smaller airport is notified of a diversion, operations staff typically notify and communicate diversion status information via a combination of means like phone, email, texts, and radio communication as well as via web-based platforms with smartphone app capabilities that can push mass notifications to internal airport staff, on-airport partners, mutual aid,

Copyright National Academy of Sciences. All rights reserved.

and network partners when needed. Smaller airports do not typically have the personnel to use social media as a communications tool to manage emergencies.

To manage routine diversions, smaller airports typically collaborate with a network of external network partners, including on-airport (e.g., FBO, airline ground handlers, Transportation Security Administration [TSA]) and off-airport (e.g., FAA) departments, organizations, and agencies—all considered within their capabilities. The list of a small airport's partners expands to first responders as a diversion situation escalates. Depending on the authority structure, first responders (e.g., ARFF, law enforcement, and medical responders) can sometimes be obtained within an airport's capabilities. When this is not the case, mutual aid, including local emergency management (e.g., city police, fire, and paramedics), regional (e.g., county emergency management and healthcare systems), and network partners such as community resources (e.g., other airports and busing companies) or federal agencies (e.g., Federal Bureau of Investigation [FBI], Customs and Border Protection [CBP]) may be called upon as diversion situations escalate from irregular operations to incident-level response to emergency situations.

Mutual aid has traditionally been defined as organizations defined by regulation to lend resources or personnel to mitigate emergency situations. These organizations include fire departments, paramedics, and law enforcement. Beyond this, industry research and real-world diversion experience have revealed that smaller airports look to other network partners to care for passengers in the first few hours after a diversion event. These network partners beyond airlines include concessionaires, ground transportation (buses), local stores, restaurants, and hotels—all of which can provide passenger care assistance during flight diversions. For ease of simplicity, all external airport partners in this synthesis are either considered network partners or mutual aid, and contracting with them is indicated as mandatory or voluntary.

While many different types of mutual aid and network partners exist, there are similarities in the way diverted flights are managed as smaller airports typically follow the Incident Command System (ICS), a major component of the National Incident Management System (NIMS), to provide a flexible and scalable system for multiple agencies and disciplines to manage emergencies as well as other unplanned events—even when they are not mandated. In addition, smaller airports use a variety of plans to manage diversions—from standard operating procedures for gas and go technical stops to IROPS plans for incident-related diversions to airport emergency plans (AEPs) for emergency-level diversions. Many non-Part 139 airports report developing AEPs or similar plans even though they are not regulated to do so to be prepared for emergencies such as certain flight diversions. Most smaller airports refer to a combination of plans to manage diversions, while some have developed stand-alone diversion plans, international diversion plans, or diversion checklists.

Smaller airports that have developed diversion checklists include these essential items in them: monitoring procedures for flight, weather, and national airspace conditions; notification triggers and procedures; communication procedures to maintain common situational awareness among mutual aid and network partners during an event; escalation triggers; coordination procedures to access resources, staff and equipment from mutual aid and network partners; and special conditions procedures like after-hours staffing procedures.

Diversion checklists are effective according to smaller airports interviewed and surveyed when they are formatted in one or more of the following ways:

- A separate, stand-alone streamlined document one to two pages in length, easy to find and access;

Copyright National Academy of Sciences. All rights reserved.

4 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

- Specific, clear, and precise instructions in an actionable bullet and dot point format, which divides complicated tasks into simple ones grouped by role and responsibility and task; or
- Tasks ordered sequentially or by escalating factors, available in a digital format or on a smartphone app, and which are verified to be current and up-to-date.

Most small airports have not developed stand-alone plans to manage passenger care needs. Since most diversions to smaller airports are gas and go stops that do not involve deplaning passengers, passenger care (e.g., food or lodging) is not typically necessary. Even for one of the most common diversion causal factors—medical emergencies—these generally result in deplaning one person requiring medical attention. In addition, research demonstrates that many airport personnel believe that most passenger care-related provisions are the responsibility of the air carriers. Nevertheless, some smaller airports do plan for some passenger care needs in their IROPS plans or in airport emergency plans (AEPs). In these, some airports designate secure space for CBP clearance or make provisions for passengers so they can recharge phones, receive information updates, find food and water, or obtain ground transportation and hotel options.

Most SH and NH airports regularly train for aviation emergencies with emergency responders from their mutual aid organizations to fulfill their Part 139 requirements, but only just a few surveyed and interviewed said that they regularly conduct diversion-specific training. Generally, GA airports report that they do not train with mutual aid or network partners for emergencies or diversions, as they are not required by regulations to do so.

Some challenges associated with diversions mainly pertain to how quickly they can escalate into more demanding situations. Some escalating factors include diversions that arrive after normal airport business hours at airports that do not staff around the clock, when non-station international diversions arrive at airports without CBP presence, or when multiple diversions arrive all at once. Construction at smaller airports can also drastically reduce parking and gate capacity, making it difficult to manage unexpected diversions.

Interviews with smaller airports revealed that beyond effective planning efforts, having relationships in place with on-airport (e.g., FBOs, fuelers, or deicers) and off-airport (e.g., FAA or FBI) partners as well as mutual aid (e.g., local first responders) and network partners (e.g., airlines or concessionaires) are critical to the success of managing and mitigating the multitude of effects of flight diversions. Small airports that collaborate with external partners by communicating regularly, sharing diversion-related and other irregular and emergency plans, training together, and conducting after-action reports (AARs) are the most successful at mitigating a variety of situations including diversions.

While trends indicate that the overall number of diversions should continue to rise in tandem with the increasing number of flights post-2020 COVID-19 pandemic recovery, there are indicators that smaller airports may experience a decline in diversions. For instance, emerging robust flight-tracking technologies are just beginning to allow air traffic controllers and pilots to see real-time airport capacities and capabilities. This is assisting them to make wiser decisions that often result in diverting aircraft to larger airports with more capabilities. Also, more regional airport planning groups have formed around major hubs, enabling airports, airlines, and government agencies to work together regularly to ensure that diverted flights are only routed to airports that can manage them at that moment.

In addition, the ramifications of the global pandemic and its associated workforce reduction combined with worldwide political uncertainties causing supply chain disruptions and steeply rising fuel costs have challenged airlines to operate as efficiently as possible. Airlines have minimal economical margins for error. As a result, airlines are reducing the number of overall flight routes, increasing pre-flight planning efforts to avoid holding patterns, and

Copyright National Academy of Sciences. All rights reserved.

only diverting to airports with available or certain capacity to handle them—all to save fuel. One major airline's objective is to avoid diverting to either smaller airports with limited resources or remote airports away from major hubs. All these factors point to potentially fewer diversions to the smallest of airports.

More research would be helpful to determine how to create a nationwide airport capacity and capability database, how to make diversion-related plans adaptable so they can be easily updated after real-world experiences, and how to keep contact lists current. In addition, more research is needed on how smaller airports recover from diversions, especially emergency-related ones.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 1

Introduction

Most flights arrive at their original destination airports; however, some flights can be diverted—most often to large or medium hub airports. However, small, non-hub, and general aviation airports sometimes receive diverted flights as well. This chapter looks at the different types of flight diversions and how they may impact smaller airports.



Flight Diversions

Flight diversions have not been the main subject matter topic of Airport Cooperative Research Program (ACRP) research until now. ACRP reports and syntheses reference diverted aircraft as contributors to irregular operations (Cogliandro, B., et al. 2016) or emergencies (Griffith, D., et al. 2014) and are considerations for airport planning efforts (Nash, J. M., et al. 2012), but diverted flights have not been a primary focus of research.

A diverted flight is "a flight which is operated from the scheduled origin point to a point other than the scheduled destination point in the carrier's published schedule" (14 CFR § 234.2). In other words, a diverted flight is an unexpected or unplanned aircraft that arrives at an unintended airport.

Diversions can be divided into three general types:



1. **Technical stop (gas and go) diversion**
   A stop commonly used to refuel the aircraft or to make unexpected essential repairs. No passengers are unloaded or loaded during a technical stop.



2. **Incident-level diversion**
   An occurrence other than an accident or emergency related to the operation of a diverted aircraft, which affects or could affect the safety of operations, and which results in landing at a non-destination airport. Examples of incident-level diversions include diverted aircraft arriving with minor mechanical issues or structural damage or aircraft arriving with disruptive passengers or passengers with medical issues. Incident-level diversions also include multiple aircraft diverting at the same time to one airport that has limited gate and/or parking availability or an international flight diverting to a non-station airport with no Customs and Border Patrol (CBP) presence.



3. **Emergency-level diversion**
   A situation in which the safety of the aircraft or of the persons on board is endangered for any reason and the aircraft is forced to land at a non-destination airport. Examples include engine failure, smoke in the cockpit, or medical emergencies like cardiac arrest.

While the focus of this synthesis is primarily on incident- and emergency-level diversions, routine technical stops are also discussed throughout.

Copyright National Academy of Sciences. All rights reserved.

Aircraft diversions are simultaneously common and complex.

They are common as they occur almost every single day across the United States and sometimes can be anticipated because of obvious circumstances like massive snowstorms.

Yet, at the same time, each flight diversion is associated with a wide range of other complex factors like the timing of the occurrence, flight crew status, compounding, and escalating issues like medical or mechanical situations, as well as the capacity and capabilities of the airport receiving the diverted flight.



All these extenuating circumstances surrounding diversions are particularly important when considering the available staffing and resources for managing them at small hub, non-hub, and general aviation airports—both internally at the airport and externally with mutual aid and network partners—which is the focus of this synthesis.

How Flight Diversions Impact Smaller Airports



This synthesis provides insight into how smaller airports manage different types of diversions—primarily incident- and emergency-level diversions with a smaller focus on routine diversions—to demonstrate when they can manage diversions with their capabilities under their authority and jurisdiction and when they need assistance from external sources such as the following.



Mutual Aid

These are first-response entities like ARFF, fire department, law enforcement, and Emergency Medical Services (EMS) when they are contracted with an airport through interlocal or joint-use agreements or are managed through another department under municipal control.



Network Partners

These are on-airport partners (e.g., airlines, FBOs, ground handlers, concessionaires), community resources (e.g., stores, hotels, busing companies), federal agencies (e.g., the FAA, CBP, and the Transportation Security Administration [TSA], and military organizations (e.g., the Air National Guard [ANG] or the United States Air Force [USAF]) when they are contracted with an airport either through verbal or written agreements.

More information about these sources can be found in Chapter 2: Defining the Stakeholders.

The term *airport capabilities* refers to the following factors:



Airport Size

Smaller airports are the subject of this synthesis and include small hub (SH), non-hub (NH), and general aviation (GA) airports. However, two other GA categories are included as defined by the FAA National Plan of Integrated Airport Systems (NPIAS), which are nonprimary commercial service (CS) and reliever (R), for additional clarity. Airport size generally indicates how many aircraft are managed daily and informs the type and amount of personnel, training, and resources these airports have on site to manage regular and irregular operations.



Airport Governance

Each of the five smaller airport designations studied in this synthesis (SH, NH, GA, CS, and R) have different governance and management structures, contractual service arrangements, and operating characteristics. These organizational differences determine whether first responders

Copyright National Academy of Sciences. All rights reserved.

8 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

like ARFF, fire department, law enforcement, and EMS are governed by the airport and can provide this capability, or they need outside assistance from mutual aid (e.g., city or county emergency management resources) to perform this critical role. More information about mutual aid can be found in the next section and in Chapter 2.



Airport Infrastructure

Infrastructure refers to the physical features of an airport that include runway type and length, instrument flight rules, terminal size, ramp space, number of common-use gates, and number of boarding bridges. As airport infrastructure is finite, its capacity can become a critical factor during diversions. For example, if an airport receives a surge, which is an influx of several diverted aircraft at one time, this can lead to more aircraft than there are gates or parking areas, and more passengers than can be managed effectively in a terminal.



On-airport Personnel and Expertise

The number of on-airport staff members employed by or contracted with an airport who are trained and qualified to manage equipment and communication technology varies among smaller airport sizes. With limited staff sizes, smaller airports often cross-train employees to perform operations and emergency functions and generally do more with less. Personnel availability can be an issue during diversions, especially if they occur after hours.



Airport Resources

Staff and others contracted by smaller airports need a variety of resources to manage diversions such as the following:

- **Supplies:** Available fuel or deicing chemicals for aircraft and items like bottled water and cots for passenger care.
- **Equipment:** Airstairs, tow bars, snowplows, and fire trucks are among the most commonly used equipment.
- **Technology:** Weather and flight-tracking websites are used to maintain situational awareness while mass notification systems are used to communicate critical information.



Airport and Partner Preparedness

Factors associated with preparedness include planning, training, collaborating, and learning from experience. Smaller airports use stand-alone diversion plans, diversion checklists, standard operating procedures (SOPs), irregular operations (IROPS) plans that include provisions for passenger care, and AEPs. However, some small airports do not reference any plans. And most only train for emergency situations as part of Part 139 compliance, not diversions specifically. The more airports create, share, and update diversion-related plans after diversion events as well as collaboratively train for aircraft diversions with mutual aid and network partners, the more prepared they are to handle the unexpected. Smaller airports that are part of a regional group or emergency working group (EWG) are often most prepared to manage diversions.



Notification

Factors associated with notification include tracking, timing, and airport accessibility. Notification tracking relates to how smaller airports use available technology or establish relationships with network partners like larger hubs in their region to be aware of changing flight conditions. Notification timing impacts how well airports can prepare for a diversion's arrival, whether during or after airport hours. It impacts whether an airport has enough personnel, infrastructure, and resources available and ready when notified of a diversion—whether internal to the airport or from mutual aid or network partners. Accessibility is the ability of decision makers, typically air traffic controllers and airlines, to have accurate 24/7 contact information to notify the correct personnel (e.g., a director or operations manager) at smaller airports through the correct means (e.g., email, radio frequency, or text number).

Copyright National Academy of Sciences. All rights reserved.

Communication



Communication within the airport and among mutual aid and network partners during a diversion event impacts how well the event is managed and mitigated. Communication success depends on whether the airport has predetermined communication channels, frequencies, and procedures (e.g., email, text, or chat feature) for use within the airport and with external partners, and whether the airport can shift to designated alternative (backup) systems if networks become overloaded to continue to coordinate efforts. In addition, communication depends on the use of commonly used and understood language to ensure clarity among all involved in mitigating the effects of a diverted aircraft.

Hours of Operation



Hours of operation can play a big role in determining whether an airport can manage a diversion with its own resources or needs assistance. This is also an important factor when considering common escalating factors that can turn an incident into an emergency or make an emergency even more challenging, such as the following:

- **Off-hours:** When airports receive an aircraft and/or deplaned passengers after an airport's normal business hours. This can pose issues for deplaning, processing (international flights), and rebooking passengers. It also can pose issues meeting basic passenger needs like communication, nourishment, and space to rest.
- **Extended delay:** These mainly occur during long-term weather events when passengers and aircraft are grounded at an airport for longer periods than expected. This can pose issues for flight crews that might time out, for airport supplies that might diminish, and for people who might lose patience.

Considering these factors, an issue with just one, like preparedness or notification, can impact the way a smaller airport can manage a flight diversion that arrives, whether routine or with moderate to severe issues. A greater number of factors that are negatively impacted increases the odds that a smaller airport needs to ask for outside assistance.

Scope of Work



This synthesis consists of published literature, an airport survey, airport interviews, and case examples of current practices from SH, NH, and GA airports that have managed diverted flights from the routine to the emergency-response level. The synthesis documents practices and lessons learned from these smaller airports as well. It also provides several practical sample diversion-related checklists from airports interviewed as well as from other ACRP guidance that can support airport operators and emergency response personnel during emergency and non-emergency flight diversions. It also includes a sample tabletop training exercise and a sample memorandum of understanding (MOU).

Synthesis Overview



The synthesis describes the wide range of services that smaller airports may need to provide during diversion events and gives current guidance on how to plan for and augment staff and resources with local emergency, mutual aid, and network partners. The intended audience for this synthesis report includes airport operators and emergency response personnel at small hub, non-hub, and general aviation airports.

In this synthesis, Chapter 2 defines the stakeholders involved at smaller airports—internal and external to the airport—that manage incident and emergency level responses to flight diversions.

Copyright National Academy of Sciences. All rights reserved.

10 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Chapter 3 provides the methodology for obtaining the data needed to complete this synthesis, which involved a literature review, a survey, interviews, and case examples. Chapter 4 digs deeper into an analysis of the literature review, focusing on how smaller airports manage flight diversions before, during, and after a diversion, including lessons learned and successful practices. Chapter 5 describes and summarizes survey and interview results from smaller airports and how they plan for and manage diverted aircraft. This chapter also provides various perspectives on flight diversion response, from the airport, pilot, FAA, FBO, technology provider, and regional network vantage points. Chapter 6 provides in-depth synopses from seven airports from each NPIAS designation (small hub, non-hub, and general aviation) on their experiences planning for, managing, and mitigating flight diversions associated with incident and emergency response in collaboration with their respective mutual aid networks. Chapter 7 looks at the benefits of collaborative planning associated with diverted flights, while Chapter 8 lists the challenges diversions present. A summary of findings is provided in Chapter 9, and the remaining sections provide a glossary, acronyms, and references. A series of appendices provide interview- and survey-related information and sample diversion checklists, including a Grab and Go diversion checklist (for airport supervisors).

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 2

Defining the Stakeholders

To identify and understand who is involved with managing flight diversions associated with incidents and emergencies at smaller airports, stakeholders have been divided into the following two broad categories:



- **Internal Airport Stakeholders:** People, departments, or divisions employed by an airport organization.
- **External Airport Stakeholders:** People, departments, divisions, organizations, or agencies outside of an airport's authority, either on or off the airport. These are divided into the following two main categories:



  - **Mutual Aid:** These are first-response entities (e.g., fire, medical, or law enforcement) that are usually contracted with an airport through interlocal or joint-use agreements or are managed through another department under municipal control. Sometimes these arrangements are called mutual aid agreements. A mutual aid agreement is defined as: "a written agreement between agencies, organizations, or jurisdictions to lend assistance across jurisdictional boundaries. It agrees to assist by furnishing personnel, equipment, and expertise in a specified manner at requisite time" (USLegal.com 2022). Responsibilities related to these agreements are usually outlined in an AEP.



  - **Network Partners:** These are on- and off-airport partners (e.g., airlines, airport tenants, regulatory agencies, or community resources) that can provide services and resources for aircraft movement and maintenance, passenger and cargo screening and clearance, customer service, and passenger care, among other things. Network partners are contracted with an airport either through verbal or written agreements to define the personnel, equipment, or expertise that will be offered during normal operations as well as during irregular-, incident-, or emergency-related situations. Responsibilities related to these agreements are typically outlined in an airport's SOPs or IROPS plan.



Internal Airport Stakeholders



To create one internal airport stakeholder list common to smaller airports, the "Expanded List of Stakeholders" from *ACRP Report 153* was first examined and compared against industry knowledge, working with airports to edit the list to align with the typical roles, departments, and divisions present at small, non-hub, and general aviation airports (Cogliandro, B., et al. 2016).

Beyond this, airport stakeholders at smaller airports are inconsistent both within their NPIAS designation (e.g., among the 21 surveyed out of 69 small hub airports) or across them (e.g., among the 24 surveyed out of 335 non-hub versus 16 out of 3,242 general aviation airports, and seven out of 611 reliever and five out of 451 nonprimary commercial service airports).

Copyright National Academy of Sciences. All rights reserved.

In addition, no two airports are alike in terms of governance and management structures, contractual service arrangements, and operating characteristics. For instance, according to *ACRP Report 116*, airports can be owned and operated by the following jurisdictions: airport authorities, counties, municipalities, joint county–city commissions, park districts, port authorities, bi-state authorities, and private owners (Price, M. 2014). Airports can also share staff and resources with their governing body (e.g., maintenance staff or equipment). Therefore, the list of internal stakeholders and mutual aid and network partners differ for each airport.

However, there are general key roles and responsibilities that are common to these smaller airports despite differences in the semantics of naming positions or departments. Therefore, this list was created with exceptions noted and examined below.

Typical Internal Airport Stakeholders at Smaller Airports (e.g., SH, NH, GA, CS, R)

- Airport Director and General Manager
- Operations Manager
- Public Safety and Emergency Manager and Duty Manager
- Maintenance Manager
- Public Affairs and Public Information Officer
- ARFF*
- Law Enforcement*
- Medical*

**See note below about first responders for a discussion on various authority structures.*

Typical Airport Stakeholder Roles during Diversions

Most smaller airports indicate that their airport director or operations manager takes the lead in managing diversion events and securing mutual aid assistance as needed. However, sometimes smaller airports designate this authority to other personnel like public safety managers, emergency managers, or duty managers at airports with larger staffing capabilities. Even though the names of the roles differ for smaller airports, leaders in these roles all take on similar responsibilities during diversion events as described in Figure 1.

Other major roles and responsibilities during diversions at smaller airports include:

- Maintenance Manager–Manages the airport terminal or facility, maintaining passenger and customer care (family assistance areas, restrooms, trash collection, etc.).

Lead Airport Stakeholder Role	Diversion Responsibilities
• Airport Director/General Manager • Operations Manager • Public Safety Manager • Emergency Manager • Duty Manager	• Tracks potential diversions and local weather conditions. • Communicates with the Air Traffic Control (ATC) and/or airlines about diverted aircraft. • Notifies internal personnel of incoming diversion(s) and contacts external stakeholders (e.g., mutual aid and network partners) as needed. • Records information related to diversion aircraft. • Continues to communicate with airport personnel, mutual aid, and network partners to manage and mitigate the effects of the diverted aircraft. • Manages or assists with resource coordination to ensure aircraft (gate, parking, deplaning, etc.), passenger and crew (passenger care, information, screening, clearance, etc.), and emergency (fire, medical, etc.) needs are met.

Figure 1. Typical lead airport stakeholder responsibilities for diversions.

Copyright National Academy of Sciences. All rights reserved.

- Public Affairs and Public Information Officer–Coordinates communication and information related to the diversion event either directly with media and passengers or cooperatively in conjunction with other mutual aid organization Public Information Officers (e.g., city or county emergency management agencies).

In addition, many of the roles mentioned earlier may be combined at smaller airports, and many small airports cross-train their employees to fill one or more roles, including first responders.

A Note About First Responders: The authority over first responders is inconsistent at smaller airports, as determined by interviewing and surveying smaller airports for this synthesis. Sometimes ARFF, law, and medical responders are considered under the airport's authority either using in-house staff or third-party contracted staff. In other cases, first responders are considered mutual aid and are contracted through interlocal or joint-use agreements or are managed through another department under municipal control. Generally, about half of smaller airports (e.g., SH, NH, GA, CS, R) have authority over ARFF, while about less than half say they have authority over law and medical services. A deeper look at the differences can be found in the survey chapter. Therefore, because these roles are split somewhat evenly across smaller airports, they can be considered either an internal or external stakeholder. This disparity impacts smaller airports in their management of diverted flights and is highlighted throughout the synthesis.

External Airport Stakeholders

To create an external stakeholder partner list common to smaller airports, lists of stakeholders were taken from ACRP guidance on IROPS planning (Cogliandro, B., et al. 2016) as well as those from emergency planning (FAA 2009) as these provided lists of stakeholders associated with irregular operations like flight diversions with incidents and with flight diversions with emergencies (Table 1).

Roles of Mutual Aid and Network Partners During Diversions

Mutual Aid and First-Response Partners

- **ARFF:** On-airport first responders that provide safety and emergency medical duties as well as combat and extinguish fires associated with diverted aircraft.
- **Fire Department:** Off-airport community first responders that provide the same duties as ARFF.
- **EMS:** Provides paramedic services for diverted aircraft passengers and crew.
- **Law Enforcement:** On- or off-airport resource that provides control and response efforts related to a diversion. Sometimes this is referred to as a Law Enforcement Officer (LEO).
- **Local Emergency Management (EM) Agency and Office:** A local office that coordinates first responders during an emergency-related diversion.
- **Local Healthcare and Hospitals:** Partners that provide medical needs for passengers and crew from a diverted flight.
- **Local Public Health Department:** A partner that can detain, medically examine, and release persons arriving on a diverted domestic flight who are suspected of carrying communicable diseases.

Copyright National Academy of Sciences. All rights reserved.

Table 1. Typical mutual aid and network partners at smaller airports.

Mutual Aid	Network Partners	
First-Response Partners • Aircraft Rescue and Firefighting (ARFF)* • Law Enforcement* • Emergency Medical Services (EMS)* • Fire Department • Local Emergency Management (EM) Agency/Office • Local Healthcare/Hospitals • Local Public Health Department (domestic flights) **Sometimes under airport authority*	**On-Airport Partners** • Airlines • Fixed-base Operators (FBOs) • Fuelers • Deicers • Ground Handlers • Hangar/Aircraft Owners • Concessionaires • Other Airport Tenants • Ground Transportation Companies (rental cars, shuttle buses, rideshare) • Airport or Other Volunteer Group **Military** • ANG, U.S. Air Force (USAF), or other military joint use on airport	**Agencies** • Federal Aviation Administration (FAA) • Federal Bureau of Investigation (FBI) • Customs and Border Patrol (CBP) • Centers for Disease Control and Prevention (CDC) (international flights) • Transportation Security Administration (TSA) • National Transportation Safety Board (NTSB) • U.S. Marshals • Air Marshals **Community Resources** • Ground Transportation (buses) • Local Stores • Local Restaurants • Local Hotels • Convention and Visitors Bureau • City/County Management

**See note (on previous page) about first responders for a discussion on various authorities.*



On-Airport Partners

- **Diverted Aircraft Airline:** Works with air traffic control (ATC) to notify and coordinate diverted flight information with the diversion airport. Communicates incident- or emergency-related diverted aircraft or passenger needs with a diversion airport and/or first responders when necessary. Provides customer service to passengers as needed (e.g., rebooking). Provides families with "an appropriate level of services and assistance" as mandated in emergency-related diversion situations that involve a fatality (49 U.S. Code § 41113). Sometimes shares resources and equipment (e.g., airstairs for deplaning) through agreements with the diversion airport.
- **FBOs/Fuelers/Deicers/Ground Handlers:** Assist with the diverted aircraft's movement, positioning, preparation, fueling, and deicing as well as customer service and ramp service according to any agreements with the airport or airline.
- **Hangar/Aircraft Owners:** Assist with temporary parking needs and share equipment or staffing to assist diverted aircraft according to any agreements with the airport.
- **Concessionaires:** Include restaurants, stores, and snack stands. Accesses extra supplies of food, water, and special provisions (e.g., diapers) and provides after-hour staffing if needed according to any agreements with the airport.
- **Other Airport Tenants:** Provide extra staffing or equipment or supplies according to any agreements with the airport.
- **Ground Transportation:** Includes rental cars, taxis, shuttle buses, local mass transit, and ride-share companies. Can provide increased services according to agreements with the airport.
- **Airport or Other Volunteer Group:** Provides staffing to assist the diversion airport with customer care.



Community Resources (Passenger Care)

- **Ground Transportation:** Provides vehicles or buses to transport passengers to local hotels or other destinations according to any agreements with the airport. These can come from busing companies or from local school districts.

Copyright National Academy of Sciences. All rights reserved.

- **Local Stores:** Provide supplies like food, water, prescriptions, and family needs (e.g., baby formula) according to any agreements with the airport.
- **Local Restaurants:** Provide food and water according to any agreements with the airport.
- **Local Hotels:** Provide special lodging arrangements or rates according to any agreements with the airport.
- **Convention and Visitors Bureau:** Provides information related to available lodging in an area.
- **City and County Management:** Provides resources as specified according to agreements with the airport.

Agencies



These agencies provide mandated services to airports, but airports can make additional agreements with these agencies depending upon unique needs, especially those related to diversion aircraft. Considerations for additional agreements are discussed in more detail in Chapter 4 Literature Review.

- **FAA:** Air traffic controllers work with airlines and pilots as they make diversion-related decisions.
- **FBI:** Provides federal law enforcement and intelligence to support airports with diversions that involve unruly passengers or other security concerns.
- **CBP:** Provides traveler processing procedures for diverted flights and procedures for deplaning passengers to secure areas at diversion airports.
- **Centers for Disease Control and Prevention (CDC):** Can detain, medically examine, and release persons arriving on a diverted international flight who are suspected of carrying communicable diseases.
- **TSA:** Provides traveler screening services for those who deplane from a diverted flight and leave an airport's secure area, requiring re-screening.
- **U.S. Marshals and Air Marshals:** These law enforcement officers are authorized to protect aircraft by deterring and countering unlawful activity (e.g., unruly passengers), sometimes involving diverting aircraft.
- **National Transportation Safety Board (NTSB):** Investigates every civilian aviation accident in the United States. The NTSB determines the probable cause of the accidents and issues safety recommendations aimed at preventing future accidents.

Military



The military often shares space on joint-use airports, and airports can make additional agreements with military units depending upon unique needs, especially related to diversion aircraft. Considerations for additional agreements are discussed in more detail in Chapter 4.

- **ANG, USAF, or other military joint use on airports:** Military forces on joint-use airports provide firefighting and medical staff, resources, and equipment to assist airports to manage diverted aircraft.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 3

Methodology

This synthesis study consisted of a literature review, a survey for smaller U.S. airports, one-on-one interviews, and case examples.



Literature Review

Literature researched for this synthesis focused on four main topics associated with how smaller airports collaboratively plan for, manage, and learn from diverted flights involving technical stops, incidents, and emergencies. These include

- Background information on diversions,
- Planning and training before a diversion event,
- Considerations during a diversion event, and
- What happens after a diversion event.

Searches were performed on both the open web using search engines like Google.com and the deep web using the Transport Research International Documentation (TRID), the Transportation Research Board (TRB) database, and ProQuest. Most peer-reviewed literature about diverted flights was limited to understanding processes for selecting diversion airports and the operating costs associated with medical diversions; however, a few pertinent documents were discovered related to the impacts of diversions on airport operations. Literature, other than ACRP guidance, specifically related to small, non-hub, and general aviation airports was also limited. Previous ACRP research and synthesis reports did provide essential operational and management background information for these types of airports, while several FAA, United States Department of Transportation (U.S. DOT), Federal Emergency Management Agency (FEMA), and NTSB documents provided more emergency management-related information. The result of the literature review is presented in more detail in Chapter 4.



Survey

An online survey was created for SH, NH, and GA airports, including nonprimary CS and R airports. The questionnaire asked questions about preparing for and managing flight diversions involving incidents and emergencies. As Table 2 indicates, a survey was sent to approximately 300 small hub, non-hub, and general aviation airports in the United States (71 survey responses and 24% response rate).



Interviews

Interviews were conducted with airport directors, operations, and emergency managers at a mix of small, non-hub, and general aviation airports with a variety of diversion-related planning

Copyright National Academy of Sciences. All rights reserved.

Table 2. U.S. survey participants by NPIAS designation.

Airport Size (NPIAS Designation)	Number Surveyed (U.S.)	Total Airports in Designation	Percent of Designation Surveyed
Small hub airport (SH)	19	69	28%
Non-hub primary (NH)	24	335	7%
Nonprimary commercial service (CS)	5	451	1%
Reliever (R)	7	611	1%
General aviation (GA)	16	3242	<1%

Source: FAA 2020b.

and management experiences over the past few years. Interviews were conducted to gain a broad understanding of the planning they do to prepare for diversions, how and who they coordinate with in terms of their mutual aid and network partner resources, what plans they have developed and how practical they are, and what current practices and lessons learned they can offer from their experiences. The choice of airports to interview was determined in part by industry knowledge of recent diversion events or through the literature review process where other related ACRP research revealed which smaller airports recently managed diversion aircraft. The literature review process also included web research (e.g., Google) that revealed applicable diversion events at smaller airports. A list of airports interviewed for this synthesis can be found in Figure 2, and Figure 3 identifies a map that highlights where these airports can be found geographically.

SMALL HUB AIRPORTS

1. Dane County Regional-Truax Field, Madison, WI
2. The Eastern Iowa Airport, Cedar Rapids, IA
3. Des Moines International Airport, Des Moines, IA
4. Portland International Jetport, Portland, ME
5. Halifax Stanfield International Airport, Goffs, Nova Scotia, Canada
6. Sioux Falls Regional Airport, Sioux Falls, ND
7. Punta Gorda Airport, Punta Gorda, FL
8. Palm Springs International Airport, Palm Springs, CA

NON- HUB AIRPORTS

1. Lincoln Airport, Lincoln, NE
2. Duluth International Airport, Duluth, MN
3. Rochester International Airport, Rochester, MN
4. Grand Forks International Airport, Grand Forks, ND
5. Hector International Airport, Fargo, ND
6. Central Wisconsin Airport, Mosinee, WI
7. Sioux Gateway/Brig Gen Bud Day Field, Sioux City, IA
8. Rick Husband Amarillo International Airport, Amarillo, TX
9. St. John's International Airport, St. John's, Newfoundland, Canada

GENERAL AVIATION AIRPORTS

1. Iqaluit International Airport, Iqaluit, Nunavut, Canada
2. Ocala International Airport, Ocala, FL
3. Flying Cloud Airport, Eden Prairie, MN (R)
4. Airlake Airport, Lakeville, MN (R)
5. Anoka County-Blaine Airport, Blaine, MN (R)
6. Crystal Airport, Crystal, MN (R)
7. Lake Elmo Airport, Lake Elmo, MN (R)
8. St. Paul Downtown Airport, St. Paul, MN (R)

Figure 2. Airports interviewed for this synthesis.

Copyright National Academy of Sciences. All rights reserved.

18 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports



Figure 3. Map of small, non-hub, and general aviation airports interviewed for this synthesis.

In addition to interviewing smaller airports, research also included discussions with large hub (LH) and medium hub (MH) airports that are part of a regional network, including Minneapolis-St. Paul International Airport (MSP), Dallas/Fort Worth International Airport (DFW), Denver International Airport (DEN), and Eppley Field (OMA).

To get a broader understanding of others involved with managing diversion, specifically a few of the major mutual aid partners called upon to provide additional resources to airports, interviews were conducted with the following:

- Pilots at an airline as well as Airlines for America (A4A) to gain an understanding of how they make diversion-related decisions and how they notify and communicate intentions to airports
- One FBO to understand the support and staffing they provide during all types of diversions
- Four technology partners to see what they provide airports to use to get synchronous information is on the horizon
- The FAA to understand the ATC perspective in diversion decision-making efforts

Interview questions are included in Appendix A, and Appendix B summarizes the survey responses.

Copyright National Academy of Sciences. All rights reserved.

Case Examples

The following airports are highlighted as case examples in this synthesis and bring knowledge gained from managing and mitigating flight diversions associated with incident and emergency response in collaboration with their respective mutual aid and network partners:

- Dane County Regional-Truax Field, Madison, WI — SH
- Punta Gorda Airport, Punta Gorda, FL — SH
- St. John's International Airport, Newfoundland, Canada — NH
- Lincoln Airport, Lincoln, NE — NH
- Rick Husband Amarillo International Airport, Amarillo, TX — NH
- Iqaluit International Airport, Iqaluit, Nunavut, Canada — GA
- Flying Cloud Airport, Eden Prairie, MN — GA/R

Case example write-ups can be found in Chapter 6.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 4

Literature Review

The literature review introduces flight diversions and discusses why diversions occur, how often diversions occur in the United States, and how diversion decisions are made. It reviews how smaller airports plan and train for diversions of varying complexity, how they manage and mitigate diversions, and how they restore operations after a diverted flight. A list of lessons learned and good practices that smaller airports are currently following is also provided in this chapter.

Introduction to Flight Diversions

Why Diversions Occur

A lot of research and information exists to explain *why* various diversions occur (Ball, J. 2008; Ryerson, M. S. 2018). Causal factors have been simplified into the following three categories:

- Factors outside the aircraft are considered anything exterior to the plane that directly contributes to a need to divert. The most common of these external factors is weather conditions. Anything from severe or prolonged storms, to fog, to high winds can cause flights to be rerouted and diverted. Other common factors are construction projects, heavy airspace traffic volume, or safety issues (e.g., power outages, runway damage) at an originally intended destination airport or even unmanned aircraft vehicles (e.g., drones) interrupting airspace around a destination airport.
- Factors associated with the aircraft include low fuel levels, minor mechanical issues (e.g., oil leaks, warning lights, etc.) to emergency issues (e.g., fires, structural damage, mechanical failures, etc.). In addition, the recent rollout of the fifth generation (5G) technology for standard wireless networks revealed how upgrades to communication technology impacted vital airborne and ground-sensing systems on aircraft, posing a critical safety risk for aircraft landing at certain airports during inclement weather conditions (FAA 2022a). Therefore, aircraft technology upgrades and even new technology that impacts aircraft can also be considered contributing diversion factors.
- Factors inside the aircraft include passengers who are disruptive or cause security issues. These have increased dramatically over the past few years—from 146 FAA investigations of unruly passengers in 2019 to 189 investigations in 2020 and 1,099 investigations in 2021 (FAA 2022b). However, the main cause of diversions (behind weather as indicated above) are passengers or crew members with medical issues ranging from minor (e.g., passing out) to emergency level, the most common being cardiac related (Epstein, C. R., et al. 2019).

How Often Diversions Occur

According to the most recent U.S. DOT data available and as demonstrated in Table 3, from January 2012 through November 2021, out of approximately 60 million flight operations during

Copyright National Academy of Sciences. All rights reserved.

Table 3. Diversion flight trends over the past 10 years.

Year	Annual Number of Diversions	Annual Total Operation	Diversions as a Percentage of Total Operations
2012	12,519	6,096,762	0.21%
2013	14,160	6,369,482	0.22%
2014	14,449	5,819,811	0.25%
2015	15,187	5,819,079	0.26%
2016	13,652	5,617,658	0.24%
2017	12,530	5,674,621	0.22%
2018	17,859	7,213,446	0.25%
2019	18,880	7,422,037	0.25%
2020	7,744	4,688,354	0.17%
2021	14,117	5,995,397	0.24%
TOTAL	**141,097**	**60,716,647**	**0.23%**

Source: Bureau of Transportation Statistics 2022.

this 10-year period, 141,097 flights were diverted, which equates to 0.23% of all flights in the United States. (Note the diversion flight decreases in 2020 and 2021 account for the reduced flight operations because of the COVID-19 global pandemic.)

The average number of annual diversions over the past 10 years is approximately 14,000 flights. However, peak delays occur in the United States during the months of June and July and the least occur in November and March, as demonstrated with numbers from 2019 in Figure 4 (Spak, M., and P. Olexa 2020).

Out of the approximate 18,000 diversions that occurred in 2019, SH, NH, and GA airports received 40% of these (7,154 flights). SH and NH airports received the most diversions out of all the smaller airport categories in 2019 (96%) as demonstrated in Table 4.

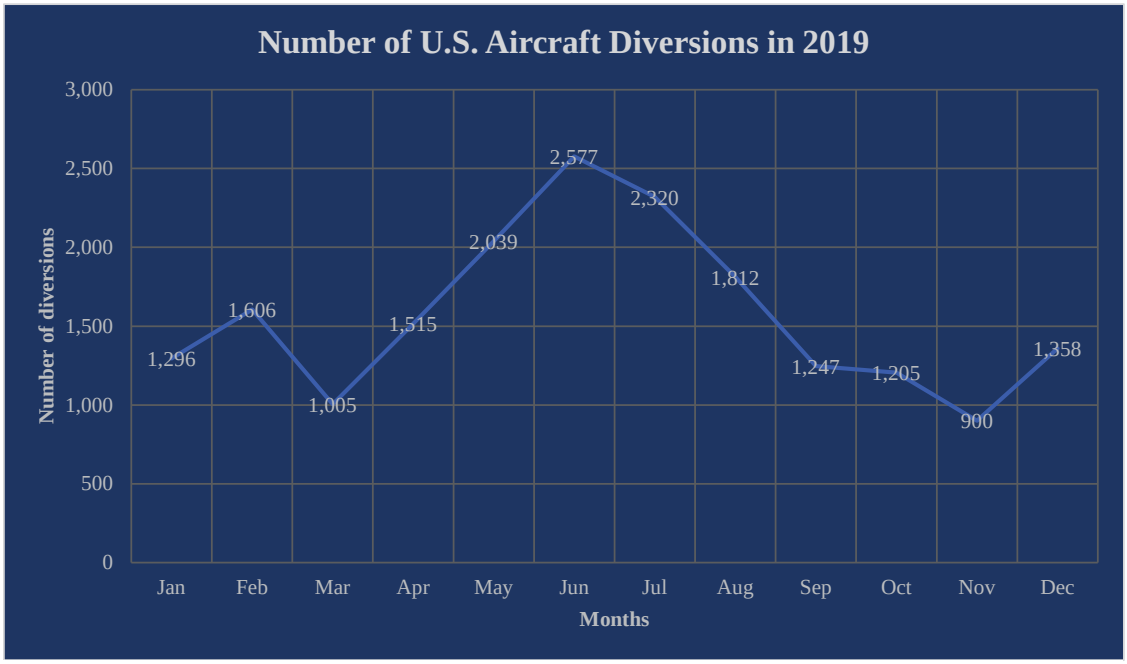


Figure 4. Number of diversions that occurred in 2019 by month (Source: Bureau of Transportation Statistics 2022).

Copyright National Academy of Sciences. All rights reserved.

22 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Table 4. Diverted flights received at smaller airports in 2019.

NPIAS Category	Number of Diversions Received in 2019	Number of Airports that Received Diversions (Out of Total Airports in Each Category)	Percent of Smaller Airports that Received Diversions (Out of Total Airports in Each Category)
Small Hub (SH)	4,294	69 (out of 69)	100%
Non-hub(NH)	2,582	156 (out of 335)	46%
General Aviation (GA)	34	21 (out of 3,242)	<1%
Commercial Service (CS)	84	5 (out of 451)	1%
Reliever (R)	160	34 (out of 611)	5%
TOTAL	**7,154**	285 (out of 4,708)	-

Source: FAA n.d.

How Diversion Decisions Are Made

While outside the scope of this synthesis, a quick look at the decision-making process associated with choosing a diversion airport reveals the complexity of this process and how it focuses on balancing safety and urgency (e.g., mitigating the incident or emergency) with manageability and practicality (e.g., fastest turnaround time and minimal disruption to passenger service).

The diversion decision-making process begins before each flight when the pilot-in-charge and airline flight dispatcher create a flight plan. During this pre-flight planning stage, particularly for long-haul oceanic flights, a few "diversion points," which are potential diversion airports, are assigned along the expected flight route and included in the flight plan. Flight plans are then submitted to the FAA air traffic controller (ATC) for approval.

For the most common type of diversion, weather-related, pilots will coordinate with airline dispatch and then ask an FAA ATC to divert to an airport near the destination airport. This is a common practice based on optimism that the flight can quickly reach its intended airport more quickly once the weather clears.

For most non-emergency situations, diversion airport decisions hinge upon the remaining fuel left in the diversion aircraft (real-time data) and the capabilities of the diversion airport (static data). The fuel gauge enables the pilot-in-command and airline dispatch to make calculations and narrow down the pool of potential diversion airports (Malandri, C., et al. 2020). Other decision-making factors for non-emergency diversions beyond runway length, apron facilities, and capacity that may be considered by dispatch include the status of

- Navigation equipment,
- Aircraft parking,
- Gate availability,
- Customs capacity,
- Refueling assets,
- Deicing assets,
- Jetway and airstair access,
- General ramp operations,
- Security, and
- Station airline and/or code share airline.

However, in emergency instances, a pilot may request to deviate from the potential diversion airports listed on the original flight plan. In these cases, the pilot will typically request this change from an en-route ATC located in an air route traffic control center, or from an Approach Controller if in terminal airspace. When these requests occur, ATCs rarely deny a change in diversion airports.

Copyright National Academy of Sciences. All rights reserved.

For mechanical and medical emergencies, proximity becomes the primary factor no matter what the airport may or may not be equipped to handle. For medical-related diversions, which are near-daily occurrences, these often necessitate immediate landing (Ryerson, M. S. 2018).

Once the alternative diversion airport is selected, an ATC updates the flight plan, clears the aircraft to the new airport, and serves as a conduit for providing capacity-related information at the diversion airport. The dispatcher at an airline's operations center coordinates flight and service needs such as ground services with a diversion airport.

It is not clear if real-time data is necessary for effective diversion management or whether information gathered during pre-flight planning provides enough data on potential diversion airports. Nevertheless, no nationwide database of all airport capacities and capabilities exists for pilots, dispatch, and air traffic control to potentially base diversion-related decisions on "live" real-time data (Spak, M., and P. Olexa 2020). Interviews with technology firms and pilots as part of this synthesis project show that this may be changing as new web-based technologies continue to be developed to improve and provide this type of data.

However, while technology platforms allow for the collection of certain data, significant challenges remain in terms of consistent metrics, staffing to keep real-time data current, integration across vendors, and assimilation into airline operation decision-making. In addition, these databases may not necessarily solve the airport notification timing issue, nor give a full understanding of an airport's capability. For instance, the database may not capture information like which airport staff members may be on vacation or may have gone home ill, which equipment (like a snowplow) may be out of commission for servicing, or even how much time it may take to secure additional personnel and resources, particularly those in rural or remote areas.

The Role Smaller Airports Play in Managing Diversions



Smaller airports often serve as a safety net in the National Airspace System by regularly managing aircraft diversions. These airports fall under multiple FAA NPIAS classifications such as SH, NH, GA, CS, and R airports (FAA 2020a). The largest group of airports in the United States is GA airports and these airports perform one of the five most important aeronautical functions according to the FAA because they serve as emergency diversionary airports. This extensive system of GA airports gives pilots alternatives during incidents and emergency situations. Their availability reduces accidents related to these types of flights (FAA 2012).

All airports, including even the smallest GA airport, are expected to accept diverted flights. According to 49 U.S. Code §47107, airports are required to be open and available for all flights. However, the impacts of diverted flights on larger versus smaller airports vary widely. For instance, smaller airports may not have a lot of congestion to contend with during a diversion event compared with a larger airport, but they may only serve one or two airlines, which may make it harder to manage the diversion if an airline does not have a base of operations there or if the airline only operates a few days out of the week. While smaller airports do not have the infrastructure (e.g., gates or parking areas), resources, or staffing of larger airports, they can access resources and personnel when they plan effectively with others, including mutual aid and network partners. More on this is discussed in the next few sections.

Before a Diversion: Planning, Training, and Coordinating



While the focus of this synthesis is primarily on incident- and emergency-level diversions, routine technical stops are also included as they form the baseline of diversion-related planning efforts. The type of planning, training, and coordination that smaller airports currently use during diversions is highlighted in this section.

Copyright National Academy of Sciences. All rights reserved.

Table 5. Airport capabilities needed for technical stops.

Capability Needed	Stakeholder Involved	Capability Examples
Physical Infrastructure	Airport Operations	Runways, aprons, ramps, tarmac, hard stands, gates
Communications/ Coordination	Airport Operations	Technology for flight tracking and for contacting/notifying stakeholders (text)
Aircraft Handling	FBO* or Ground Handler*	Tow bars or other related equipment for aircraft movement/positioning/preparation, fueling/deicing, customer service and ramp service
Supplies	FBO*, Fueler* or Deicer*	Fuel and deicing chemicals (during winter operations)

**On airport network partner.*



How Small Airports Plan for Routine Technical Stop Diversions

Technical stops (gas and go) are flights that divert to an airport to refuel or to make unexpected essential repairs. No traffic is unloaded or loaded during a technical stop. These gas and go diversions are considered a routine procedure that airports can manage under their own authority and with their own resources. Diversion-related capacity and capabilities needed for unplanned, yet typical "gas and go" technical stops at a diversion airport are shown in Table 5.



The important role of FBOs and ground handlers: Aside from airports providing the physical infrastructure, staff, and technology needed to communicate and coordinate diverted flights, services and equipment provided by FBOs and ground handlers are critical needs of diverted flights. The types of diversion-related equipment and services they provide can be found in Tables 5 and 6. Ground handlers and FBOs are both categorized as part of an airport's capabilities as they play an integral and day-to-day part of airport operation. However, they can also be categorized as an on-airport network partner. (See Chapter 2 for more on Stakeholder Definitions.)

The International Air Transport Association (IATA) estimates that airlines outsource more than 50% of the ground handling that takes place at the world's airports (IATA 2013). Ground handling at smaller airports is typically provided through service-level agreements between FBOs and airports. However, airlines sometimes provide ground handling via their employees or third-party contracted personnel.



Airport-Related Plans for Technical Stops: Small airports are not mandated to use specific plans or procedures to manage technical stop flight diversions. However, smaller airports typically use SOPs in their planning efforts. SOPs are used for diversion flights just as they are for regularly planned flights. These typically include activating written agreements with local ATC and airport authorities to coordinate these diversions and with FBOs and fuelers to provide fuel and/or mechanical assistance as needed.

Note about plans: While the various plans are described in this chapter, the number of airports that use these plans can be found in Chapter 5.

Copyright National Academy of Sciences. All rights reserved.

How Small Airports Plan for Diversions with Incidents



According to 49 CFR § 830.2, an incident means an occurrence other than an accident, associated with the operation of an aircraft, which affects or could affect the safety of operations. Incident-related flights can no longer fly to their destination airport, but the immediate safety of the aircraft and people on board are not in danger. This typically occurs because of situations outside the aircraft (e.g., weather), with the aircraft (e.g., minor mechanical issues), or within the aircraft (e.g., passenger posing a security hazard).

Plans for incident-related diversions: These types of flight diversions fall between a routine technical stop and an emergency, depending on their severity and/or compounding conditions. Some of the main plans smaller airports use to manage incident-related diversions include:

- **IROPS Plan:** Some smaller airports use IROPS plans to manage incident-related flight diversions. They can also be applicable for technical stops that escalate into incident-level diversions. Airport IROPS plans were introduced in *ACRP Report 65* along with provisions for planning for and managing diverted flights, including the following:
  - Flight and weather tracking, collaborative notification, and communication procedures;
  - Ramp and tarmac parking plans for diverted flights;
  - Provisions for diversion airports without CBP presence;
  - The development of diversion airport capacity and capabilities (resources and equipment) lists; and
  - The coordination of passenger care needs, including nourishment, transportation, lodging, and special needs such as medication and baby supplies (Nash, J. M., et al. 2012).

  ACRP Report 153 (Cogliandro, B., et al. 2016) was created in 2016 as a follow-up to *ACRP Report 65*. It was created in part to provide additional diversion-related practices, including understanding and sharing airport resources within a geographic area and the use of flight diversion technology. Both resources have been widely supported by industry organizations and have helped shape many airport IROPS plans.
- **Airport Diversion Plan:** Some smaller airports have developed their own stand-alone, dedicated plans for diversions. These include procedures for a diverted flight notification, communication, and coordination among the airport as well as mutual aid (first-response) and network partners.
- **International Flight Diversion Plans:** Some airports have developed diversion plans specific to international flights, which tend to involve bigger aircraft (e.g., widebody jets), many passengers, and CBP passenger processing regulations. These plans cover a unique gate, runway, and parking pad needs of larger aircraft; aircraft deplaning; crowd control; after-hour provisions; and contingency planning for non-station airline flight diversions. An international diversion plan template is included in Appendix C. In addition, the CBP posted in a memo the proper diversion guidance for commercial carriers to address unscheduled arrivals, including flight diversions and technical fuel stops and response procedures for international flights diverted to airports without a Federal Inspection Station (CBP 2019).
- **Airport Diversion Checklist:** A handful of smaller airports have developed diversion checklists as practical and actionable plans for use in real time. *ACRP Report 65* provides a sample diversion checklist that is also provided in Appendix C (Nash, J. M., et al. 2012). This diversion checklist is grouped by roles and includes procedures to use before, during, and after a diversion event. Also included in Appendix C is a Grab and Go Diversion Checklist (for airport supervisors) and a Sample Airport International Diversion Checklist (from MSN Airport).
- **Extended Delay Contingency Plan and Emergency Contingency Plan:** These plans are mandated for airlines, small hub, and non-hub airports, and operators of airports used for diversions by air carriers that provide "covered air transportation." In Section 432301 of the FAA Modernization and Reform Act of 2012, airlines are to develop and coordinate their

Copyright National Academy of Sciences. All rights reserved.

26 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

extended delay contingency plans with both the scheduled airports they serve and their diversion airports. In addition, airport operators must develop and post emergency contingency plans with the DOT and upload updated plans every 5 years. Airports must describe how, "following excessive tarmac delays," airport operators will "to the maximum extent practicable: (1) Provide for the deplanement of passengers; (2) Provide for the sharing of facilities and make gates available at the airport; and (3) Provide a sterile area following excessive tarmac delays for passengers who have not yet cleared United States Customs and Border Protection (CBP)." While these plans are more focused on protecting airlines from violating the 3-hour Domestic and 4-Hour International Tarmac Delay Rules, these do provide a few key elements that can be used in incident- and emergency-level diversions that pertain to deplaning, sharing resources, and managing international diversions as described above.

- **Business Continuity Plan (BCP) or Continuity of Operations Plan (COOP):** Smaller airports also use these as part of their overall diversion-related planning efforts. However, these plans are not used to manage diversion events, but rather to assist with assessing risks (e.g., reliability of communication systems) related to future potential situations such as flight diversions. Information from BCPs and COOPs can be used to revise or update diversion-related management plans.
- **Airport Emergency Plan:** These plans have specific provisions for aircraft incidents. See the emergency section for a complete discussion about how this plan is used for both incidents and emergencies related to diversions.

While recent literature, mainly ACRP-related guidance, and directives from industry organizations like Airports Council International-North America (ACI-NA) and the American Association of Airport Executives (AAAE), support the development and use of IROPS and diversion-related plans for managing diversion aircraft, no research exists on the efficacy of these plans or how well they help airports mitigate diversion events.



How Small Airports Plan for Diversions with Emergencies

Emergencies develop on aircraft because of one or more factors related to outside the aircraft (e.g., lightning strike), with the aircraft (e.g., fire), or within the aircraft (e.g., medical emergency). Flights with emergencies usually can no longer fly to their destination airport because of the immediate safety needs of the aircraft and because the people on board are in danger.

When diverted flights arrive with emergency-level issues, needs for capacity and capabilities expand and typically more external stakeholders are needed to manage these types of situations. This list in Table 6 continues from Table 5 (gas and go example) and expands to include more potential needs.



Plans for flights with emergencies: To manage diversion-related emergencies at airports, both mandated and non-mandated plans apply.

- **Standard Operating Procedures:** Specific SOPs are developed for emergency response in accordance with the AEPs and other emergency plans.
- **Airport Emergency Plan:** As mentioned previously, the AEP makes provisions for both incidents and emergencies. An airport emergency plan is mandated for all airports holding an FAA 14 CFR, Part 139, operating certificate. While all small hub airports and 74% of non-hub airports are classified "Part 139" certified, only 3% of all general nonprimary commercial service, reliever, and general aviation airports are required to have an AEP. However, according to AC 150/5200-31C, these smaller airports must follow the general guidelines prescribed by Homeland Security Presidential Directive 5 (HSPD-5), Management of Domestic Incidents and Homeland Security Presidential Directive 8 (HSPD-8), National Preparedness (FAA 2009; U.S. Department of Homeland Security 2003a; U.S. Department of Homeland Security 2003b). In the development of new AEPs, the FAA also suggests that all airports

Copyright National Academy of Sciences. All rights reserved.

Table 6. Expanded airport capabilities needed for incident- and emergency-related diversions.

Expanded Capability Needed	Potential Stakeholders Involved	Resource Examples
Firefighting/ emergency response	• ARFF* • Fire Department* • Airport Operations Staff • Airlines • NTSB	• ARFF vehicles • Fire engine/ladder trucks/mini pumpers/command apparatus • Foam apparatus
Medical response	• EMS* • Local healthcare providers • Airlines • Local public health department (domestic flight) • CDC (international flight)	• Ambulance • Oxygen/nitrogen cart • Defibrillators • First aid kits
Security	• Law enforcement* • Airport security coordinator • TSA • FBI • Military (on joint-use airports) • U.S. Marshals and Air Marshals	• Police vehicles • Operations vehicles • Safety cones, stakes, flags, signs, barriers • Secure/sterile/isolation areas
Deplaning	• Airlines • FBO • CBP (international flight) • TSA (rescreening)	• Pushback tug/tractor, wheel chocks • Tow bar (universal or aircraft specific) • Stairs (pushup/motorized, truck, aircraft-specific) • Americans with Disabilities Act (ADA) equipment (mechanical lifts, ramps) • Passenger buses/shuttles, vans or trucks
Aircraft mechanical	• Airlines • FBO • Hangar/aircraft owners • Airport tenants	• Mobile maintenance stands • Fuel truck • Deicing unit • Air start unit • Hydraulic jacks • Ground power unit/mobile generator • Tools/parts
Passenger care provisions	• Airport volunteer group • Airlines • Concessionaires • Ground transportation • Local stores, pharmacies • Local restaurants • Local hotel • Convention and Visitors Bureau	• Lift truck • ADA equipment (wheelchairs) • Food/water/ice/paper products • Baggage tugs, carts and belt loaders • Potable water cart • Portable air conditioning or heater cart • Lavatory cart/service truck • Cots/mats/blankets/pillows • Designated areas: family play areas, lactation/nursing areas, quiet areas, pet relief areas • Information–website or brochure (lodging, concessions, special areas, special services like infant needs, security instructions) • Passenger buses
Communication/ information	• Airport • Airlines • Lead Agency Incident Commander	• Airport or Emergency Operations Center (AOC or EOC) • Information display system • Public announcement system • Airport website and/or social media platforms • Email, phone, text, chat, dashboard

**Airport or Mutual Aid responsibility.*

Copyright National Academy of Sciences. All rights reserved.

(certificated and non-certificated) use as guidance certain provisions in the National Fire Protection Agency Standards, including 424, 1500, 1561, and 1600.

AC 150/5200-31C also stipulates what airports should include in their plans, such as defining the roles and responsibilities of internal stakeholders (FAA 2009). It is a document that outlines how mutual aid agreements are defined and specifies ways information and resources can be shared during events like diversions that are associated with incidents and emergencies. It follows the ICS, a major component of the National Incident Management System (NIMS), to provide a flexible and scalable system for multiple agencies and disciplines to manage emergencies as well as planned events. More on this topic is discussed in the Diversion Management section on page 34. Part 139 Airports are required to review their AEP internally and externally with organizations with responsibilities in the plan, once per year. A lot of research has been done on the efficacy of AEPs and many airports have reported that AEPs are difficult to use when response operations are underway either because they have too much or too little information. They have also said that it is hard to keep these plans updated (Murphy, S., et al. 2021).

- **AEP Annex and Hazard-Specific Plans (e.g., weather or pandemic):** According to *ACRP Synthesis 115*, airports have found it beneficial to create and develop other hazard-specific plans and annexes as a supplement to the AEP (Murphy, S., et al. 2021). These include plans for diversions involving incidents and emergencies. They can be pulled out of a more extensive AEP to provide practical guidance during an event.
- **Aircraft Emergency Response Plan (ERP):** Many smaller airports have developed and use these to manage emergency-related diversions. They are planned to be specific, like an incident or emergency-level diversion, although it is not known if smaller airports have developed ERPs to specifically plan for diversions. According to the International Civil Aviation Organization (ICAO), an ERP is a comprehensive, operational-level document outlining specific roles, a set of actions, and timeframes to respond to unexpected situations, disruptions, or potential disruptions. These are to be integrated with other emergency and related plans with interdependencies (ICAO 2022).
- **Airport Emergency Operations Center (EOC) Plan:** Some small airports use this plan to detail where their EOC will be set up and how it will command and control events like emergency diversions. It is a plan that defines which mutual aid resources will share a space with the airport to mitigate an emergency. For smaller airports, the EOC is sometimes managed by the local jurisdiction that has authority over the airport.
- **Crisis Communication Plan (CCP) and Emergency Communications Plan (ECP):** To a much lesser degree, smaller airports say they use these types of plans to manage communication associated with diversions. The low number may be in part because ECPs are often embedded into the AEP and may not be considered a stand-alone plan. This finding is also aligned with research from *ACRP Synthesis 73* that discovered that airports have not transitioned from traditional ECPs to CCPs that can provide communication plans for specific events like an emergency diversion (Smith, J. F., et al. 2016).

While most small airports print out and store plans on office shelves, more plans are digital and can be viewed in the field on laptops or smartphones. Some small airports use smartphone apps to access their diversion-related plans.



How Small Airports Train for Diversions

Only some smaller airports train specifically for diverted aircraft situations. It is not known if these airports complete their diversion-related training as part of mandatory emergency management training or in addition to it. The type of required training applies to Part 139 certified airports (although some smaller airports participate in this type of training with other larger airports in their region) and includes emergency management training; however, there is no national training standard for this. The only stipulation per AC 150/5200-31C is

Copyright National Academy of Sciences. All rights reserved.

that airports train for emergency events annually with their internal and external stakeholders (primarily first responders) and conduct an exercise with a full-scale functional exercise (called a "recertification or triennial exercise") once every 3 years (FAA 2009). The triennial exercise now almost always includes a mass casualty component that could be applicable to incident- or emergency-related flight diversions.

Again, while the exact type of training that small airports conduct regularly is unknown, smaller airports do report that, aside from first responder partners, they also train with on-airport and external mutual aid and network partners that play key roles in the management of all types of aircraft diversions. These include training with airlines, FBOs and fuelers, other airport tenants, concessionaires, ground transportation companies, local hotels, and counselors.

There are a few diversion-related training exercises available to smaller airports. *ACRP Report 153* developed a modifiable tabletop exercise for international diversions that can be scaled for smaller airports, and this is included in Appendix C (Cogliandro, B., et al. 2016). *ACRP Report 153* also provides other tabletop exercise templates to help airports train for weather-related situations (e.g., ice, winter operations, thunderstorms, and hurricanes) and other issues (e.g., power outages), and these can also be modified to include a diversion element for training purposes.

In addition, some suggested practices for training with mutual aid come from *ACRP Synthesis 115* (Murphy, S., et al. 2021). This guidance encourages smaller airports to train first responders on specific things like defining access points, identifying staging areas, and making sure roles are defined in advance. This guidance also suggests smaller airports have staff complete Incident Command System (ICS) training so that staff gain an understanding of EM concepts and principles that may be practical for diversion events with an emergency component.

How Small Airports Coordinate with Mutual Aid and Network Partners



Small airports coordinate with their mutual aid and network partners through a variety of agreements, including the following:

- **Mutual aid agreements:** First responders, whether under the authority of the airport or mutual aid organizations, are required to act under statutes or regulations. Mutual aid agreements are commonly made between smaller airports and first responder and emergency response organizations, such as local firefighting agencies, local law enforcement, and emergency medical services. Written agreements for first-response mutual aid are made directly with each of these responders when airports do not have authority over them. However, some airports hire third-party contracted staff to perform these services under their authority. In addition, some smaller joint-use airports have written agreements with their Air National Guard or an on-airport or nearby Air Force Base to provide ARFF and other first responder services (Smith, J. F., and K. Kenville 2013). Local emergency management agencies and local healthcare agencies and hospitals are also considered part of this group. FAR Part 139 and Advisory Circular 150/5200-31C require defining mutual aid agreements with emergency responders and implicitly encourage written mutual aid agreements (FAA 2009). When first responders are considered mutual aid, smaller airports contract them through interlocal or joint-use agreements or access them via another department under municipal control.
- **Network partner agreements:** Smaller airports also enter into voluntary agreements with airport organizations including FBOs/fuelers/deicers, ground handlers, hangar and aircraft owners, concessionaires, airport tenants, and ground transportation companies (e.g., taxis, rental car agencies) as well as community organizations like local stores, restaurants, busing companies, and hotels. There are a variety of voluntary written agreements that smaller airports

Copyright National Academy of Sciences. All rights reserved.

enter, including service-level agreements, letters of agreement, MOUs, joint-use agreements, and interlocal agreements, among others. See Appendix C for a sample MOU. These typically define what resource or assistance is needed, when assistance is activated, the methods for providing the resource, general time parameters for assistance (e.g., 8 or 12 hours), and include the major roles and responsibilities of each party involved. Some smaller airports use verbal agreements to attain certain resources or services, like calling in additional staff.



During a Diversion: Managing and Mitigating

Diversion Notification



Minimal research considers or analyzes how airports, especially smaller ones, are notified of aircraft diversions. The typical notification process is outlined in a previous section and focuses on the process by which diversion decisions are made. A brief overview of this process reveals that discussions about potential diversions decisions occur among the ATC, aircraft pilot, and airline stations. This means that smaller airports are not usually brought into these early discussions and rather receive notification after a diversion decision that impacts them is made.

However, some small airports use tracking technology (discussed in a later section in this chapter) to keep abreast of potential diversions and some small airports that are part of a regional group, typically formed around a major hub, are included in communications that notify them of potential unexpected aircraft. Notification communication comes in various formats but typically can come in the form of a telephone call (landline or cell phone), a text, or an alert on a virtual dashboard shared with a regional group via one of several types of critical event management software.

The timing of diversion notification is critical for airports so that they can adequately prepare internal staff and activate agreements with mutual aid and on- and off-airport network partners as needed. However, during emergencies, there is often little time available for notification. Notification can be a problem for GA airports if diversions occur after hours, as many of these types of airports have minimal full-time staff members or are not staffed 24 hours per day. This can be compounded when a small airport does not have a tower, which can make notifying emergency services challenging (Murphy, S., et al. 2021). Smaller airports are also challenged with having to take time to ensure contact lists are updated regularly to verify that the right people receive diversion notifications.



Diversion Management

Smaller airports typically manage gas and go diversions under the direction of airport management. However, for those diversions with incident- or emergency-level issues, most smaller airports use either an on-site AOC to manage diversion incidents or give control over to their city or county's EOC to manage an emergency. City and county EOCs follow management and command structures outlined in the NIMS, the ICS, and sometimes Unified Command (UC). Much literature exists that discusses the rationale for and effectiveness of NIMS and ICS, and how these are required protocols to include in AEPs per AC 150/5200-31C, *Airport Emergency Plan* (31C) (FAA 2009). In addition, *ACRP Report 103* devotes an entire chapter to GA airports and the methods they can use to structure and implement NIMS/ICS (Stambaugh, H., et al. 2014).

NIMS is a resource management framework that brings multiple jurisdictions, mutual aid, agencies, and organizations together to plan for and mitigate an emergency. The NIMS

Copyright National Academy of Sciences. All rights reserved.

framework organizes emergency operations under a lead emergency management agency at an EOC off-site from an emergency. An ICS is often used in conjunction with NIMS. ICS is a tool to access the proper strategic and tactical response to mitigate an emergency like the immediate first responders at a scene. ICS also forms the authority structure, establishes lines of command and control, and applies a common terminology as discussed in the next section.

Creating UC during an emergency event is an industry-recommended practice for airports (Smith, J. F., K. Kenville, and J. M. Sawyer 2015). UC can be set up in the field or an EOC and is a way to reach beyond NIMS and ICS responders and manage other critical network partners during an emergency. UC can bring in on-airport organizations like FBOs, airlines, and concessionaires; agencies such as the TSA and CBP; and community organizations such as ground transportation, local stores, and even volunteer groups. Using NIMS, ICS, and UC together can standardize management and communication and may improve the overall coordination of diverted flights at smaller airports.

Diversion Communication



Little peer-reviewed or airport industry research exists on how effective airports, especially smaller ones, are at communicating with internal staff as well as with mutual aid and on- and off-airport network partners during aircraft diversions. Nor is there much information on how much they are implementing newer communication technology such as social media to manage diversions. More information on the technology used in communication during diversions can be found in the next section.

Smaller airports use terms from NIMS, ICS, and UC—all described in the previous section—as well as endeavor to use common language that is understood by all parties (airport, mutual aid, and network partners) to ensure clarity during irregular- and emergency-level events like diverted aircraft.

However, there are numerous industry practices related to improving overall communication efforts at airports during irregular or emergency conditions that can be applied to diversion situations, including ensuring that smaller airports designate redundant and interoperable communication that includes predetermined communication channels, frequencies, and procedures (e.g., email, text, virtual dashboard, etc.). Small airports are also encouraged to ensure they can shift to designated alternative (backup) systems if there is a power outage or networks become overloaded to continue to coordinate efforts.

ACRP literature advocates for developing comprehensive CCPs for response and recovery for responding to events like aircraft diversions, making them more extensive than what is required in an AEP or comparable plan for non-Part 139 airports to be effective (Smith, J. F., et al. 2016). Smaller airports are also encouraged to leverage the communications capabilities of their emergency partners whenever possible.

An example of leveraging emergency partners comes from *ACRP Synthesis 73*, where non-hub airport Rochester International Airport added social media to its triennial emergency training exercises to test crisis communications—especially related to public messaging during a crisis—with a much smaller staff than larger airports (Smith, J. F., et al. 2016). In their efforts, they learned how to leverage public information resources from other municipal departments to help the airport manage social media via the UC and Joint Information Center (JIC). The UC/JIC is established when mutual aid is activated. They also found it is important to have a single point of contact for social media and a single communication plan CCP. In addition, they also discovered how important it is to determine ahead of time how to manage communication in the field when computers and landlines are out of reach.

Copyright National Academy of Sciences. All rights reserved.

32 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports



Diversion-Related Technology

Smaller airports use a variety of technology to track potential flight diversions as well as to communicate and manage the resources and personnel needed for mitigating them. Some of the technology that is available according to related ACRP research includes the following:

- **Tracking weather:** A variety of public and privately funded weather-tracking websites are available online.
- **Tracking aircraft:** There are many online sources available to track aircraft, including federal, state, and commercial options. More details about additional flight-tracking technology can be found in the "Technology Perspective" section of Chapter 5.
- **Accessing diversion-related plans:** Apps and digital versions of diversion-related plans can be used to access these plans on laptops and phones to ensure situational awareness for all partners, especially those in the field.
- **Communication and coordination:** In terms of communicating situational awareness and coordinating resources, traditional technologies like telephone, texting, email, and two-way radio are still typically used to communicate between the airport and mutual aid and network partners in real time. Some smaller airports that are part of a regional group sometimes have access to collaborative critical event management software products that larger airports or municipalities procure to get up-to-date information about response efforts related to incident and emergency diversions. Social media is an emerging trend to use in emergency management (Barich, F., et al. 2020), but smaller airports typically do not have the staffing needed to monitor and create messaging on various social media platforms during a flight diversion event. They typically rely on city or county emergency management agencies (mutual aid) to assist with this type of messaging during emergencies.

Lists of other technologies that can be used for communication and coordination efforts during diversion events can be found in *ACRP Report 65* (Tool 10) and *ACRP Report 153* (Appendix A: Data Sources) (Nash, J. M., et al. 2012; Cogliandro, B., et al. 2016).



Passenger Care

While air carriers are responsible for providing food and beverage as well as lavatory services to passengers while they are onboard an aircraft, literature advocates that airports also proactively plan to meet various passenger care needs during situations like flight diversions. This is particularly important for passengers on international flights diverted to airports that do not have CBP presence. Those passengers need to remain on board until they can depart. If the delay is long, the aircraft may run out of food and beverages. Situations like these may cause diversion airports to need interim assistance until an air carrier can source its aircraft, especially if the aircraft belongs to a non-station airline and the airline needs to call in staff from another location, potentially taking a few hours to more than a day. In other circumstances, where passengers from a diverted flight are deplaned and experience a prolonged delay in an airport's terminal, these passengers will have additional needs.

ACRP Report 65 states that passengers have five basic needs during irregular operations events such as diversions (Nash, J. M., et al. 2012). These include food and water, facilities, special services, lodging, and information. While there are no mandates for airports to provide these amenities to passengers, the aviation industry does have a collective reputation, meaning that if a need is not met by a traveler, then both the airport and the airlines experience the blame.

IROPS plans from several small airports include some of the following ideas for meeting passenger needs and making the terminal safe and secure for all passengers:

- **Food and water:** Ask for extra staffing and supplies from concessions, create pre-orders with local pizza stores (per number of passengers), and curb alcohol sales in the terminal.

Copyright National Academy of Sciences. All rights reserved.

- **Facilities:** Designate quiet areas (dim lights for sleeping), pet relief areas, and lactation and nursing areas. Ensure extra maintenance staffing for restrooms and waste bins in the terminal.
- **Special Services:** Secure buses from local bus companies or school districts to transport passengers to local hotels or their flight's intended destination. Consider special needs such as language support (e.g., interpreters) and family support such as retailers stocking extra diapers or formula.
- **Lodging:** Contact the Convention and Visitors Bureau for current hotel availability. Create a budget and space for storing mats, blankets, and pillows.
- **Information:** Print brochures with lodging and transportation contact information as well as local pharmacies where prescriptions can be filled.

In addition, small airports can create an EWG to coordinate passenger care needs during certain flight diversions. *ACRP Synthesis 99: Emergency Working Groups at Airports* highlights developing airport–airline collaborative arrangements like this to fulfill airline-mandated family assistance needs among various stakeholders during emergency situations, primarily aircraft crashes (Smith, J. F. 2019). EWGs could also be leveraged to plan, train, and maintain not only mutual aid but also network partners to support passenger care-related tasks during a variety of diversion-related situations mentioned in this synthesis.

The EWG approach is also suggested in AC 150/5200-31C-*Airport Emergency Plan* (31C) (FAA 2009). It advocates forming an EWG with all air carriers serving an airport so that they can collectively make plans to respond to and assist with any aircraft that arrives at an airport during the first 12 hours. This guidance also suggests that the air carrier group make plans to assist charter carriers or emergency diversions that involve non-tenant air carriers. All these suggestions may improve the industry's ability to meet passenger expectations and needs in the early hours of an incident or emergency. A Sample Checklist for Creating an Emergency Working Group (EWG) is provided in Appendix C.

Crowd, Traffic, and Access Control



During incident- and emergency-related diversion situations, controlling access to certain airport areas is important for safety and security reasons. According to a sampling of small airport AEPs, these plans designate either an airport's security personnel or a city or county's law enforcement agency as the primary stakeholder that manages crowds, traffic, and overall airport access as situations warrant. A command post may be set up to manage crowds, manage traffic, and access control and other police duties when necessary. This command post may be set up in the security office of the airport or in emergency situations at the city or county law enforcement station. Additional off-duty personnel can be called in if a situation escalates.

Airport security personnel are also in charge of ensuring airport identification standards are maintained under 49 CFR § 1542: Airport Security. They maintain a list of credentialed (badged) individuals who have escorted or unescorted access to secure areas. This list can be used during emergency events to help verify which personnel can access secure airport areas.

However, spontaneous volunteers, information seekers, curious onlookers, and even exploiters may converge at an airport to either volunteer to help, just watch, or try to profit from a situation. In an unplanned situation at a smaller airport, it can be helpful to establish protocols for managing these various types of people. But these plans need to be made ahead of time, as things like assessing qualifications, running background checks, and providing training all need to be accomplished before volunteers can assist with on-airport incidents and emergencies (IEM, Inc., et al. 2012).

Local and state emergency operation plans often include provisions for these various types of people under sections entitled Unaffiliated Volunteer Management. In addition, the FEMA

Copyright National Academy of Sciences. All rights reserved.

34 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

published *Managing Spontaneous Volunteers in Times of Disaster: The Synergy of Structure and Good Intentions* (FEMA n.d.). FEMA's publication suggests that airports consider liability issues related to volunteers and suggests setting up a volunteer coordination database.

ACRP Report 95: Integrating Community Emergency Response Team (A-CERTS) at Airports provides guidance on how to involve community volunteers who have a response interest and who are willing to supplement airport staff during times of need (IEM, Inc., et al. 2013). This guidance provides training tools and exercises to develop an Airport Community Emergency Response Team (A-CERT).

However, during an emergency event, there is a need to manage onlookers and volunteers and increased traffic. Law enforcement personnel can set up an area to collect names of those interested in future training as well as assign officers to manage crowds in and around an airport terminal and traffic to and from the airport as needed.



After a Diversion: Restoring Operations

How Smaller Airports Recover from Diversions

According to *ACRP Synthesis 60: Airport Emergency Post-Event Recovery Practices*, "an airport's ability to recover effectively is the measure of its resiliency" (Smith, J. F., K. Kenville, and J. M. Sawyer 2015). Resiliency in this synthesis refers to the ability to recover quickly from one or more flight diversions to maintain regularly scheduled flight schedules and operations.

Much emergency-related research states that recovery begins while airports are in the process of managing an incident or emergency, like certain flight diversions. However, there is little research available specifically related to flight diversion events that assesses small airports' resiliency and ability to recover from diversion situations while managing the events.

To continue normal operations while managing or coordinating diverted aircraft means that airports understand and plan for the various types of impacts from diversion situations on business operations before they occur. Research recommends that airports "consider a separate set of plans, SOPs and checklists to cover this activity" (Atlason, G., et al. 2014). Some smaller airports have developed such plans to ensure that recovery is efficient. These include Business Continuity Plans which assist airports in maintaining operations during disruptions like incidents and emergencies, and Continuous Operations Plans that help airports perform their essential functions during disruptions to normal operations. These plans focus on the consequences of an event and not the causal factors of an event. They also consider potential investments (e.g., duplicate or backup equipment) that can be made to reduce the impacts of future disruptions. And they focus on the ability to sustain response efforts during long-term events in ways that prevent staff from burning out. However, literature related to these plans does not directly discuss how to continue operations during diversions.

Efforts related to diversion recovery are also found in many small airport AEPs and IROPS plans. AC 150/5200-31C recommends that recovery plans consider things like

- Final damage assessment (written, pictorial, including video),
- Public information announcements,
- Facility repair,
- Supply inventory and restoration,
- Cost documentation,
- Economic impact,
- Documentation of actions taken,
- Personnel utilization by time on duty,

Copyright National Academy of Sciences. All rights reserved.

- Critical incident stress debriefing requirements, if necessary,
- Equipment utilization documentation,
- Overall cleanup activities, and
- Air operations area inspections, if appropriate (FAA 2009).

The literature reviewed suggests that it is beneficial to create airport-specific forms and checklists to guide recovery efforts that are accessible and implementable during an actual incident or emergency. To that end, a sample diversion-specific recovery checklist for airport supervisors is included in Appendix C.

How Smaller Airports Share and Learn



Minimal research was found that relates specifically to sharing and learning from flight diversion events. However, industry guidance identifies various ways that smaller airports can keep diversion-related plans current as well as update them after real-world experience. For instance, ACRP guidance associated with IROPS plans and AC 150/5200-31C stipulates that AEPs should include procedures for conducting after-action reports (AARs) with all stakeholders involved in an event to debrief what went well and what needs improvement (FAA 2009). Any knowledge that is gained from reviews like these can be transformed into new policies and procedures and included in updated plans. Literature also suggests circulating plan updates to internal airport staff, mutual aid, and network partners, communicating the changes and any new training on updates to these stakeholders.

ACRP Synthesis 50 encourages airports and first responder agencies and organizations to review each other's emergency management plans, programs, and procedures in the form of a peer review (Smith, J. F. 2014). This report also advocates for the self-review of an airport's or agency's own emergency procedures considering an actual incident at another airport.

ACRP Synthesis 115 suggests "cross-walking" multiple plans at airports (e.g., IROPS plans or AEPs) that could affect the airport with plans developed by surrounding jurisdictions (e.g., mass casualty plans, evacuation plans, or severe weather plans) (Murphy, S., et al. 2021). This can be valuable to ensure plans do not conflict or have duplicate information and enhance the airport's ability to respond to and recover from incidents and events. In addition, a needs assessment or gap analysis can help illuminate where plans may need additional information. *ACRP Synthesis 115* also advocates for developing specific AEP annexes that can provide future guidance on new or unique circumstances, also providing training using metrics as needed.

Lessons Learned and Good Practices



Lessons Learned

Some lessons learned from the literature review include the following:



- **Challenges with plans and checklists:** Plans like AEPs, and sometimes checklists, can be too lengthy to be useful when applied or are not set up in an order that flows with a typical diversion incident. Plans and checklists that are out of date and do not contain accurate information are not useful. Also, layout and formatting issues that, for instance, cut off portions of vital information can make it hard to use them in practice.
- **Notification:** It can be challenging for smaller airports to ramp up for diversion events when they are not given advanced notification. Response efforts can be improved when smaller airports are given more time to prepare, especially those in rural areas that draw from mutual aid response network partners from a considerable distance from them. It can also be challenging

Copyright National Academy of Sciences. All rights reserved.

to keep contact information for airport staff updated and current with various mutual aid and network partners.

- **Airlines:** *ACRP Report 153* recommends that diversion airports implement working agreements with each domestic and foreign carrier that identifies an airport as a diversion airport (Cogliandro, B., et al. 2016). These types of agreements should specify that the airline needs to notify the diversion airport of a flight diversion decision as that decision is made. The timely sharing of information between stakeholders gives the airport a chance to prepare for incoming aircraft and passengers.



Good Practices

Some good practices identified through the literature review were used to inform the survey conducted as part of this synthesis. These good practices include

- Using stand-alone diversion checklists,
- Diversion plan organization,
- Notification procedures,
- Diversion plan contingency practices,
- Coordination practices with mutual aid and network partners, and
- Coordination procedures with airlines and federal agencies.

More details on these good practices can be found in Chapter 5 in Tables 18 through 22.

Another good practice identified in the literature is for smaller airports to use a regional approach to help with diversions. The regional approach is recommended by various research, including *ACRP Report 65* and *ACRP Report 153* (Nash, J. M., et al. 2012; Cogliandro, B., et al. 2016). By defining a region around a larger hub that includes smaller airports as well as all mutual aid and network partners, it expands a region's resource pool to help manage irregular or emergency events. There are a few major well-established networks in the United States, like the Minneapolis-St. Paul Regional IROPS Network, the North Texas Regional Network, and the Denver International Airport Network—all of which include all smaller airports (e.g., SH, NH, GA, CS, and R) in that region on regular calls and/or meetings. More on the regional approach and perspective can be found in Chapter 5.

EWGs can also be formed not only to help fulfill passenger needs during emergencies related to diversions among various airport, airline, and community resources, but they can also be a practical way to update, share, and coordinate various plans among the different organizations. It can also be a way to bring these groups together to train for specific special events that might result in aircraft diversions. When various airport and emergency-related organizations come together often, it can improve collaboration, trust, and coordination and creates a forum for learning which translates into better plans. Overall, it can result in improved continuity of operations and resiliency (Smith, J. F. 2019).

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 5

Survey and Interview Responses

This synthesis required an understanding of how small hub, non-hub, and general aviation airports currently manage diversions. This type of information was collected through a survey and interviews with smaller airport participants.

Survey Responses



An online survey was created for smaller U.S. airports to obtain information about how they prepare for and have managed flight diversions involving incidents and emergencies. The survey targeted approximately 300 small hub, non-hub, and general aviation airports in the United States (71 survey responses and 24% response rate), primarily airports affiliated with small airport committees at the ACI-NA and AAAE.

The survey was open for 4 weeks (January 24, 2022 to February 21, 2022). Table 7 illustrates the geographical distribution of responding airports, which represents 30 states in all regions of the United States. Table 8 shows the distribution of the respondents among the nine FAA regions. Figure 5 shows the states that are in each FAA region.

Figure 6 provides survey respondent participation by the following FAA NPIAS categories:

- **SH - small hub** airports (27%/19 out of 71)
- **NH - non-hub** airports (33%/24 out of 71)
- **GA - general aviation** airports including R - relievers and CS - nonprimary commercial service (40%/28 out of 71)

Plans That Smaller Airports Use During Flight Diversions



Several survey questions asked smaller airports about the plans they currently use to manage diverted aircraft at their airports. Respondents chose from a list of potential applicable plans and could write in any other relevant plans as well, as indicated in Table 9.

Findings from Table 9 include the following:

- Most smaller airports use IROPS plans, AEPs, or Aircraft ERPs to manage diversions.
- Only SH and NH airports report using stand-alone, dedicated airport diversion plans.
- All but GA and R airports respondents use IROPS plans for diversion management.
- Not all of the 36% of airports that use AEPs to manage diversions are mandated to do so. Some of the GA, CS, and R airports use them even though they are not Part 139 certified.
- Twenty-three percent report that they do not have plans for diversions; these are mainly GA and R airports.

Copyright National Academy of Sciences. All rights reserved.

Table 7. Geographic distribution of survey responses.

State or Province	Responding Airports	State or Province	Responding Airports	State or Province	Responding Airports
Alaska	1	Maine	1	Oregon	1
Arizona	5	Michigan	1	Pennsylvania	1
California	2	Minnesota	1	South Carolina	2
Colorado	1	Missouri	2	Texas	4
Florida	14	Nebraska	3	Utah	1
Georgia	1	New Hampshire	2	Virginia	3
Idaho	1	New York	1	Washington	2
Illinois	2	North Carolina	1	Wisconsin	5
Iowa	3	North Dakota	2	**TOTAL**	**71**
Kentucky	2	Ohio	3		
Louisiana	2	Oklahoma	1		

Table 8. FAA regions of survey responses.

State or Province	Responding Airports
Alaskan	1
Central	6
Eastern	6
Great Lakes	14
New England	2
Northwest Mountain	6
Southern	22
Southwest	7
Western-Pacific	7
Total	**71**

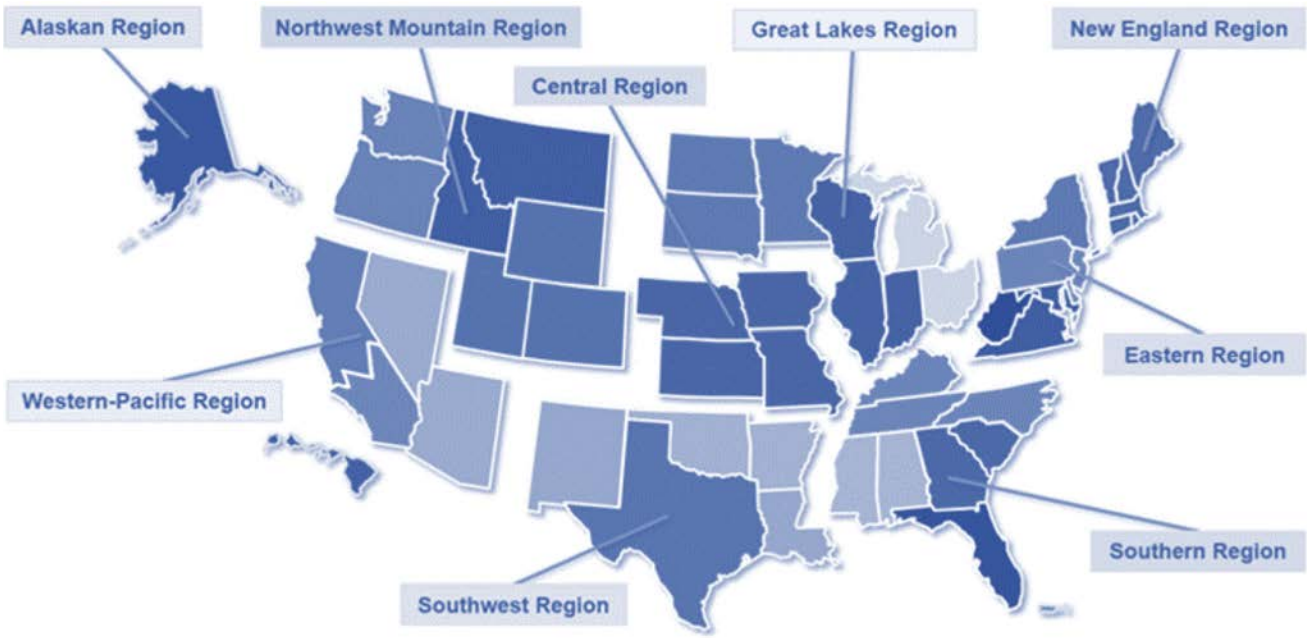


Figure 5. FAA regions (Source: FAA 2021a).

Copyright National Academy of Sciences. All rights reserved.

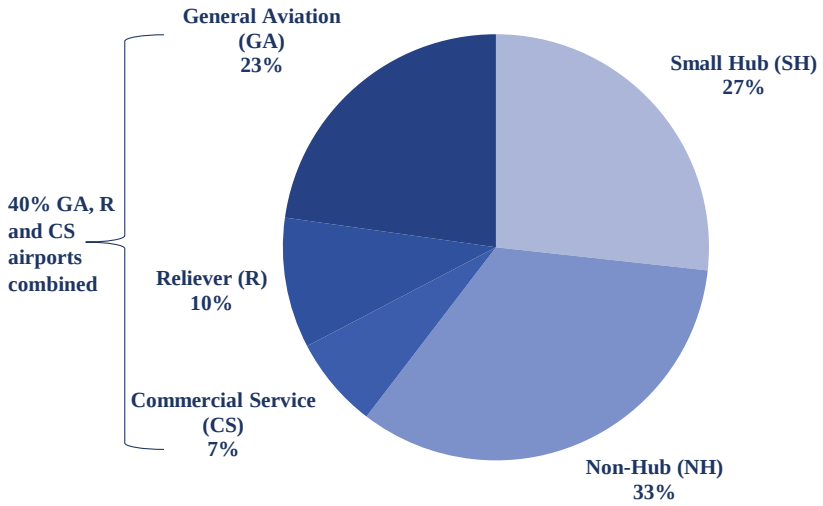


Figure 6. Airport survey respondents by NPIAS categories.

- Only NH airports (13%) indicated that they have specific stand-alone plans that address a major focus area for this synthesis, mutual aid plans.
- Regarding write-in comments, one R airport stated it developed an AEP list for diversions, while two NH airports noted that they use DOT emergency contingency plans to assist with some of their diversion-related responses. One NH airport stated that it uses an emergency response letter of agreement with its Air Traffic Control Tower (ATCT) and contracted responders.

As shown in Figure 7, most smaller airports indicate that diversion-related procedures can be found in more than one of their airport plans, while around one-third use just one plan to manage diversions, and 23% responded that they do not use a written plan to manage diverted aircraft.

Table 9. Types of plans used by all of the smaller airports surveyed.

Type of Plan	Total (69)	SH (19)	NH (24)	GA (14)	CS (5)	R (7)
Irregular Operations (IROPS) Plan	41%	74%	54%	0%	20%	0%
Airport Emergency Plan (AEP)	36%	53%	38%	21%	40%	14%
Aircraft Emergency Response Plans (e.g., Standard Operating Procedures)	29%	21%	33%	21%	40%	43%
Airport Diversion Plan	14%	16%	29%	0%	0%	0%
City/County/Local Emergency Plan	7%	5%	8%	0%	20%	14%
Hazard-Specific Plans (e.g., weather, pandemic etc.)	7%	11%	8%	0%	20%	0%
Airport Training and Exercise Plan	6%	5%	4%	7%	0%	14%
Airport EOC Plan	6%	16%	4%	0%	0%	0%
Business Continuity Plan (BCP)/Continuity of Operations Plan (COOP)	4%	5%	4%	0%	20%	0%
Crisis Communication Plan/Emergency Communications Plan	4%	5%	8%	0%	0%	0%
Family Assistance Plan	4%	0%	8%	0%	20%	0%
Mutual Aid Plan	4%	0%	13%	0%	0%	0%
Unsure	3%	5%	0%	7%	0%	0%
AEP Annex	1%	0%	4%	0%	0%	0%
My airport has no plans for diversions	23%	0%	13%	57%	20%	57%
Other Plan (please specify)	4%	0%	8%	0%	0%	14%

Number of responses per airport category in parentheses.

Copyright National Academy of Sciences. All rights reserved.

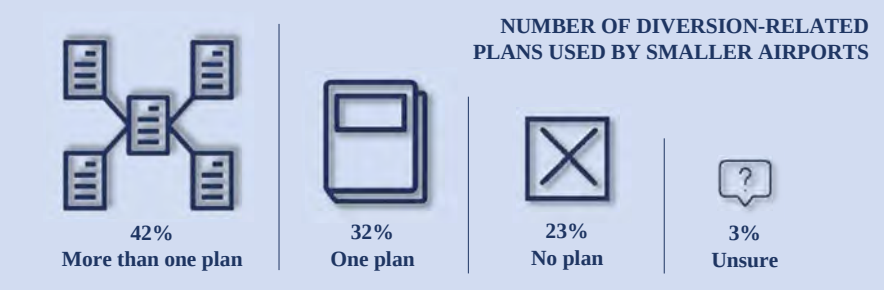


Figure 7. Number of diversion-related plans in use by smaller airports.

Of the 42% of respondents who listed more than one diversion-related plan, these airports indicated that they use plans associated with both emergencies (e.g., AEPs and Aircraft Emergency Response Plans) and irregular operations (e.g., IROPS or COOPs). While the survey did not ask airports to specify what type of diversion-related planning is included in each type of plan, it can be inferred from industry knowledge that many use AEP-type plans to plan for diversion emergencies and IROPS- and COOP-related plans to plan for routine diversions and diversions associated with minor incidents.

Overall, smaller airports approach diversion planning in unique ways. Some combine diversion planning with one or more emergency or irregular operations plans while some have developed stand-alone plans for diversions. Many of the smallest airports (CS and R) do not use plans. Some have developed diversion-related checklists as described next.

Stand-Alone Diversion-Specific Checklists and SOPs

As shown in Table 10, little more than half of 51 airports surveyed report they have developed stand-alone practical and actionable SOPs or checklists to use specifically for flight diversion events. While an average of 60% of all SH, NH, and GA airports report that they have some sort of actionable plan to use during diversions, none of the CS or R airports reported having these in place. This may be attributed to the low number of diversions received by these smallest airports. It is also important to note that just half of all smaller airports surveyed do not have designated plans for diversions.

Industry Recommendations for Diversion Checklists: Of the 51% that indicated they currently use stand-alone diversion-specific checklists (as demonstrated in Table 10), many of their checklists and plans include industry-recommended elements (as demonstrated in Table 11).

Table 10. Use of stand-alone diversion checklists or SOPs.

Diversion Checklist or SOP	All (51)	SH (17)	NH (21)	GA (5)	CS (5)	R (3)
Yes–do use	51%	65%	57%	60%	0%	0%
No–do not use	49%	35%	43%	40%	100%	100%

Copyright National Academy of Sciences. All rights reserved.

Table 11. Use of industry-recommended elements in diversion checklists.

92%	**Notification procedures** (e.g., how to contact mutual aid and network partners)
80%	**Notification triggers** (e.g., when to call upon mutual aid and network partners)
72%	**Communication procedures** to maintain common situational awareness among mutual aid and network partners during an event (e.g., email, text, virtual dashboard)
72%	**Coordination procedures** to access resources, staff, and equipment from mutual aid and network partners (e.g., sharing agreements in place)
72%	**Special conditions procedures** (e.g., after-hours staffing procedures, aircraft/passenger surge management, prolonged event considerations)
68%	**Escalation triggers** (e.g., defines when an incident becomes an emergency)
68%	**Monitoring procedures** (e.g., weather, flight status, and regional/national airspace status)
WRITE IN RESPONSES:	**Cancellation procedures**, **aircraft parking procedures** (e.g., priorities), and **media interaction procedures**

Diversion-Specific Checklist Attributes: Twenty-five survey respondents indicated that they currently use stand-alone diversion-specific checklists. The types of attributes that they indicated made them easy to use include the following:

Organization

- Bullet-point format
- Grouped by role and responsibility
- Grouped by task
- Tasks ordered sequentially or by escalating factors
- Specific, clear, and precise instructions
- Divides complicated tasks into simple ones
- Verified to be current and up-to-date (e.g., includes time stamp)

Accessibility

- Easy to find (e.g., bulletin board)
- Easy to access (e.g., tear-out sheet)
- Quick reference (1–2 pages)
- Separate stand-alone document
- Digital plan (laptop and smartphone accessible)
- Available on a smartphone app

Most of the comments equated ease of use with a plan's shorter length, separation from other plans and procedures, and organization by roles and responsibilities. None of the airports surveyed use a smartphone app to access their diversion-related plan. Other respondents noted in "write-in" comments that their plans are easy to use either because they are part of the AEP or can be found on their airport's website.

The Impact of First Responders on Airport Capabilities during Diversions

When diversions arrive at an airport, airports assess their real-time capabilities to see if they can meet a variety of potential non-emergency and emergency needs with their own resources

Copyright National Academy of Sciences. All rights reserved.

or with the help of mutual aid or network partner resources. This survey section demonstrates who has authority over first responders and how this impacts an airport's diversion response efforts.



First responder authority at smaller airports: Table 12 demonstrates the jurisdiction over services provided by first responders, specifically ARFF, EMS, and law enforcement among the various NPIAS designations. The following results demonstrate how authority over first responders varies widely among survey respondents, and that SH and NH tend to have more in-house first-responder capabilities than other smaller airport designations.

The overall disparity among all smaller airports can most likely be attributed to the broad-ranging governance and management structures, contractual service arrangements, and operating characteristics of all the airports surveyed—both within NPIAS designations and across. Some "write-in" comments from some SH and NH airports, such as the following, further clarify the various mutual aid agreements that these airports use for first responder assistance:

- "ARFF and Medical is under a Joint Use Agreement with the Department of Defense (DOD). Medical is also supported by City Fire and EMS. Law Enforcement is under agreement with the County Sheriff's Office"–Small hub airport
- "The airport is owned and operated by the city which provides ARFF and law enforcement." –Small hub airport
- "We contract with local Sheriff's Department for Law Enforcement Officers (LEOs) and with on-field Air National Guard for ARFF and medical."–non-hub airport
- "We manage these functions under our own authority, but contract this out."–non-hub airport

There were few survey respondents representing GA, R, and CS airports, however, a few trends did emerge from this modest pool of the smallest airports. For instance, GA airports (82%) and R airports (86%) reported that they do not have airport authority over first responders. A few comments from these airports include the following:

- "We have interlocal agreements for ARFF and law enforcement."–General aviation airport
- "City interlocal agreement with ARFF and Law enforcement"–General aviation airport
- "The city manages the ARFF and Law Enforcement"–Nonprimary commercial service airport



Airport Coordination of Additional Assistance

As shown in Table 13, when survey respondents were asked who coordinates with mutual aid and network partners at their airports, half of all respondents said that their airport's operations manager works one-on-one to obtain mutual aid and network partner assistance. Approximately 43% say that their airport director or general manager handles this task, while 17% say that it is handled by a local emergency management agency. A few write-in responses stated that it falls under the chief or director of public safety, a combination of operations and public safety, or the fire chief, and four mentioned the duty manager or incident commander on duty.

Table 12. First responders under smaller airport authority.

First Responders Under Airport Authority	All (71)	SH (19)	NH (24)	GA (16)	CS (5)	R (7)
ARFF	49%	58%	71%	13%	80%	13%
EMS (Medical)	23%	37%	33%	6%	0%	0%
Law Enforcement	38%	68%	50%	6%	0%	14%
None	41%	32%	17%	81%	0%	86%
Other	10%	0%	13%	13%	40%	0%

Copyright National Academy of Sciences. All rights reserved.

Table 13. Airport points of contact with mutual aid and network partners.

Airport Point of Contact	All (58)
Airport Operations Manager	50%
Airport Director or General Manager	43%
Local EM Agency (city, county, or local multi-county regional)	17%
Airport Emergency Manager	5%
State EM Agency	1%
Other (Chief/Director of Public Safety, Fire Chief, Duty Manager, etc.)	14%

Additional Assistance from Mutual Aid and Network Partners



Who assists: As Table 14 indicates, mutual aid and network partners were defined in the survey as "not just the agencies that your airport has formal agreements with to assist you during both emergency and non-emergency events but also [. . .] local organizations that support your airport when your capabilities have been exceeded."

Table 14. Organizations considered mutual aid and network partners.

Mutual Aid and Network Partners	All (55)	SH (16)	NH (22)	GA (9)	CS (2)	R (6)
First Responders/Emergency Management						
Law Enforcement	87%	69%	95%	89%	100%	100%
ARFF	38%	50%	41%	22%	0%	33%
Fire Department	87%	81%	91%	78%	100%	100%
EMS (Medical)	84%	75%	86%	89%	100%	83%
Local EM Agency/Office	75%	81%	77%	56%	100%	67%
Local Healthcare/Hospitals	64%	75%	73%	22%	100%	50%
Federal Agencies/Military						
FAA	56%	81%	59%	22%	50%	33%
TSA	60%	75%	86%	11%	50%	0%
CBP	35%	50%	41%	0%	0%	33%
FBI	51%	75%	59%	11%	0%	33%
CDC	15%	38%	9%	0%	0%	0%
NTSB	24%	31%	23%	11%	0%	33%
U.S. Marshals	5%	6%	9%	0%	0%	0%
Air Marshals	7%	13%	9%	0%	0%	0%
Military (joint use on airport)	25%	38%	27%	11%	50%	0%
Mutual Aid and Network Partners	**All (54)**	**SH (16)**	**NH (22)**	**GA (8)**	**CS (2)**	**R (6)**
On-Airport Partners						
Airlines	57%	75%	77%	0%	67%	0%
FBOs/Fuelers	70%	75%	73%	50%	67%	67%
Hangar/Aircraft Owners	28%	31%	14%	50%	0%	50%
Concessionaires	31%	56%	36%	0%	0%	0%
Other Airport Tenants	30%	38%	18%	38%	33%	33%
Ground Transportation	28%	56%	27%	0%	0%	0%
Community Resources						
Local Stores	4%	6%	5%	0%	0%	0%
Local Restaurants	6%	6%	5%	0%	0%	17%
Local Hotels	20%	38%	18%	0%	0%	17%
Airport or Other Volunteer Group	11%	19%	5%	13%	0%	17%
Red Cross	43%	63%	50%	13%	0%	17%
Counselors	7%	19%	5%	0%	0%	0%

Copyright National Academy of Sciences. All rights reserved.

44 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Findings from Table 14 include the following:

Mutual Aid and Emergency Management: First responders such as law enforcement, fire departments, EMS (medical), and emergency management providers such as local EM agencies are all consistently considered as aid providers across all smaller NPIAS airport designations, according to Table 14. This generally aligns with earlier survey responses related to airport authority over law enforcement and EMS (medical). Specifically, the more these roles are considered under airport authority, the less they are considered mutual aid and vice versa. However, two anomalies occur for law enforcement at SH and NH airports. Close to 70% of SH airports reported in Table 12 that they have authority over law enforcement, and this aligns with Table 14 findings as 69% of SH say law enforcement is mutual aid. The same situation exists for NH airports as well. ARFF is less likely to be seen as a mutual aid partner because it is most often managed under airport authority. Local healthcare and hospitals are also considered a partner for half to two-thirds of airports interviewed, except for GA airports.

On-Airport Network Partners: According to Table 14, most smaller airports representatives surveyed (70%) consistently viewed FBOs and Fuelers as partners across all NPIAS designations. Airlines are considered partners by more than half of all SH, NH, and CS airports respondents surveyed. However, a few of the GA and R airports surveyed do not view airlines as partners. All but CS airports consider hangar and aircraft owners as partners. SH and NH consider concessionaires as partners, while GA, CS, and R airports do not, most likely because these smallest airports typically do not offer concessions at their airports. Thirty percent of all airports surveyed consider other airport tenants as partners. Only some SH and NH (28%) view ground transportation (e.g., rental cars or shuttle buses) as mutual aid, while none of the GA, CS, and R airports do.

Community Resource Network Partners: Only small percentages of respondents indicated that local stores (4%) and local restaurants (6%) are network partners. Twenty percent of the airports surveyed consider hotels and 11% consider airport volunteers as part of their mutual aid network. The Red Cross and counselors were included in the survey as they appeared as stakeholders in numerous emergency-related ACRP research projects. However, their roles are applicable to aircraft accidents, which makes them outside of this synthesis project.

Federal Agency and Military Network Partners: The FAA and TSA are viewed as network partners across all smaller NPIAS airport designations except for R airports. Reliever airports most likely do not consider TSA as a partner because TSA does not typically provide screening at these airports. The FBI is considered a partner for a majority of SH and NH airports, but for fewer GA and R airports and for none of the CS airports interviewed. The CBP is considered a partner by less than half of SH, NH, and R airports, and by none of the GA and R airports surveyed. Only eight respondents from SH and NH airports consider the CDC as mutual aid. Finally, GA, CS, and R airports do not view other federal agencies listed on the survey as part of their mutual aid network. It is also rare for smaller airports to view the military as mutual aid. The NTSB was included as it appeared in numerous emergency-related ACRP research projects, but its role is a more investigatory one (after an aircraft accident), which makes it outside of this synthesis project.



Airports That Share and Coordinate Diversion-Related Plans

Who airports share and coordinate plans with: Table 15 demonstrates where smaller airports collaborate with mutual aid and network partners by sharing their diversion-related plans. This table reveals the gaps where smaller airports do not share plans. It is not known whether this

Copyright National Academy of Sciences. All rights reserved.

Table 15. Airports that share and coordinate diversion-related plans.

Mutual Aid and Network Partners	All (55)	SH (16)	NH (22)	GA (9)	CS (2)	R (6)
First Responders/Emergency Management						
Law Enforcement	36%	63%	32%	33%	0%	0%
ARFF	31%	50%	32%	22%	0%	0%
Fire Department	27%	44%	23%	33%	0%	0%
EMS (Medical)	25%	44%	18%	33%	0%	0%
Local EM Agency/Office	20%	38%	14%	22%	0%	0%
Local Healthcare/Hospitals	18%	38%	9%	22%	0%	0%
Federal Agencies/Military						
FAA	27%	56%	23%	11%	0%	0%
TSA	25%	56%	23%	0%	0%	0%
CBP	15%	19%	18%	11%	0%	0%
FBI	15%	38%	5%	11%	0%	0%
CDC	2%	0%	0%	11%	0%	0%
NTSB	2%	0%	0%	11%	0%	0%
U.S. Marshals	2%	0%	0%	11%	0%	0%
Air Marshals	2%	0%	0%	11%	0%	0%
Military (joint use on airport)	9%	6%	14%	11%	0%	0%
Mutual Aid and Network Partners	**All (54)**	**SH (14)**	**NH (22)**	**GA (8)**	**CS (2)**	**R (6)**
On-Airport Partners						
Airlines	48%	75%	64%	0%	0%	0%
FBOs/Fuelers	54%	75%	55%	38%	0%	33%
Hangar/Aircraft Owners	9%	0%	9%	25%	0%	17%
Concessionaires	22%	38%	23%	13%	0%	0%
Other Airport Tenants	13%	13%	14%	13%	0%	17%
Ground Transportation	17%	31%	14%	13%	0%	0%
Community Resources						
Local Stores	0%	0%	0%	0%	0%	0%
Local Restaurants	2%	0%	0%	13%	0%	0%
Local Hotels	7%	6%	9%	13%	0%	0%
Airport or Other Volunteer Group	2%	0%	5%	0%	0%	0%
Red Cross	19%	50%	5%	13%	0%	0%
Counselors	4%	6%	0%	13%	0%	0%

means that smaller airports do not have the staff or time needed to share plans with other outside organizations. It is not known if the authority structure of first responders impacts this finding, for instance, if a county department oversees first responders, their plans may not be shared with the airport and vice versa. It is not known if this finding demonstrates silos that may exist in the industry.

Findings from Table 15 include the following:

Mutual Aid and Emergency Management: According to Table 15, less than half of the respondents from SH, NH, and GA airports interviewed report that they share their diversion-related plans with first responders. Thirty-six percent of these airports report sharing diversion-related plans with law enforcement, 31% with ARFF, 27% with fire departments, 25% with EMS, 20% with local EM agencies, and 18% with local healthcare and hospitals.

On-Airport Network Partners: Table 15 indicates that approximately half of SH and NH airports report sharing plans with airlines (48%). A little more than half of all

Copyright National Academy of Sciences. All rights reserved.

46 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

respondents indicated that they share plans with FBOs and Fuelers (54%). Nine percent of the survey responders indicated they coordinate with hangar and aircraft owners, while 22% indicate that they coordinate plans with concessionaires, 13% with other airport tenants, and 17% with ground transportation companies.

Community Resource Network Partners: Only a small percentage indicated that they coordinate plans with local hotels (7%), local restaurants (2%), and airport volunteer groups (2%). None of the airports indicated that they share and/or coordinate plans with local stores.

Federal Agency and Military Network Partners: Twenty-seven percent of SH, NH, and GA airports report sharing diversion-related plans with the FAA, while 15% share with the CBP and with the FBI, respectively. Only 25% of SH and NH share and coordinate with the TSA. Most SH, NH, and GA reported that they do not share plans with the CDC (2%), U.S. Marshals (2%), Air Marshals (2%), or military partners (9%). Most CS and R airports reported earlier that they do not have diversion-related plans so, therefore, it aligns that they do not have anything to share with public safety and government agencies.

A note on commercial service airports: While only two CS airports filled out this survey, they both indicated that they do not share or coordinate diversion-related plans. It can be inferred from other survey questions that these airports do not have diversion-related plans to begin with.



Airports That Train with Mutual Aid and Network Partners

Collaborative Training: Table 16 indicates that most smaller airports train primarily with their first responders. SH and NH say they also train with a few network partners they rely upon for diversions, like airlines and FBOs as well as a few agencies like the FAA, TSA, CBP, and FBI.

Findings from Table 16 include the following:

Mutual Aid and Emergency Management: Out of 55 responses, 42% of the airports surveyed said they train with law enforcement, 44% with fire departments, 35% with EMS, 27% with local EM offices, and 20% with local healthcare and hospitals. While SH, NH, and GA report training collaboratively with ARFF (27%), three respondents from R and CS airports indicated that they do not, possibly because CS airports consider ARFF under their own authority. Training with these first responders is most likely because of the Part 139 certification process (described in the literature review) that mandates airports train for emergency events annually with their internal and external stakeholders (primarily first responders) and conduct an exercise with a full-scale functional exercise (called a "recertification or triennial exercise") once every 3 years.

On-Airport Network Partners: Out of 54 respondents, 30% of SH and NH airports reported training with airlines. Thirty percent of SH, NH, and GA airport survey respondents indicated that they train with FBOs and Fuelers, while only 7% from these airports indicated that they train with other airport tenants.

Community Resource Network Partners: Nine percent of SH and NH airports indicate that they train with concessionaires and 6% with ground transportation companies. Only 6% of SH airports indicated that they train with local hotels. None of the survey responders (0%) indicated they coordinate with hangar and aircraft owners, local stores, local restaurants, or airport volunteer groups. While only two CS airports and six R airports filled out this survey, they all indicated that they do not train with local organizations.

Copyright National Academy of Sciences. All rights reserved.

Table 16. Smaller airports that train collaboratively with partners.

Mutual Aid and Network Partners	All (55)	SH (16)	NH (22)	GA (9)	CS (2)	R (6)
First Responders/Emergency Management						
Law Enforcement	42%	50%	41%	22%	50%	33%
ARFF	27%	38%	32%	11%	0%	0%
Fire Department	44%	56%	41%	33%	50%	33%
EMS (Medical)	35%	56%	23%	22%	50%	33%
Local EM Agency/Office	27%	50%	32%	0%	0%	0%
Local Healthcare/Hospitals	20%	50%	9%	11%	0%	0%
Federal Agencies/Military						
FAA	11%	25%	9%	0%	0%	0%
TSA	24%	38%	32%	0%	0%	0%
CBP	13%	19%	18%	0%	0%	0%
FBI	16%	31%	18%	0%	0%	0%
CDC	0%	0%	0%	0%	0%	0%
NTSB	2%	6%	0%	0%	0%	0%
U.S. Marshals	0%	0%	0%	0%	0%	0%
Air Marshals	2%	0%	5%	0%	0%	0%
Military (joint use on airport)	11%	19%	14%	0%	0%	0%
Mutual Aid and Network Partners	**All (54)**	**SH (16)**	**NH (22)**	**GA (8)**	**CS (2)**	**R (6)**
On-Airport Partners						
Airlines	30%	44%	41%	0%	0%	0%
FBOs/Fuelers	30%	44%	27%	38%	0%	0%
Hangar/Aircraft Owners	0%	0%	0%	0%	0%	0%
Concessionaires	9%	25%	5%	0%	0%	0%
Other Airport Tenants	7%	13%	5%	13%	0%	0%
Ground Transportation	6%	6%	9%	0%	0%	0%
Community Resources						
Local Stores	0%	0%	0%	0%	0%	0%
Local Restaurants	0%	0%	0%	0%	0%	0%
Local Hotels	2%	6%	0%	0%	0%	0%
Airport or Other Volunteer Group	0%	0%	0%	0%	0%	0%
Red Cross	17%	38%	14%	0%	0%	0%
Counselors	2%	6%	0%	0%	0%	0%

Federal Agency and Military Network Partners: Some SH, NH, and GA airports train with federal agencies while CS and R do not. Twenty-four percent of SH and NH airports report training with the TSA, 16% with the FBI, and 11% with the FAA. Eleven percent of SH and NH airports indicate that they train with a military organization. Most CS and R airports report not training with federal agencies and military-related agencies.

Airports That Train Specifically for Diversions With Partners



Diversion-Specific Training: Table 17 reveals that very few smaller airports conduct diversion-related training.

Copyright National Academy of Sciences. All rights reserved.

Table 17. Smaller airports that train for diversions with partners.

Mutual Aid and Network Partners	All (58)	SH (16)	NH (22)	GA (9)	CS (2)	R (6)
First Responders/Emergency Management						
Law Enforcement	11%	25%	9%	0%	0%	0%
ARFF	18%	25%	27%	0%	0%	0%
Fire Department	4%	13%	0%	0%	0%	0%
EMS (Medical)	4%	13%	0%	0%	0%	0%
Local EM Agency/Office	4%	13%	0%	0%	0%	0%
Local Healthcare/Hospitals	0%	0%	0%	0%	0%	0%
Federal Agencies/Military						
FAA	5%	13%	5%	0%	0%	0%
TSA	11%	25%	9%	0%	0%	0%
CBP	9%	19%	9%	0%	0%	0%
FBI	2%	6%	0%	0%	0%	0%
CDC	0%	0%	0%	0%	0%	0%
NTSB	0%	0%	0%	0%	0%	0%
U.S. Marshals	0%	0%	0%	0%	0%	0%
Air Marshals	0%	0%	0%	0%	0%	0%
Military (joint use on airport)	5%	6%	9%	0%	0%	0%
Mutual Aid and Network Partners	**All (54)**	**SH (16)**	**NH (22)**	**GA (8)**	**CS (2)**	**R (6)**
On-Airport Partners						
Airlines	15%	25%	18%	0%	0%	0%
FBOs/Fuelers	15%	25%	14%	13%	0%	0%
Hangar/Aircraft Owners	0%	0%	0%	0%	0%	0%
Concessionaires	9%	25%	5%	0%	0%	0%
Other Airport Tenants	6%	13%	5%	0%	0%	0%
Ground Transportation	6%	13%	5%	0%	0%	0%
Community Resources						
Local Stores	0%	0%	0%	0%	0%	0%
Local Restaurants	0%	0%	0%	0%	0%	0%
Local Hotels	2%	6%	0%	0%	0%	0%
Airport or Other Volunteer Group	0%	0%	0%	0%	0%	0%
Red Cross	2%	6%	0%	0%	0%	0%
Counselors	0%	0%	0%	0%	0%	0%

Findings from Table 17 include the following:

Mutual Aid and Emergency Management: Eighteen percent of SH and NH respondents report training for diversions with ARFF and 11% with law enforcement. Beyond this, a few SH and NH respondents train with their local fire department, EMS, or local EM agency.

On-Airport Network Partners: Out of the SH and NH airport respondents, 15% of these airports report training for diversions with airlines, 15% with FBOs and Fuelers, 9% with concessionaires, 6% with other airport tenants, and 6% with ground transportation companies.

Community Resource Network Partners: Only 2% of SH airports reported training for diversions with local hotels. CS and R airports interviewed conduct diverted flight training

Copyright National Academy of Sciences. All rights reserved.

Table 18. Notification-related good practices.

Notification-Related Good Practices	Percent of Airports Using Good Practices (55 responses)
• Update and test your 24/7 mutual aid and network partner contact list regularly (at least annually)	64%
• Identify both primary and secondary contacts at all network partners with various communication methods identified for each (e.g., office phone number, cell phone number, and email address)	51%
• Establish communication protocols (e.g., email, text, dashboard, etc.) that all partners can access for sharing notifications and real-time, accurate, and consistent updates throughout the event	33%
• Use common definitions and acronyms following National Incident Management System (NIMS) and Incident Command Structure (ICS) during events, using common/plain language, and avoiding unique codes/phrases whenever possible	58%
• Test your airport's notification system (technology) via "live" tests throughout the year	47%
• Develop a backup plan for technology during power failures, cyber attacks, and network overloads	31%
• Unsure	5%
• Do not perform these good practices	15%

with their local organizations and other than one GA that trains with FBOs, GAs did not indicate that they conduct diversion-related training either.

Federal Agency and Military Network Partners: Only a small percentage of SH and NH airports report conducting diversion training with agencies involved in diversions such as the FAA, TSA, CBP, and the FBI. None of the GA, CS, or R airports surveyed conduct diverted flight training with their public safety and government agencies.

How Smaller Airports Implement Good Practices with Mutual Aid Partners



The survey looked at whether various smaller airports implement diversion-related industry good practices related to their mutual aid network, as indicated in Tables 18–22. These good

Table 19. Contingency-related good practices.

Contingency-Related Good Practices	Percent of Airports Using Good Practices (55 responses)
• Specific aircraft-related practices (e.g., plans for widebody, international flights, and non-station airline needs)	36%
• Surge of passengers (plans that delineate overflow areas) or aircraft (plans that delineate hard stand areas and parking areas)	33%
• After hours conditions (e.g., plans/agreements for on-call or late shift staffing)	45%
• Additional impacts to airport facilities unrelated to the diversion, including landside impacts (e.g., plans for power outages, weather-related damages) and airside impacts (e.g., alternate plans for damaged runway)	42%
• Compounded communication issues (e.g., plans for network overload or power outages)	16%
• Extended delay situations (e.g., plans for preventing equipment overuse)	35%
• Long-term weather event	29%
• National Airspace System conditions (e.g., natural disaster in one part of the country causing flight diversions)	16%
• Unsure	2%
• Do not perform these good practices	27%

Copyright National Academy of Sciences. All rights reserved.

50 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Table 20. Coordination-related good practices.

Coordination-Related Good Practices	Percent of Airports Using Good Practices (55 responses)
• Meet regularly with your mutual aid and network partners via meetings, workshops, or training sessions	55%
• Ensure the airport's diversion-related plans are coordinated and aligned with mutual aid partner plans (e.g., regional/municipal contingency plans, mutual aid operational plans, emergency response plans, etc.)	27%
• Establish, update, and share a centralized list of local equipment and resources for use in flight diversion events	36%
• Establish and clarify funding sources for incident- and emergency-related costs associated with diversions (e.g., food and water for stranded passengers)	11%
• Identify and establish alternative or multiple service or supply vendors	15%
• Conduct an AAR with mutual aid and network partners, assigning personnel with follow-up responsibilities to ensure accountability	47%
• Update diversion-related plans and training based upon findings from AAR	27%
• Disseminate updated diversion-related plans both internally at your airport and to mutual aid and network partners	24%
• Unsure	2%
• Do not perform these good practices	25%

Table 21. Airline coordination good practices.

Airline Coordination Good Practices	Percent of Airports Using Good Practices (55 responses)
• Understand each airline's priority airport list for flight diversions	27%
• Understand each airline's aircraft towing/gate and deplaning requirements (e.g., widebody aircraft) and available equipment on site at your airport	44%
• Understand typical daily staffing availability as well as whether they use any third-party staffing vendors and how they communicate with them	35%
• Establish agreements to clearly define the assistance airlines can provide to flight diversions associated with non-station airlines at your airport	15%
• Establish agreements to clearly define who (airport or airline) manages passenger assistance/victims' assistance and what resources, supplies, and staffing they will provide in various incident- and emergency-situations	25%
• Establish equipment-sharing agreements with airlines (e.g., airstairs for deplaning, etc.)	22%
• Unsure	2%
• Do not perform these good practices	40%

Table 22. Agencies and airport tenant coordination good practices.

Agencies/Airport Tenant Coordination Good Practices	Percent of Airports Using Good Practices (55 responses)
• FAA - Establish and regularly review capacity constraints policy	15%
• TSA - Establish and regularly review security procedures for stranded passengers	38%
• CBP - Establish and review deplaning plans for diverted international flights	36%
• VARIOUS - Establish and regularly review after-hour staffing plans with TSA, CBP and concessions	17%
• Unsure	13%
• Do not perform these good practices	38%

Copyright National Academy of Sciences. All rights reserved.

practices come from ACRP guidance and industry knowledge and include notification practices, contingency practices, and general coordination practices. The survey also looked at specific practices related to how airports coordinate with airlines, federal agencies, and airport tenants.

- **Notification:** This category covered how well airports update and test contact information, create communication backup contacts, and use common language to ensure the correct people are contacted during diversion events and clear communication occurs. Many implement these good practices, but 15% of respondents do not. It is unclear why this occurs; however, it may be in part because of survey respondents reporting that they do not have plans in place and therefore would not have notification information to update. It could also be related to the small staff available at airports to manage this type of information.
- **Contingencies:** This category covered how well airports plan for the unexpected in terms of aircraft accommodations and associated conditions like surge (influx of many aircraft or passengers), after hours, and long-term events as well as others that can compound or escalate diversion-related conditions. Less than half of respondents say they implement many of these good practices while 27% say they do not.
- **General Coordination with Mutual Aid and Network Partners:** This category covered more details on how often airports meet with mutual aid and network partners and how well they coordinate plans, share resources, manage funding, and update plans with them. More than half of the respondents meet with their mutual aid and network partners, but less than half of respondents say they implement other listed good practices, while 25% say they do not.
- **Airport Coordination Practices with Airlines:** This category covered more details about industry recommended practices with airlines, like understanding how airlines make decisions and how they manage staff and equipment. Table 21 demonstrates that many smaller airports surveyed do not currently use many of the industry-recommended practices with airlines.
- **Specific to Agencies and Airport Tenants:** This category covered more details about industry recommendations currently in practice at airports related to agencies and airport tenants. Less than half of respondents say they implement many of these good practices while 38% say they do not.

Frequency of Mutual Aid and Network Partner Assistance



The survey asked respondents to state how many times in the past 12–18 months they asked their mutual aid and network partners for assistance with incident- or emergency-related flight diversions. As shown in Figure 8, of the 57 respondents who answered this question, 18 airports responded that they did not have to ask for diversion-related support from their partners, while 29 airports reported that they called upon partners from one to five times over this timeframe (mainly SH and NH airports). Three airports reported needing partner assistance for diversions from five to 10 times over this period, and four airports called for assistance more than 10 times over this period.

Recent Diversion Experience Exceeding their Capabilities



When respondents were asked if their airports received any incident- or emergency-related flight diversions in the past 12–18 months that exceeded their airport capabilities, 89% of the respondents said no. This demonstrates that while smaller airports may ask for assistance from mutual aid and network partners (see the previous section) to help manage diversions, it does not necessarily mean that they have exceeded their capacity when this happens. This may be because of the way mutual aid is defined and the various governing structures that smaller airports have over first responders

Only six airports (10%) said that they experienced a major diversion in the past 2 years requiring mutual aid assistance. This small number may be in part because of pandemic conditions that

Copyright National Academy of Sciences. All rights reserved.

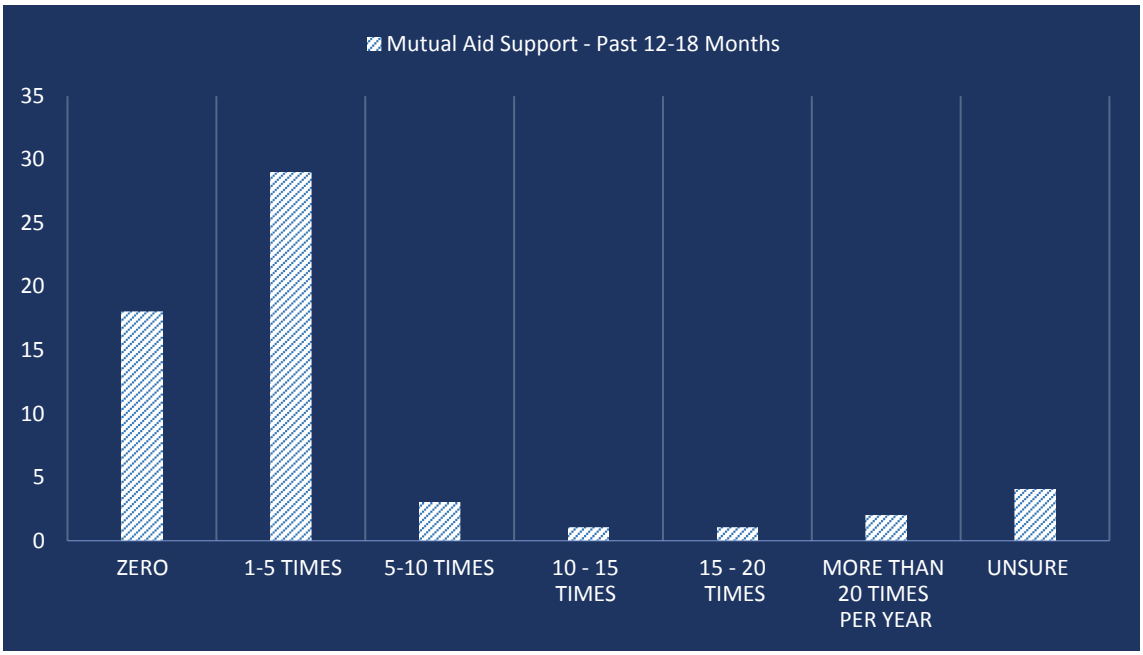


Figure 8. Mutual aid support asked for by 57 smaller airport survey responses.

reduced overall enplanements over the past few years. It also may be related to the relatively small number of flights that are diverted annually. Using 2020 figures, less than 1% (0.23%) of all flights in the United States are diverted annually (Bureau of Transportation Statistics 2022). Finally, it could also be explained by a combination of newer web-based flight-tracking technology, regional planning group efforts, and airline business policies focused on efficiency and fuel conservation, all possibly contributing to fewer diversions to smaller airports.

Of the six airports that recently received diverted flights, a little over half said these were because of poor weather conditions. Three of the diversions involved aircraft emergencies and two involved medical emergencies, while another three involved an incident-related diversion. Many of the airports indicated compounding factors like multiple diversions at once, passenger deplanement, and diversions arriving after normal airport hours.

In following up with the six airports that indicated in the survey that they had received a recent diversion requiring mutual aid assistance, Ocala International Airport (OCF) reported that they have not had emergency diversions in the last 2 years despite what was indicated on their survey. This was most likely because of the survey participant referring to events that happened more than 2 years ago. OCF did report that they currently do not have ARFF and emergency services on the field so they schedule Part 139 operations in advance with each carrier so that the ARFF vehicle and crew can be positioned shortly before a diverted aircraft's arrival. OCF also mentioned that they do use an FAA-approved AEP during diversion events and have mutual support agreements in place with city and county agencies to assist in mitigating them.

Because Shreveport Regional Airport (SHV) is a primary and secondary diversion airport for DFW and IAH (George Bush Intercontinental Airport), and it regularly contends with moderate and severe thunderstorms, it handles an average of 75 to 150 diversions annually. Its 2017 IROPS plan (posted online to comply with DOT tarmac delay contingency plan rules) includes detailed provisions for diversion planning while its AEP covers medical and mechanical incidents and emergencies (Shreveport Airport Authority Operations Division n.d.). The IROPS plan clearly states roles and responsibilities for airport operations and mutual aid and network

Copyright National Academy of Sciences. All rights reserved.

partners related to communication and notification, security, screening, passenger amenities, and sanitation. The plan also outlines responsibilities for FBOs related to ramp parking, passenger deplaning (domestic and international), and ground transportation. SHV has EMT-trained personnel at the airport's ARFF station but also calls upon the City of Shreveport's Fire Station 16 which has EMT-trained firefighters who have direct airport operating area (AOA) access to the commercial aviation ramp and all concourse gates. Station 16 participates in SHV's emergency alert system and AEP. SHV also has an IROPS Working Group with many network partners such as airlines, FBO, concessions, CBP, ATC, TSA, rental car agencies, and sanitation contractors.

Lincoln Airport and Punta Gorda Airport both provided additional details on their various emergencies and incidents over the past few years, which can be found in Chapter 6. Attempts were made to follow up with other airports in Table 23 to obtain more information about their survey answers, but these respondents indicated that they did not wish to be contacted for any follow-up conversations.

Interview Responses



Various interviews were conducted with airports, airline pilots, FBOs, and technology companies to all provide their perspectives related to diversions.

Airport Perspective



Routine technical stops to incident-level diversions: According to survey responses and interviews with SH airports like the Eastern Iowa Airport in Cedar Rapids, Iowa, and NH airports like St. John's International Airport in Newfoundland, Canada, flight diversions that involve an incident can typically be managed effectively using an airport's own capabilities. This is especially true for smaller joint-use civil-military airports that often have a higher ARFF index level such as Dane County Regional Airport-Truax Field. Unless they are landlocked or experience an excessive amount of flight diversions at one time like Hector International Airport in Fargo, North Dakota, did when it received 20 diversions at once, SH and NH airports report that they have the capacity and capability to handle most incident-level diversions.



However, smaller GA airports tend to exceed their capabilities faster than SH and NH airports. GA airports typically need assistance from mutual aid and a network of community partners to manage incident-level diversions because of their limited hours of operation, very small staff size, limited or no federal agency presence such as CBP or TSA, and fewer (if any) in-house emergency response resources. However, this does not hold true for nonprimary CS airports as they service larger aircraft and are required by law to have an ARFF presence, which gives them in-house emergency resources. However, since GA airports tend to have fewer incoming and outgoing daily flights, diversion-related emergencies can be managed without impacting normal flight operations. GA airports have a network that typically includes other nearby larger airports in their region from which they can share equipment and other resources if needed and local emergency response organizations.

Smaller airports typically do not receive early notification of flight diversions; they hear from the ATC or an airline pilot about a diversion just before or just as they witness it landing on their runway. One airport reported that when the operations director arrived at the airport one morning, he found an unplanned aircraft parked on their runway. For smaller airports located in rural or remote locations, incident- or emergency-related diversions can be challenging to manage without adequate warning when they rely on first responders from mutual aid organizations located miles away. However, with a variety of newer flight-tracking technology websites, smaller airport staff stay abreast of changing conditions whether they are on or off the airport. They also use websites to track weather as well as regional and national airspace.

Copyright National Academy of Sciences. All rights reserved.

54 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Table 23. Survey respondents with recent diversions that exceeded their capabilities.

Name and/or Designation	Causal Factors/ Diversion Type	Compounding Conditions	Size of Aircraft Involved	Planning Involved	What Went Well	What Did Not
1. Punta Gorda Airport, Punta Gorda, FL Small Hub	• Poor weather conditions • Minor mechanical incident	• Multiple aircraft diversions • Passenger deplanement	• Large commercial aircraft	• AEP • Diversion checklist	• Early notification • Contacting/gaining assistance from Mutual Aid • Information sharing among mutual aid • Crisis communication • Use of technology • Following a plan • Incident management • Passenger care	None identified
2. Shreveport Regional Airport, Shreveport, LA Non-hub	• Poor weather conditions • Incident	• Multiple aircraft diversions • Passenger deplanement • After hours	• Large commercial aircraft	• Diversion Plan • IROPS Plan	• Contacting mutual aid • Information sharing	• Early notification • Passenger care
3. Lincoln Airport, Lincoln, NE Non-hub	• Minor mechanical • Aircraft emergency • Medical emergency • Incident + emergency	• Passenger deplanement	• Large commercial aircraft • Military aircraft	• AEP • IROPS Plan • Mutual aid plan	• Early notification • Contacting/gaining assistance from mutual aid • Information sharing with partners • Social media used • Crisis communication • Use of technology • Following a plan • Preparation/training • Incident management • Passenger care	• None identified
4. Watertown International Airport, Watertown, SD Non-hub	• Minor mechanical • Incident	• None	• Regional jet	• Airport diversion plan	• Early notification • Contacting mutual aid • Information sharing among mutual aid • Crisis communication • Social media • Following a plan • Understanding capacity • Understanding roles and responsibilities • Response time from mutual aid • Adequate passenger care • Victim's assistance from mutual aid	• Use of technology • Preparation (diversion-related training)
5. Ocala International Airport, Ocala, FL General Aviation	• Poor weather conditions • Aircraft emergency • Medical emergency	• Medical and aircraft emergency	• Large commercial aircraft • Corporate/ private jet	• AEP with diversion checklist • Training/ exercise • Aircraft Emergency Response Plans	• Communication • Planning • Incident management • Passenger care • Assistance from mutual aid partners	• None identified

Copyright National Academy of Sciences. All rights reserved.

Table 23. (Continued).

Name and/or Designation	Causal Factors/ Diversion Type	Compounding Conditions	Size of Aircraft Involved	Planning Involved	What Went Well	What Did Not
6. Springfield-Beckley Municipal Airport, Springfield, OH General Aviation	• Poor weather conditions • Aircraft emergency • Emergency	• After airport normal business hours	• Smaller commuter aircraft	• AEP • Diversion checklist	• Early notification • Information sharing among mutual aid • Following a plan • Preparation/training • Understanding capacity • Understanding roles and responsibilities • Response time from mutual aid • Adequate passenger care • Victim's assistance from mutual aid	• None identified

One airport, Palm Springs International Airport (PSP), reports that it receives good diversion notifications from the San Diego Terminal Radar Approach Control facility. The airport also reports that its operations staff as well as its concessionaire use AeroCloud Systems to understand the number of anticipated passengers arriving from diverted flights. The airport also uses this technology to manage gates and aircraft parking and relies heavily on long-term relationships with and the expertise and diverse resources of its four ground handlers to marshal and coordinate diverted flights efficiently.

Once a smaller airport is notified of a routine or incident-level diversion, operations staff typically notify and communicate diversion status information via a combination of means like phone, email, texts, and radio communication as well as via newer web-based platforms with smartphone app capabilities that can push mass notifications to internal airport staff and on-airport and off-airport partners. These primarily include FBOs and airline ground handlers to manage gas and go technical stops. Fixed-base operators provide things like fuel, deicing chemicals, or labor and parts for any aircraft maintenance needed. Airline ground handlers provide any labor and equipment needed to move an aircraft such as tow bars and deplaning equipment such as airstairs. A sample list of potential equipment needed during diversions can be found in Appendix C diversion checklist from ACRP Guidance.

For incident-level diversion response, a small airport's ability to handle it with its own capabilities depends on the authority structure of first responders. When ARFF, law enforcement, and medical responders are under an airport's authority, they typically find they can manage incidents on their own. When this is not the case, they call upon mutual aid from local emergency management providers (e.g., city or county police, fire, or paramedics) or first responders from other airports in their region if they belong to a regional group for assistance according to plans developed before the event (see more on this topic in the "Regional Perspective" section in this chapter). These first responders provide the expertise and equipment needed such as firetrucks and firefighting capabilities, ambulance and medical supplies, and personnel for crowd control and management.



Airport capabilities at the time of the diversion also play a large role in determining how effective an airport can be at managing them. Some smaller airports have limited aircraft parking spots and gates. This becomes even more challenging during capital improvement projects. Dane County Regional-Truax Field Airport (MSN) noted that their gate capacity is currently cut in half because of construction and the staging area for materials is also impacting the number

Copyright National Academy of Sciences. All rights reserved.

56 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

of available parking spots. They have informed their regional diversion planning group, which includes local airline station managers, about their capacity restrictions. Local airline station managers inform airline network operations about the restrictions, enabling the airline to make informed decisions to temporarily reduce the number of diversions to their airport. And Palm Springs International Airport (PSP) is adding two common-use gates in 2022 to help manage high traffic volume and ensure the airport remains flexible when diverted flights arrive.

Most SH and NH airports refer to using one or more plans to manage diversions, ranging from SOPs for gas and go technical stops to IROPS plans, extended delay contingency plans and emergency contingency plans, and even airport emergency plans to manage incidents associated with diverted flights. To a much lesser degree, BCPs or COOPs are used for diversions. Some SH and NH airports have developed stand-alone diversion plans, diversion checklists, and specialized diversion plans, including international diversion plans. Many GA airports indicate that they do not have plans for diversions in place. Only a few GA airports have developed Aircraft ERPs and AEPs as they are not mandated to create these plans.

Halifax Stanfield International Airport (YHZ), a small hub airport in Nova Scotia, Canada, focused on developing a diversion plan over the past 3 years. They found they needed to formalize diversion-related solutions for the diverted flights they receive from various international airlines, as they are a last-stop location for overseas flights from North America. Their experience with diversions revealed that they needed solutions for numerous international airlines to effectively use their sophisticated common-use technology as well as additional provisions for passengers in the terminal and on the landside. The foundation of YHZ's new diversion plan is based on its Crisis Management Plan, Business Continuity Plan, and on *ACRP Report 65* and *ACRP Report 153* (Nash, J. M., et al. 2012; Cogliandro, B., et al. 2016). YHZ also undertook an IROPS and BCP information exchange with Washington National Airport (DCA) and Dulles International Airport (IAD) where they spent 4 days sharing and learning from one another to improve plans and response efforts associated with diversions and other continuity-related issues.

Also, in many small airports' IROPS plans are passenger care guidance for situations that may arise during diversion situations. Some provide guidance for diversions that may occur after typical business hours or cause passengers to be stranded for a prolonged time. Some make passenger care provisions with concessionaires to call in workers to stay open later or contract with local restaurants, typically pizza places or big-box stores, who can provide food and beverages to stranded passengers. Some contract with ground transportation companies and hotels to provide buses and lodging for passengers. Others make either verbal or formal agreements with hangar and aircraft owners to provide parking spaces or lend equipment for situations like diversions.

Most SH and NH airports regularly train for various types of emergencies with emergency responders from mutual aid organizations to fulfill their Part 139 requirements, but only a few surveyed and interviewed said that they conduct diversion-specific training on a regular basis. Generally, GA airports report that they do not train with mutual aid or other network partners for emergencies or diversions, as they are not mandated to do so.

Incident-level diverted flights are managed by following the ICS, a major component of NIMS, to provide a flexible and scalable system for multiple agencies and disciplines to manage incidents—even when they are not mandated.

Some challenges associated with diversions mainly pertain to how quickly they can escalate into more demanding situations. Some escalating factors include diversions that arrive after normal airport business hours at airports that do not staff around the clock, when non-station international diversions arrive at airports without CBP presence, or when multiple diversions arrive all at once, especially when capacity is restricted because of airport construction projects. Challenges can also arise for smaller airports if they rely on first responders from their city

Copyright National Academy of Sciences. All rights reserved.

or county emergency management agencies and they receive an incident- or emergency-level diversion while those resources are busy responding to another community emergency.

Response to emergency-level diversion: It is uncommon for emergency flight diversions to cause SH and NH airports to exceed their capabilities for many of the same reasons explained in the previous section. Again, this is not the case for GA airports for the same reasons listed above for incident-level diversions, as they rely more on mutual aid resources.



However, when emergencies do exceed a smaller airport's capabilities, some use previously made agreements to obtain more resources. These agreements are typically found in their AEPs or other specific plans such as AEP Annex and Hazard-Specific Plans, Aircraft ERPs, or Airport EOC Plans. Some GA airports rely on regional plans in place like a county emergency plan instead of having their own plans. It is rare for smaller airports in all NPIAS designations to use a CCP and Emergency Communications Plan (ECP).

Emergency-level diverted flights, like incident-level diversions, are also managed by following the ICS, a major component of NIMS, to provide a flexible and scalable system for multiple agencies and disciplines to manage emergencies. Although using social media as a communication tool during emergencies is a newer trend, smaller airports do not typically have the personnel to do this.

Findings indicate that emergency-level diversions do not legally require the activation of Family Assistance Plans. Even for medical emergencies, one of the most common diversion causal factors, these generally result in the deplaning of only one person in need of medical attention. However, even though some airports believe that passenger care provisions are primarily the responsibility of air carriers, some smaller airports do make some provisions for passenger care needs in their IROPS plans or AEPs. They include designating space for passengers to congregate or for CBP clearance, to recharge phones and receive information, and to obtain food and water, among other things. Some even contact their airport volunteers to assist with passenger care needs during emergencies.

Airline Pilot and Customer Experience Perspective



According to an airline pilot for a major U.S. carrier, the evaluation factors used to make flight diversion decisions are airspace, cost of fuel, staffing, aircraft size (e.g., widebody vs. narrow body), and data that helps inform shared situational awareness. The biggest challenge this pilot reports related to diversions is managing the current labor shortages in the aviation industry.

This airline has a diversion plan that is coordinated with its airline's headquarters and has an Airport Diversion Chart that recommends which airports offer full-service capabilities, specifically things like ground handling, FBO, and station support. Pilots at this airline plan well in advance for diversions and communicate with dispatch representatives to assess diversion options if things like weather or medical emergencies arise. This airline also has developed a flight diversion policy for non-station diversions, and the airline's partners (e.g., regional carriers) have similar diversion policies as this major carrier. This airline works directly with the ground handlers, FBO, and contracted staff at diversion airports.

For medical emergencies, this airline reports that it has medical doctor contacts at station airports. Medical situations are evaluated on board with a medical assistant over radio and support from a medical doctor at a station airport. The decision to divert is the pilot's decision based on information from the medical team involved.

According to an airline's customer experience representative, this airline has a kit they regularly use during diversions called "Care While Waiting," which is a cart with water and snacks that is wheeled out to passengers on the aircraft. The airline is always mindful of the 3-hour and 4-Hour DOT Tarmac Delay Rules and monitors the situation to ensure passenger needs are

Copyright National Academy of Sciences. All rights reserved.

58 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

met. If passengers are deplaned, the airline calls their catering company to make further food and beverages available.



Technology Perspective

Several technology companies were interviewed for this synthesis. These companies provide a wide range of air traffic management, flight tracking, collaborative decision-making (CDM), and decision support tools, among other things.

According to one technology provider, its systems focus on surface tracking and resource management at diversion airports, particularly monitoring equipment like flight stairs and tow bars. Their products are used by large-hub airports to send out emails to smaller airports in their region to inform them of potential flight diversions. This company also reports that more airlines are becoming interested in tools for shared situational awareness and diversion-tracking capabilities as they want to save time and money. They have found that airlines also want more specifics related to the capabilities and capacity of airports, including the availability of CBP staff. This company is currently challenged with providing streamlined information to pilots on their tablets, like a comparison of deicing pad availabilities at potential diversion airports, without competing with communications from the pilot's airline dispatcher. Technology affordability is one issue for smaller firms as surface tracking technology can be expensive for smaller airports. This company recommends that a smaller airport ask several important questions when deciding on technology to implement like: What are the right diversion tools for my airport? How can the technology be modified for my airport size? How do I get broad distribution? How deep should I go with data? What is the cost-benefit analysis for the various types of technology options?

Another technology provider offers a chat feature with multi-airport communication capability so real-time conversations and decision-making can take place. This provider also provides CDM technology and tools that include a flight operations communication platform with global data feeds to use in daily operations. Data involved with their diversion-related technology includes passenger demand, airline scheduling modifications, or asset management. They also can provide notification tools for delays, congestion, and constraints so airports can adjust and minimize passenger and airline disruptions while ensuring efficient airside operations.

Canadian airports are using a web-based platform to manage daily airport operations and safety. The system was developed with airports in mind, and it can be accessed on laptops and smartphones. It features an electronic log that tracks all types of different diversion-related data inputs, which is then used to create AARs. It can pull data from an incident or emergency and tailor it specifically for different user types.

Newer technology CDM platforms have also been developed that can integrate various technologies to improve overall airport response to diversions. For instance, airports are using a variety of software packages that can provide a centralized dashboard to combine flight tracking with alert notification procedures. These software platforms also become virtual EOCs with data that can be instantly archived for after-action reporting. By using integrated technology, airports along with their city and county government agencies as well as city and county agencies are all notified and kept up-to-date on the latest diversion-related information.

From discussions with several airports and from talking with several technology companies, it is clear that there are several challenges or pain points associated with implementing new technologies that all focus on timing issues, including the following:

- **Staffing:** The time it takes to implement new software varies widely from airport to airport. Timing is dependent on many different factors such as the amount of upskilling and training

Copyright National Academy of Sciences. All rights reserved.

needed as well as other staffing challenges, such as COVID-19 impacts. Training time can be considerably lengthier with new staff, as reported by several airports.
- **IROPS:** Several airports reported that migrating to a new system during seasons where there are a lot of irregular operations events can be challenging.
- **Data entry:** Airports report that data entry can be a time-consuming process when implementing a new airside or airport-wide software system.

However, despite these challenges, airports reported successful outcomes with software implementation, as technology firms provided them with proper training and instruction and spent ample time with staff to ensure they were ready for full implementation.

Many of these real-time flight-tracking providers report that the costs related to data have been reduced and that diversion planning has become a major focus for airlines. However, they note that airlines are concerned that they only get fragmented data from providers and not a consolidated picture of the entire national airspace system for use in their diversion decision-making efforts. This comment is also included in the "Further Areas of Study" section of Chapter 9.

FBO Perspective



According to one FBO, training is the most important factor needed in successfully planning to manage diversions as they often occur without much notification. This FBO conducts ongoing personnel training so they can manage a high level of activity while remaining calm and professional. The FBO interviewed mentioned that because they coordinate a variety of local services at an airport (e.g., aircraft fueling, maintenance, and deicing; and passenger and crew catering, concierge, conference support, and chauffeuring services), they must be staffed well around the clock. The biggest challenge this FBO has is securing enough ground transportation for passengers whether they are dealing with a diverted flight to their airport or following up with a flight that was to arrive at their airport but diverts to another airport.

Regional Perspective



More and more airports are seeing the value of collaboration and are forming regional networks to improve operations. Most regions are centered around a larger hub airport and include a variety of all other airport sizes as well as airlines and federal agencies. They meet regularly to openly talk about lessons learned and good practices from recent events and discover where collaboration and sharing resources can improve future performance.

Three well-established networks in the United States, like the Minneapolis-St. Paul Regional IROPS Network, the North Texas Regional Network, and the Denver International Airport Network were consulted for this synthesis. All the networks include smaller airports (e.g., SH, NH, GA, CS, and R) on regular calls and/or meetings. These meetings focus on understanding capacity and capability at airports in the region, lessons learned from recent irregular or emergency events, and anticipation and preparation for upcoming major events.

The North Texas Regional Network meets twice per year via conference call or Zoom call. They currently have 26 airports in the network. They report that their region gets many different types of diversions regularly and that three airports in their region bear the brunt of diversions. Only one of these three, Will Rogers World Airport in Oklahoma City, is a small airport. This airport was able to tell the regional group in the most recent conference call that they are starting runway construction, which impacts their capacity. The regional network is taking this into account for all future diversion planning.

Copyright National Academy of Sciences. All rights reserved.

60 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Many of the airports in the North Texas region communicate via the Passur Diversion Tool, as the large hub in the region, Dallas/Fort Worth International Airport (DFW), and its hub airline, American Airlines, use it. When DFW diverts flights, it uses the Passur Diversion Tool's chat feature with American Airlines and potential diversion airports to understand the capability of the ground crews at each potential diversion in real time.

The Minneapolis-St. Paul (MSP) Regional IROPS Network has also met bi-annually over the past 11 years. They have had a variety of in-person workshops, conference calls, and Zoom calls over the years to collaborate. Most recently, they met on April 12, 2022, and many airports discussed their diversion-related planning and experience managing diverted aircraft. Most airports reported that they only receive a few diversions per year, and many have been able to manage them well with their own staff. Several of them, like Central Wisconsin Airport, mentioned how they welcome them as long as they receive early diversion notification as diversions provide airports with additional revenue. However, Sioux Falls Regional Airport reported that diversions are stressful for them as they are contending with threadbare staffing levels and do not have diversion-related plans in place. Their biggest challenge is managing airline crew time-outs as these can result in an extended delay situation that is hard to manage with few staff members on hand. Hector International Airport also agreed that they are leery about getting a lot of diversions as their gate space is limited. The MSP Network agreed that the biggest factor in successfully managing diversions is having the connections and relationships in place to obtain the information needed and get the resources in place to mitigate them.

MSP Regional IROPS Network developed a Playbook for their region to use that includes profiles for each diversion airport in the region to share with other airports in the region as well as airlines so that everyone is kept up-to-date on their capacity and capabilities. Smaller airports in MSP's Regional IROPS Network have found that airline station managers appreciate these profiles as it helps them inform their diversion preparedness. A sample profile is included in Figure 9.



Challenges

Challenges have been identified through the literature review, survey results, and interviews conducted for this synthesis. Some of the major challenges associated with small airports managing diversions include the following:

- **Coordination**
  - Lack of plan coordination, possibly because of the authority structure of first responders, limited staffing, and time available to do this or to a lack of plan development.
- **Training**
  - Limited diversion-specific training and exercises in use.
- **Resources**
  - Supply chain issues, like jet fuel, as occurred in the spring of 2022 at Austin-Bergstrom International Airport, resulting in massive flight cancellations and stranded passengers.
  - Available labor force to manage diversions at airlines and airports of all sizes.
- **Understanding of Responsibilities**
  - Blurred lines between airlines and airport responsibility over providing passenger needs such as food and water.
- **Technology**
  - Cost-prohibitive tracking and communication technology that could help smaller airports provide real-time capacity and capabilities to airlines and larger airports needing to divert flights.
  - Numerous types of communication technologies in use by various smaller airports create challenges for creating a regional or national approach to collaborative diversion planning.

Copyright National Academy of Sciences. All rights reserved.



Figure 9. Sample airport profile from MSP's Regional IROPS Network Playbook (Source: MSN Airport).

- **Safety and Risk**
  - Risk management issues associated with receiving incident- and emergency-level diversions.
  - Safety and security issues associated with managing unruly passengers before agencies like the FBI can get on site.
- **Escalation**
  - A myriad of factors exists that can escalate diversion events from simple gas and go operations to emergency status. These include prolonged severe weather events, supply chain issues for products like deicing chemicals and fuel, and receiving multiple diversions at once, among many others.
  - International diversions at smaller airports without CBP presence can delay passenger processing and potentially increase tarmac delay times.
  - Ground transportation is overlooked as a network partner. This has been a problem for smaller airports when multiple diversion events occur and there is no way to get many passengers to hotels or other accommodations expediently.

Copyright National Academy of Sciences. All rights reserved.

62 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

- **Air Service**
  - The impacts to passengers on low-cost carriers that only operate one or two flights per week from a smaller airport. Passengers can be stuck for days before the airline can rebook them on another flight.
  - Impacts from less hub-and-spoke aircraft operations to more point-to-point flights can make diversion planning more challenging as smaller airports may encounter more non-stationed airlines.
- **Maintaining a State of Readiness**
  - Ensuring equipment readiness when unplanned diversions occur. This can be challenging because of maintenance schedules and equipment that can negatively be impacted in extended cold or heat snaps.
- **New Knowledge Application**
  - Finding ways to not only collect information related to lessons learned but also apply new concepts among all internal and external stakeholders involved.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 6

Case Examples

The following airports are highlighted as case examples in this synthesis and bring knowledge gained from managing and mitigating flight diversions associated with incident and emergency response in collaboration with their respective mutual aid networks:

- Dane County Regional-Truax Field, Madison, WI — SH
- Punta Gorda Airport, Punta Gorda, FL — SH
- St. John's International Airport, Newfoundland, Canada — NH
- Lincoln Airport, Lincoln, NE — NH
- Rick Husband Amarillo International Airport, Amarillo, TX — NH
- Iqaluit International Airport, Iqaluit, Nunavut, Canada — GA
- Flying Cloud Airport, Eden Prairie, MN — GA/R

Small Hub Airports

Dane County Regional Airport-Truax Field, Madison, WI (SH)

- **Level and type:** Incident level and Numerous gas and go flights
- **Diversion dates:** June 12, 2021
- **Causal factors and diversion type:** Weather
- **Compounding conditions:** Multiple diversions with limited gates and parking
- **Size of aircraft involved:** Small and large commercial aircraft
- **Interview sources:** Michal P. Junik, Airport Operations Supervisor, and MSN's website

Airport Information: Dane County Regional Airport-Truax Field (MSN) is a joint-use civil-military airport located approximately 7 miles northeast of downtown Madison, Wisconsin. The airport is NPIAS-designated with current enplanements of 1,043,185. The Wisconsin Air National Guard 115th Fighter Wing and Wisconsin Army National Guard operate from this airport. The airport is considered a department of Dane County and is led by the County Executive, an airport director, and the Dane County Airport Commission. The Airport Operations Department is led by eight airport operations supervisors who manage the day-to-day operations and oversee approximately 70 to 80 additional personnel that work in airfield and terminal operations positions. The airport has formal agreements with the Dane County Sheriff's Office to provide law enforcement personnel on site and joint-use agreements to obtain ARFF capabilities from the 115th Fighter Wing. Its FBO and TSA both operate 24/7.

Diversion Planning: As a Part 139 airport, MSN has developed an AEP to manage emergencies associated with diversions, among other things. The airport also includes provisions for diversions in its tarmac delay contingency plan; however, the airport is in the process of updating this plan to become more of an irregular operations plan. The new operations supervisor has found that

Copyright National Academy of Sciences. All rights reserved.

64 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

MSN needs a plan that formalizes cooperative planning among various stakeholders, provides more broad applications beyond aircraft delays, and focuses on the planning before and recovery from IROPS situations like diversions. MSN also has an international diversion plan as well as a stand-alone diversion checklist that is accessible in paper copy binder format and in PDF form available on iPads. This checklist is provided in Appendix C. They do not have a formal written Family Assistance Plan but are assessing how to leverage their limited accommodations. They do have passenger care provisions and store distressed passenger support items (e.g., pillows, blankets, water bottles, etc.). Other planning efforts by MSN include pre-designating CBP hold areas and ensuring concessionaires are well-stocked and are on-call after hours.

MSN does not train specifically for diversions; however, the airport's operations supervisor has recently taken ACI World's Online Learning Centre's "Airport Irregular Operations (IROPS)" course and has reviewed *ACRP Report 65* and *ACRP Report 153* as a first step in updating and revising plans associated with diversions and delays. He is seeking ways to future-proof the airport and improve overall airport resiliency when confronted by irregular and emergency situations by ensuring that all industry best practices are included in irregular planning efforts, including diversion events.

In terms of capacity, MSN has two common-use gates, remote aircraft parking for 12 aircraft with additional space for emergencies, remote aircraft fueling, available personnel for security escorts, and a computerized message dissemination system. In terms of diversion-related equipment, MSN has two airport stair trucks and has access to airline- and FBO-owned airstairs.



Diversion Tracking, Notification, and Communication: MSN typically receives diversions from O'Hare International Airport and Minneapolis-St. Paul International Airport. The airport uses a variety of technology applications to track potential diversions to MSN, including Flight Aware and Ariva. The platform used depends on the preference of the operation manager on duty. Formal notification of an incoming diversion comes either from ATC dispatchers or from air carriers, either in advance of a diversion or when an aircraft touches down on the runway. MSN is typically notified via phone or email with flight information and intentions. MSN uses the Passur Airport Communicator Platform to communicate within the airport and among mutual aid and network partners. MSN has a social media presence, and all communication is handled by its Director of Marketing and Communications for both emergency and marketing purposes. In addition, MSN has also constructed a new EOC facility, but it had not been activated at the time of this publication.



Mutual Aid and Network Partners: The airport is located 4 miles from the closest Dane County Firehouse, which provides ambulance and medical service through a formal agreement. Additional law enforcement beyond the airport's capabilities is provided via an agreement with the Dane County Sheriff's office. The airport is 7 miles away from the nearest hospital, but no formal agreement has been made between the airport and them. The airport's tarmac delay contingency plan identifies concessions as a partner in delay and diversion situations, but no formal written agreements with them are currently in use. However, both concessions and the TSA communicate well with the airport and ensure they have proper staffing available when situations warrant it, and they are forthcoming with information if issues arise. The Red Cross is also identified as a network partner in the airport's AEP, but not in its contingency plan. For international diversions, MSN has a CBP hotline to call to ensure the aircraft is pre-cleared but is looking for additional ways to collaborate with the CBP to create plans to effectively manage both multiple international flights and international flights that arrive after hours. In addition, MSN would like airlines to share their contingency-related plans with them as well. The airport is developing MOUs with each of its various mutual network partners, so the airport has an a la carte system for assistance as needed.



Diversion-Related Information and Experience: On June 12, 2021, six aircraft from four different airlines (three from Envoy, one from Republic, one from Skywest, and one from American

Copyright National Academy of Sciences. All rights reserved.

Airlines' main line) were diverted to MSN from O'Hare International Airport because of strong thunderstorms and wind, and when all these gas and go flights arrived, MSN only had four aircraft parking spots available to serve these aircraft. MSN communicated with Envoy's ground handlers so that they could coordinate the flights as efficiently as possible to ensure that none of the aircraft remained on the tarmac beyond 3 hours. MSN also coordinated with its FBO, Wisconsin Aviation, who called in extra personnel to manage the amount of jet fuel needed to refuel the six aircraft. All flights were refueled and departed from MSN in under 3 hours and all regularly scheduled flights were able to land and take off during this time as well.

However, this multiple-flight diversion event revealed the need to plan more proactively with airlines to communicate upcoming capacity issues associated with the airport's new South Terminal construction. As a result, the airport talked with its station airlines' dispatchers about ramp and movement area limitations because of construction to come and received assurances that these limitations would be considered in diversion-related decisions for the foreseeable future. This event also revealed that more coordination among mutual aid and network partners is needed and that more collaborative training would be helpful as well.

The AAR from this event revealed the following: "If technological devices (e.g., iPads) are anticipated to be a primary tool in managing, documenting, and keeping up with an event, ensure devices are properly configured to ensure resources are readily available before a diverted flight arrives. Also, flight diversions can be more effectively managed when airport operations personnel are familiar with ground crew operations. Lastly, thorough documentation and time keeping ensures that no aircraft exceeds the tarmac delay time limitation per U.S. DOT rules."

Punta Gorda Airport, Punta Gorda, FL (SH)

- **Level and type:** Incident, multiple diversions
- **Diversion date:** December 30, 2018
- **Causal factors:** Thick fog
- **Compounding conditions:** After hours
- **Size of aircraft involved:** Large commercial aircraft
- **Interview sources:** Ray Laroche, Director of Operations and Maintenance and PGD's website



Airport Information: Punta Gorda Airport (PGD) is a public airport located 3 miles east of Punta Gorda, in Charlotte County, Florida. It serves Florida's southwest coast between Sarasota and Ft. Myers. It receives a lot of spring break traffic during March and April and it also regularly contends with special events (e.g., college sports weekend travel). The airport is an NPIAS-designated small hub airport with current enplanements of 787,022. It is owned by the Charlotte County Airport Authority, run by five board commissioners. The airport's daily operation is managed by a director of operations and maintenance who oversees approximately 53 staff members as well as an FBO general manager who oversees approximately 26 personnel. Low-cost carriers Allegiant (year-round) and Sun Country (October through April) operate from the airport.



Diversion Planning: The airport plans for most of its diversions via its IROPS plan but manages medical and other emergencies through its AEP. Most of PGD's diversions come from low-cost airlines that service PGD as well as nearby St. Petersburg, Sanford, and Ft. Meyers. Challenges associated with diversion planning include the fact that the airport does not have a 24/7 tower or CBP facilities, has only 10 RON commercial ramp spaces, and is limited to Airbus A321-type and below per ground-service equipment limitations. However, all the airport's eight gates are under common-use leases to air carriers and are controlled by the airport, so PGD can make these gates available to an air carrier seeking to deplane at a gate to the extent practicable.

Copyright National Academy of Sciences. All rights reserved.

66 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

PGD trains regularly for emergencies with other mutual aid organizations to maintain their Part 139 certification, but does not train specifically for diversion events. However, PGD has a learning culture and when staff members attend conferences or workshops, they share insights and lessons as the airport is continually improving its operational response.

The airport believes that the key to successful diversion management and ensuring resiliency is proper planning, specifically focusing on regular communications with mutual aid and network partners and making sure contact information is current. The airport also uses an after-action discussion with all stakeholders, called a "hot wash," to obtain lessons learned so they can continually improve. PGD is part of the Florida Airports Council and has access to this network for training opportunities and knowledge transfer.



Diversion Tracking, Notification, and Communication: Currently, PGD does not have an operations center, but it hopes to have one in the future. For now, it handles all communications related to diversions via various technologies such as Flight Radar for tracking potential diverted flights and a new virtual tower technology called Virtower (virtower.com) that many of the airports in Florida are now using, which enables Virtower customers to track operations and gates and parking status to help determine real-time capacity.



Mutual Aid and Network Partners: Charlotte County Fire and EMS operates an ARFF station (Firehouse #7) located at the northwest corner of its airfield and has two ARFF trucks and EMS staff. The airport has formal agreements with Charlotte County to provide ARFF service and uses its multiple fire stations equipped with ambulances for ALS response. Law enforcement beyond the airport's capabilities is provided by the Charlotte County Sheriff's Office via an Off-Duty Detail contract. The airport is 4.3 miles away from the nearest hospital.

Allegiant will soon be taking over the airline's currently contracted ground services and they will have "call-in-staff" that will work with the airport staff during diversion events that exceed the airport's capabilities. The airport has been challenged with providing ground transportation during diversion events and has added ground transportation companies to its mutual aid network.

The airport noted that their county is very active and has worked with 10 to 15 different third-party services for rapid emergency response that is GPS-based so that dispatchers can monitor responders' locations on their smartphones. PGD has found that this access has improved mutual aid coordination and communication.



Diversion-Related Information and Experience: PGD occasionally receives gas and go diversions from Allegiant from Sanford Airport or St. Petersburg Airport because of daily afternoon thunderstorms throughout most of the year. The main challenge they encounter is receiving multiple diversions at one time. In some instances, four to five diverted flights show up at the same time as regularly scheduled flights, making it challenging to manage all the traffic. One of the biggest issues the airport faces when this happens is providing enough buses to transport diverted passengers off-airport.

Occasionally, PGD will receive a medical emergency and when this happens, PGD receives notification from the ATC or a pilot and, depending on the severity, it either calls its ARFF or reaches out to County Fire for support. It uses its mapping technology to know exactly where emergency response is needed and therefore which firehouse to call upon. ACISS Systems Inc. provides the system dispatchers use to inform first responders if there is a mass casualty event, giving them the ability to change gates. They also have a third-party mapping grid for 911 on site that provides real-time information for any medical flight challenges encountered.

On December 30, 2018, three Allegiant Airline flights en route to Orlando were diverted to PGD and arrived at the airport at 12:30 a.m. Approximately 300 people remained at the airport overnight and waited for at least 2 hours for their luggage as airport and ground handler staff

Copyright National Academy of Sciences. All rights reserved.

are not allowed on the apron once lightning has been detected within a 5-mile radius. Allegiant typically arranges ground transportation, but because of the holiday weekend, no ground transportation or buses were available. Passengers had to arrange their own ground transportation and submit receipts to the airline for reimbursements.

In addition, a few years ago, PGD received six widebody diversions averaging about 186 passengers per flight. It was during a Florida college tournament weekend and around 700 people arrived at the terminal. There were very limited ground transportation services as all rental cars were booked and the airport did not have any arrangements previously made with public transportation or school buses. It took hours for passengers to find transportation. This issue has been resolved through the addition of an on-site ground transportation provider that can call on additional assets from surrounding counties if needed.

St. John's International Airport, Newfoundland, Canada (NH)

- **Level and type:** International flight with multiple medical emergencies
- **Diversion date:** Sunday, January 24, 2016
- **Causal factors:** Weather and turbulence
- **Compounding conditions:** Multiple injuries on an international flight after normal business hours on a weekend
- **Size of aircraft involved:** Large commercial aircraft
- **Interview sources:** Kirk White, Director of Operations and YYT's website



Airport Information: St. John's International Airport (YYT) is equivalent to a United States small hub airport as it primarily serves Air Canada flights. It is located approximately 6 miles northwest of St. John's, a city on Newfoundland Island off Canada's Atlantic coast. The airport serves approximately 180,000 passengers annually and is considered one of the largest airports in Atlantic Canada based on civilian traffic. It is also home to Air Reserve Flight Torbay, a Royal Canadian Air Force (RCAF) transient servicing unit that provides servicing and support except for fuel to all RCAF and North Atlantic Treaty Organization aircraft upon request. Torbay Air Station is located 5 miles from YYT. The St. John's International Airport Authority owns and operates the airport and is considered a private, not-for-profit corporation. The Authority is overseen by a 12-member board of directors and a 10-member senior management team, which includes a Manager of Emergency Response and Planning. Approximately 85 additional full-time and seasonal personnel work in airfield and terminal operations positions. YYT operates 24/7 and has its own Category 6 level (equivalent to ARFF Level B or C Index) fire department as well as security and medical response capabilities. The airport also has FBOs and one deicing operator on site. However, the airport does not provide any ticketing or ground services and needs to rely on carriers for those services.



Diversion Planning: YYT has built resiliency into its diversion planning by developing a robust emergency plan that is coordinated with the St. John's City Fire Department, Eastern Health Medical, and the Royal Newfoundland Constabulary. The emergency plan includes 20 different scenarios of potential emergency and security situations including aircraft diversions. In addition, they have an Emergency Planning Committee, which includes mutual aid agencies as well as departments within the airport that have to meet one to two times per year to coordinate with one another, share lessons learned, update key data, and plan exercises to test the airport emergency plan.

YYT also uses an electronic diversion checklist that it stores and accesses on the Vortex system, which is a web-based platform used by some Canadian airports to manage daily airport operations and safety management. The system was developed with airports in mind, and it can be accessed on laptops and smartphones. It features an electronic log that tracks all types of

Copyright National Academy of Sciences. All rights reserved.

68 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

different diversion-related data inputs, which is then used to create AARs. It can pull data from an incident or emergency and tailor it specifically for different user types.

The airport is part of a national planning group called the Canadian Airports Council, a division of ACI-NA, which advocates for policies that strengthen the ability of Canada's airports to serve their travelers and communities. YYT also communicates with a local Airline Operations Committee that focuses on operational efficiencies, which include diversion-related scenarios.

YYT conducts full emergency exercises biennially with aircraft for both security and emergency scenarios to comply with Canadian Aviation Regulations. Every other year they conduct a tabletop exercise, which includes the agencies that are on the Emergency Planning Committee.



Diversion Tracking, Notification, and Communication: The airport's operations center uses Flight Aware technology to track potential diverted aircraft. Formal notification of an incoming diversion comes from NAV CANADA air traffic control (FAA counterpart) and is typically communicated anywhere from a few minutes beforehand to 40 to 60 minutes prior to receiving a diverted aircraft. YYT is typically notified via phone with flight information and intentions. YYT uses an automated notification, initiated by the airport duty manager, to advise airport management. Phone calls are also initiated by the Security Operations Center to their internal emergency response resources and their mutual aid partners.

YYT has a social media presence, but they do not use it for emergency situations, but rather for marketing and promotion purposes. The affected air carrier is responsible for any release of diversion-related information, as YYT can only comment on the operational status of the airport. If needed, a joint statement is coordinated between the airport and the carrier.



Mutual Aid and Network Partners: Additional firefighting, law enforcement, and medical assistance are provided by the City of St. John's when needed. The airport only has a formal mutual aid agreement with the City of St. John's for the provision of firefighting services. The airport is located 5 miles from the closest fire department that provides fire, ambulance, and medical services. Law enforcement is provided by the Royal Newfoundland Constabulary. The airport is 5 miles away from the nearest health center. Agreements have not been made between YYT and other agencies from nearby provinces.

YYT also has strong relationships with other partners such as the following:

- NAV CANADA air traffic control (FAA counterpart)—this agency is responsible for providing instrument flight rules and visual flight rules services to aircraft that are arriving and departing or transiting the YYT area.
- Canadian Air Transport Security Authority (CATSA) (TSA counterpart)—this agency provides passenger and non-passenger screening in accordance with Canadian Aviation Security Regulations.
- Canada Border Services Agency (CBSA) (CBP counterpart)—this agency provides customs and immigration at YYT.

The airport generally relies on air carriers to be responsible for deplaning passengers. If needed, the airport duty manager can coordinate a request to extend concessions for assistance as well.



Diversion-Related Information and Experience: YYT is referred to as a small-big airport because of its location and expertise in diversions. Because of its location on the easternmost part of the continent, it is often used as the first or last stop for flights before or after crossing the Atlantic Ocean. The airport also offers overnight parking for commercial airlines on the Airport Terminal Apron.

The airport considers domestic diversions to be easy to manage as they receive one every couple of weeks, particularly in the spring and summer, and these diversions are typically flights

Copyright National Academy of Sciences. All rights reserved.

headed to Europe. The airport generally receives gas and go, mechanical, and medical emergencies. When the airport receives an emergency diversion, they set up an emergency coordination center to communicate with all partners and mutual aid alongside the operations center. When this happens, staff need to perform two different roles at the same time: emergency coordination and continuity of operations. YYT's biggest challenge is managing international diversions as this scenario can result in many people coming off a large aircraft at one time. They are used to medical emergencies where just one person, the injured one, is deplaned and the flight continues.

On January 24, 2016, YYT received a diverted international aircraft with multiple medical emergencies after typical operating hours on a Sunday night. About 4 hours into the American Airlines flight from Miami, Florida to Milan, Italy, it encountered severe turbulence and the plane suddenly dropped and rolled to one side.

As with all emergencies, the airport duty manager received information from NAV CANADA about this diversion. Both internal airport and mutual aid and network assistance were needed for this emergency. The need to tend to the number of passengers and crew requiring medical attention involved both internal emergency services and external 911 medical services. And because this was an international aircraft, it also required CBSA assistance.

The emergency response team was notified via phone and the emergency coordination center was activated, with additional staff called in. The scenario for a medical diversion in the Airport Evacuation Plan was utilized; however, because of the number of injured, the need for more assistance was recognized. The airport's emergency response team provided initial care with the 911 medical services providing ambulances for three flight attendants and four passengers who all needed medical attention.

Since CBSA staff operate in 16-hour shifts with no backup staff available, this vital international processing service is not always available 24/7. Typically, local CBSA is informed of international diversions via phone during normal operating times; however, in circumstances like this event, a call is made to CBSA Operations in the town of Gander to make callouts for staffing and services.

Air Canada, an interline partner with American Airlines, arranged for buses to take passengers to hotels for the night. Air Canada or other airlines that operate in YYT typically provide passenger assistance.

No AAR was filed for this event. Air Canada and American Airlines covered the costs of the hotels, meals, and water, so no costs were incurred by the airport for this diversion event.

Non-Hub Airports

Lincoln Airport, Lincoln, NE (NH)

- **Level and type:** Various incident- to emergency-level diversions since 2012
- **Diversion dates:**
  - 2012: 1 diversion (international diversion)
  - 2016: 1 diversion (medical emergency)
  - 2017: 1 diversion (gas and go)
  - 2018: 3 diversions (weather and gas and go)
  - 2019: 4 diversions (weather and gas and go)
  - 2020: 3 diversions (gas and go, medical, and military)
  - 2021: 8 diversions (weather)
  - 2022: 1 diversion (mechanical emergency)

Copyright National Academy of Sciences. All rights reserved.

- **Causal factors:** Primarily weather incidents, but also medical and mechanical emergencies
- **Compounding conditions:** Multiple diverted flights
- **Size of aircraft involved:** Large commercial aircraft and regional jets
- **Interview sources:** Bob McNally, Director of Operations, and LNK's website



Airport Information: Lincoln Airport (LNK) is a commercial service civilian airport with Nebraska Air National Guard (ANG) and Army National Guard aviation units located approximately 5 miles northwest of downtown Lincoln, Nebraska. The airport is an NPIAS-designated non-hub airport with 2019 enplanements of 150,214. The Nebraska Air National Guard and its 155th Air Refueling Wing operate from this airport, which has a 12,901-foot runway. The airport is owned and operated by the Lincoln Airport Authority, which is comprised of a five-member, publicly elected board of directors. Airport senior staff members include an Executive Director, director of operations, Director of Planning and Development, Director of Finance and Administration, and Director of Communications and Customer Engagement. The Authority owns or controls 5,500 acres, of which around 2,500 acres constitute airside property. The airport has a staff of 23 operations and maintenance personnel, 11 police officers, approximately 15 customer service people, and six engineering and administrative staff members. Currently, LNK is served by United Airlines with six flights per day and primarily receives United diversions.

Because the Nebraska ANG provides primary ARFF response at the airport, and because of the response needed by the Air Force related to the KC-135 refueling operation, the routine response to any aircraft emergency is equivalent to an Index E response. Typically, most small commercial airports have Index A or B capabilities. LNK and the National Guard Bureau operate under an airport joint-use agreement that establishes the parameters under which the Air National Guard provides primary ARFF services at the airport. In addition, the ANG provides initial medical response to an emergency medical technician (EMT) level under a separate multi-agency mutual aid agreement. Under that same agreement, the local fire department provides medical and ambulance services for all emergencies. This arrangement has been in place for 25 years and, during that time, the airport, ANG, and all mutual partners have developed a collaborative culture, and all help and back up one another as needed, a key factor in the airport's ability to remain resilient during diversion situations. The airport employs its own police officers consisting of 11 full-time officers who staff the terminal facilities on a 24/7 basis. Each officer is trained in the approved diversion procedures. These officers can call upon assistance from other local law enforcement agencies under the previously mentioned mutual aid agreement. These officers have even been trained to run aircraft-loading bridges if needed as a last resort.



Diversion Planning: LNK uses an AEP for emergencies and an IROPS plan for incident-level planning. The IROPS plan addresses some non-emergency procedures while other processes are addressed in SOPs. LNK also has a Mutual Aid Plan that outlines how it works with mutual aid partners, including all local and certain state assets and personnel including fire, rescue, law enforcement, and resource providers.

LNK conducts an annual review of IROPS and emergency procedures with United Airlines station managers as well as with its FBOs, Duncan Aviation and Ross Aviation. The airport also owns resources that are often needed in diversions, such as airstairs and tow bars. These resources are available to all tenants for use during diversions.

LNK has been anticipating diversions related to unmanned aircraft systems and drones, particularly related to operations at Offutt Air Force Base located approximately 60 miles northeast of LNK. The airport has held working sessions to plan for potential diversions. To date, there have not been any diversions at LNK because of unmanned aircraft systems.



Diversion Tracking, Notification, and Communication: The airport has a 24/7 communications center. The airport uses a variety of technology applications to track potential diversions including FlightAware, Flightview, and Passur.

Copyright National Academy of Sciences. All rights reserved.

Formal notification of an incoming diversion comes either from ATC controllers or from air carriers—either in advance of a diversion or when an aircraft touches down on the runway. LNK is typically notified of emergency diversions via the Alert Notification System that connects ATCT, 911, ARFF, and airport operations. Non-emergency notifications can come from air traffic or air carriers via standard phone service with flight information and intentions. LNK uses a shared inter-agency radio network to communicate within the airport and among all mutual aid organizations.

LNK has not traditionally used social media as part of irregular operations or emergency operations, but they recently hired a Director of Communications who plans to implement more of this communication in the future.



Mutual Aid and Network Partners: Because of the resources associated with the ANG's ARFF capabilities, most diversions and other emergencies are addressed with on-site resources. In addition, the airport's FBOs, Duncan Aviation and Ross Aviation, provide services to process the diversions not handled by United Airlines.

The airport is approximately 9 miles away from the nearest hospital. Annual planning and triennial exercises involve all local hospitals and medical transportation providers. All local hospitals are participants in the local emergency operations plan, as is the airport. The airport's IROPS plan identifies Tailwind Concessions as a critical player. Tailwind provides all food service concessions in the terminal and is on-call to respond as needed 24/7 in the event of a diversion where passengers need to be disembarked. TSA communicates well with the airport to ensure TSA has proper staffing available when needed and is forthcoming with information if issues arise. The Red Cross is also identified as a network partner in the airport's AEP, but not in its IROPS plan.

For international diversions, LNK contacts the CBP via cell phone to the lead agent in Omaha to ensure the aircraft is pre-cleared. Customs has advised that they will not, as a rule, respond to LNK in the event of an international diversion.



Diversion Experience: Most of the diverted flights that LNK receives are considered gas and go operations and, currently, most are operated by United. Even though United currently has very bare-bones staffing, many of United's staff, along with airport operations personnel and airport police, are experienced employees and know how to manage diversions. With the resources and capabilities of the ANG on site, the airport can provide an initial response to emergencies associated with diversions because of their long-term employees and relationships with key mutual aid and network partners.

Most of the diversions that LNK receives are gas and go technical stops that are easily managed but can become an incident-level response effort when multiple diverted "technical stop" flights arrive at the same time. This happened in 2021 when LNK received eight diversions at one time but only had three parking spots available. It was able to manage through a lot of collaboration and coordination.

An emergency diversion situation occurred at LNK prior to the airport's formal logging of diversion information. Therefore, this incident description is based on the memory of the director of operations and occurred before 2015. In this situation, the captain of a Boeing 757 suffered a heart attack and the first officer had to land the plane at LNK. ATCT relayed the emergency declaration from the aircraft to 911 and ARFF. All responding equipment, including off-site mutual aid resources, were pre-positioned at the time of the landing. Rescue equipment arrived at the aircraft in less than 2 minutes and the captain was transferred to a city ambulance and transported to the nearest hospital. One interesting note is that the first officer, though he had landed the aircraft safely, was not authorized by the airline to taxi the plane as a solo operation. The aircraft, with passengers on board, was towed to a gate at the terminal. Passengers were deplaned into the terminal, where food and drink were provided by the airport. A replacement crew arrived on

Copyright National Academy of Sciences. All rights reserved.

the next scheduled United Airlines flight and the B757 departed with passengers approximately 4 hours after landing.

Rick Husband Amarillo International Airport, Amarillo, TX (NH)

- **Level and type:** Various incident- to emergency-level diversions since 2012
- **Diversion dates:**
  - August 2, 2019–Emergency stop because of a mechanical issue
  - August 11, 2016–Weather causes multiple flight diversions
  - August 31, 2015–Mid-air altercation and arrest
  - March 27, 2012–Medical emergency
- **Causal factors:** Medical and mechanical emergencies and unruly passenger
- **Compounding conditions:** Multiple flight diversions, extended delay
- **Size of aircraft involved:** Large commercial aircraft
- **Interview sources:** Garnett Schafer, Operations Duty Manager, and AMA's website, media reports



Airport Information: Rick Husband Amarillo International Airport (AMA) is a public airport located just 6 miles east of downtown Amarillo, Texas. The airport is an NPIAS-designated non-hub airport with current enplanements of 355,705. The airport is operated and managed by the City of Amarillo's Department of Aviation, which is comprised of a Director of Aviation, Deputy Director of Aviation, Assistant Director of Aviation, Airport Program Coordinator, Administrative Assistant, Airport Facilities Manager, Police Commander, Airport Operations Manager, and three Operations Shift Coordinators.

AMA's Department of Aviation is an enterprise fund of the City of Amarillo requiring no funding from property taxes, general funds of local governments, or special district taxes. Capital needs are generally met by operating surpluses, passenger facility charges, and FAA Airport Improvement Program grants.

The airport has an Aircraft Rescue and Firefighting Index "B," but Index "C" equipment is available upon request. AMA maintains two 1,500-gallon ARFF trucks. It has tiedown and hangar parking space. Its primary runway was built for the USAF Strategic Air Command base and at 13,502 feet is among the longest commercial runways in the U.S. AMA's runway is still used for military training. It also is home to two significant aviation businesses, Bell (formerly Bell Helicopter) and International Aerospace Coatings, one of the largest aircraft painting companies in the world.



Diversion Planning: AMA uses an IROPS plan for incident-level planning and an AEP for emergencies. The IROPS plan includes roles and responsibilities for both domestic and international diverted flights. It also includes staffing and passenger provisions for both regular-hour and after-hour diversion support including extra staffing and a designated sterile area in the Administrative Wing. AMA has also identified shared equipment support for remotely parked aircraft (e.g., mobile stairs, bus transport, special needs lift, etc.). The airport has also made plans for after-hour and extended delay situations that include support from various mutual aid resources as detailed in the mutual aid and network partner section.

According to their IROPS plan, AMA has an IROPS Committee that meets during quarterly Airline Tenant Association meetings. The meetings serve as a reminder of the committee's mission: to update stakeholder contact information, to review the responsibilities of stakeholders involved in irregular operations response, to track inventory and update changes, and to disseminate information or train members on changes or updates to specific stakeholder plans or the AMA IROPS plan.



Diversion Tracking, Notification, and Communication: The airport has an Airport Operations Center that is staffed around the clock. The AMA Operations Center and the airport's FBO use flight-tracking resources including Flight Aware to maintain situational awareness. Once an

Copyright National Academy of Sciences. All rights reserved.

airline or the ATC notifies AMA Airport Operations of a diverted flight, the AMA Operations Center launches the Veoci Virtual EOC system to document the diversion. AMA Operations communicates flight information, airport field conditions, and any other relevant information that may have an impact on the overall operation through media releases and by issuing Notices to Air Missions as appropriate. In emergency situations that are beyond the capability of AMA, the airport coordinates with the Amarillo Area Office of Emergency Management.

AMA has not traditionally used social media as part of irregular operations or emergency operations and relies on the City of Amarillo Office of Public Communications and Community Relations, which works with the AOC to monitor and post traveler advisories and information with the news media (and on social media sites) during diversion events.



Mutual Aid and Network Partners: ARFF and Fire, airport operations, and police are all considered part of the airport's capabilities and are under the AMA Department of Aviation's authority. Beyond this, AMA relies on airline ground handlers as the primary source of mutual aid equipment, with ARFF and Fire to assist with it. Its FBO is open 24/7 and provides aircraft servicing, fueling, and deicing procedures. AMA has procedures in place with its FBO to call in additional staff after normal business hours during times when AMA experiences diverted flights. The FBO also can help with aircraft parking on their ramp during diversion events.

Fire Station 10, ARFF, is located on airport property and provides fire and rescue and medical services. Law enforcement is provided by the Amarillo Airport Police Department and Amarillo Police Department. The airport is 15 miles away from the nearest health center, Northwest Texas Healthcare System. Ambulance service is provided by AMR and dispatched through ARFF and/or AOC.

Through its IROPS plan and AEP, AMA secures additional assistance from its retail tenants, primarily equipment. It also has outlined after-hour support and extended delay situations (e.g., additional staff) from the TSA, CBP, ground transportation, FBO, and concessions. It also maintains a list of "big box" stores and hotels where they can also secure additional provisions as needed.

AMA also considers the airports that are part of the North Texas Regional Network as part of their mutual aid network. This group meets regularly to discuss issues related to diversions, delays, capacity, and capability in the region.



Diversion-Related Information and Experience: AMA typically receives diverted flights from major hub airports in Denver, Dallas-Fort Worth, Dallas, Houston Intercontinental, and Houston Hobby. While many of these diverted flights are commercial aircraft from one of the three providers at AMA (American Airlines, Southwest Airlines, and United Airlines), sometimes flights are associated with one of their code-sharing partners or a non-station airline, which can complicate communication efforts if contact information is not readily known or accessible. In addition, large-scale weather events can affect several airports in the region, resulting in multiple simultaneous diverted flights to AMA. As a small airport, AMA has limited capacity and resources, with the capability to handle three diverted flights at one time to limit disruptions to scheduled service. In an extreme circumstance, AMA could support a diverted flight at each of the seven available gates. Gates at AMA are under common-use gate leases, permits, or agreements to air carriers and are controlled by the airport.

AMA's Operations Center uses the Veoci Virtual EOC system to maintain a record of each diversion event. Capacity constraints of the airport, including those associated with aircraft (gate and equipment availability) and passengers (deplaning, concessions capacity) are communicated through Passur. Aircraft that are known diversions to AMA are tracked and monitored by AMA Operations in a Diversion Log on the Veoci system. AMA coordinates with the FBO and ARFF to provide transportation, fuel, standby fire department trucks (if needed), and airstairs, among other equipment and supplies as requested.

Copyright National Academy of Sciences. All rights reserved.

If a diverted flight can be handled by the airline, ATC directs aircraft to the requested gate. If the airline does not operate at AMA or if staff is not available to support a diverted flight, ATC notifies AMA Operations to request support. The FBO coordinates with the airlines to position all equipment in a way that maintains fuel trucks' access to the ramp.

On August 2, 2019, an American Airlines flight from Dallas to Seattle diverted to AMA because of an emergency-related mechanical issue. The ATC notified AMA Operations of the diversion and AMA used its Veoci system to notify ARFF and Fire. Additional crews were brought in from the City of Amarillo's Fire House #7 as an extra precaution because of the number of passengers on the commercial flight.

The flight landed safely around 9:00 a.m. and all 182 passengers were able to deplane using the airport's airstair truck, which was arranged through FBO, airline, and AMA operations communications. Passengers were taken to a secure area designated in AMA's IROPS plan. The airline and FBO worked diligently on mechanical issues, however, because of the situation's severity, American Airlines brought in a different aircraft for the continuation to Seattle. Passengers spent about 6 hours in the secure area, where they were provided with food and beverages courtesy of American Airlines. The new flight departed at 3:30 p.m.

On August 31, 2015, a Southwest Airlines flight from San Diego to Chicago diverted to AMA because of a mid-air altercation. The airline pilot discussed the situation with ATC and notified AMA that the flight needed to land because of approximately 10 people fighting on the plane. The ATC requested police officers to meet the flight when it landed around 10:30 p.m. Because of the timing of the event, which occurred after typical business hours, AMA Operations notified the Amarillo Police Department as well as the FBI to provide law enforcement assistance.

Once the flight landed, AMA operations directed the pilot to an available gate, where law enforcement escorted the numerous passengers involved in the altercation to a secure area for questioning. Six passengers ended up being detained and taken to the Randall County Jail. All other passengers were deplaned, and some were questioned by law enforcement. The airline provided food and water for the passengers as they waited for the aircraft to re-board. Six passengers besides the ones taken into custody decided to remain in Amarillo. The rest of the passengers reboarded and departed around 1:00 a.m.

On Tuesday, March 27, 2012, a JetBlue flight from New York to Las Vegas was diverted to AMA because of a medical emergency. The captain of the flight began acting erratically and left the cockpit where he was subdued by passengers. The co-pilot locked the cockpit door and contacted the ATC to request an emergency diversion to AMA. Another captain, traveling off duty, entered the flight deck before landing at Amarillo and took over the duties of the ill captain once on the ground.

AMA operations were notified of the diversion and contacted ARFF and Fire, Amarillo Paramedics, AMA's Airport Police Department, Amarillo Police Department, the FBI, and TAC Air for support. Once the aircraft landed at Amarillo around 10 a.m., it was met by local law enforcement and the FBI, who were able to secure the captain without incident. He was transported by local ambulance for medical evaluation at Northwest Texas Hospital, which paramedics informed ahead of time.

The aircraft was towed to the terminal building and the passengers were safely deplaned from the aircraft. Passengers were provided food and beverages by the airline and reboarded the flight within a few hours where they continued to Las Vegas.

General Aviation Airports

Iqaluit International Airport, Iqaluit, Nunavut, Canada (GA)

- **Level and type:** Mechanical emergency
- **Diversion date:** February 1, 2017

Copyright National Academy of Sciences. All rights reserved.

- **Causal factors:** Engine failure
- **Compounding conditions:** Extreme cold, remote location
- **Size of aircraft involved:** Commercial aircraft
- **Interview sources:** John Hawkins, Assistant Deputy Minister for Economic Development and Transportation with the Government of Nunavut, media accounts, and Winnipeg Airport Services Corporation website



Airport Information: Iqaluit International Airport (YFB) is located about 1.24 miles from the center of the City of Iqaluit on the southern tip of Baffin Island in Canada, which is an extremely remote community. The airport is owned by the Government of Nunavut and operated, under a 30-year contract, by Nunavut Airport Services. The company is a subsidiary of Winnipeg Airport Services Corporation (WASCO), which in turn is a subsidiary of Winnipeg Airports Authority. WASCO, in collaboration with Arctic Infrastructure Partners, provides airport management, airport operations, facility maintenance, and technical services for the Iqaluit International Airport.

YFB hosts scheduled passenger service from Ottawa, Montreal, Rankin Inlet, and Kuujjuaq on carriers such as Canadian North, and from smaller communities throughout eastern Nunavut. It is also used as a forward operating base by the RCAF as it has an 8,605-foot-long asphalt-coated runway. The airport provides Category 5 ARFF during published hours and can provide Category 7 ARFF with prior notice.

The airport is classified as an airport of entry by NAV CANADA and is staffed by the CBSA. CBSA officers at this airport can handle general aviation aircraft only, with no more than 15 passengers. YFB serves as a diversion airport for polar routes. The airport was expanded and refurbished as a part of the airport's $298.5 million improvement project in 2017 to include a new air terminal building and a new Combined Services Building for ARFF firefighting trucks and heavy equipment. The new air terminal building provides additional international holdroom capacity up to 105 passengers, expandable to 300 passengers for short periods of time if needed.



Diversion Planning: YFB created a Wide Body diversion plan in 2017 that is updated annually. The plan was developed because the airport handles many diversions with widebody aircraft such as the B777, B747, A340, and A380. The plan is intended to provide guidance when an aircraft larger than the critical aircraft for YFB arrives at the airport. This document outlines ground handling (contact information and equipment on site), aircraft movement (ground marshaling, designated taxi routes, and remote stand and apron parking locations), and passenger considerations that may arise in the event that such an aircraft arrives at the airport, including numerous temporary passenger holding areas both on-airport (e.g., the maintenance garage, hangars, and the terminal) and off-airport (local schools, lodges, and community centers), ground transportation options, and terminal and off-airport backup catering services.



Diversion Tracking, Notification, and Communication: Formal notification of an incoming diversion comes from NAV CANADA air traffic control (the FAA counterpart). The Wide Body diversion plan provides all the phone numbers needed to manage diversions including operations, ARFF on duty, station airlines, NAV CANADA, Canadian Border Services Agency (cell, office, and pager), CATSA, FBO, fuelers, city fire department, security, community services (hotels and buses), city offices, contractors, heavy equipment managers, and electricians. Incidents are documented through the airport's maintenance and facility tracking system (Maximo) and through the airport's ERP.



Mutual Aid and Network Partners: Iqaluit International Airport Fire Department operates one crash tender for local fire suppression and has support from Iqaluit Fire Department located 1.12 miles from the airport when needed. Category 7 ARFF, including 2 crash tenders, is available upon prior notification.



Diversion-Related Information and Experience: Over the years, several international flights have been diverted to Iqaluit because of technical reasons, unruly passengers, and on-board

Copyright National Academy of Sciences. All rights reserved.

medical emergencies. Most recently, a British Airways 787-900 diverted because of mechanical issues, and a United Airlines 787-900 diverted because of medical issues.

On February 1, 2017, a Swiss International Air flight from Zurich to Los Angeles diverted to Iqaluit International Airport when one of its 777-300 engines shut down as it was technically designed to automatically do when a fault is indicated. The flight crew diverted the aircraft with 217 passengers on board to Iqaluit for a safe landing about an hour and a half after the engine shut down.

The airport was notified of the diversion and called in ARFF resources as well as medical personnel as a precaution. Once the plane landed, ground handlers initially could not turn around the aircraft on the runway. After about 45 minutes a tow tug was able to push the aircraft back onto the apron.

Passengers were kept on board and the airline provided dinner, as CBSA in Iqaluit can only process international flights with up to 15 passengers. At the time of the incident, it would have taken hours to deplane and process 217 people through immigration.

Swiss Air arranged for another reliever aircraft from New York to pick up passengers. Airport operations had to be resourceful and called in all school buses from the City of Iqaluit to transfer the passengers from the diverted flight to the relief aircraft. The airport only had one airstair capable of handling both the original diverted flight (Boeing 777) and the reliever flight (A330). The ground crews had to position the stairs on the diverted flight, deplane all passengers onto the buses, then reposition the stairs to the reliever aircraft. Once all passengers boarded the reliever flight, they proceeded to New York, where they were rebooked onto direct flights to their ultimate destinations.

Flying Cloud Airport, Eden Prairie, MN (GA/R)

- **Level and type:** Incident and diversion
- **Diversion date:** July 15, 2021
- **Causal factors:** Aircraft mechanical emergency
- **Compounding conditions:** During an annual tabletop exercise, construction on the airfield
- **Size of aircraft involved:** Small GA aircraft—PA46
- **Interview sources:** Blaine Peterson, Manager of Flying Cloud Airport, Metropolitan Airport Commission (MAC) Reliever Airport Roundtable Discussion, airport website



Airport Information: Flying Cloud Airport (FCM) is considered a general aviation and reliever airport located approximately 10 miles southwest of Minneapolis in Eden Prairie, Minnesota. It is the busiest reliever airport in the MAC, which is the organization that owns and operates Minneapolis St. Paul International Airport (MSP) and six reliever airports in the Minneapolis area. FCM has over 100,000 takeoffs and landings per year. The airport also features three flight schools.

The airport is overseen by a 12-member board of directors and a 10-member senior management team, which includes a Manager of Emergency Response and Planning. Approximately four additional personnel work in airfield operations positions. FCM operates 24 hours per day and has a mutual agreement with the City of Eden Prairie for emergency services including police, fire, and medical response. The airport also has three FBOs and one deicing operator on site. However, the airport does not provide any ticketing or ground services and needs to rely on carriers for those services.



Diversion Planning: The airport is also part of a regional planning group called the MSP Regional IROPS Diversion Network and this group discusses and manages diversion response among other things. FCM and the other MAC reliever airports all recently updated their

Copyright National Academy of Sciences. All rights reserved.

emergency plans. FCM also participates in emergency exercises that are typically tabletop exercises. As a general aviation and reliever (GA/R) airport, they are not mandated to conduct annual emergency training exercises, but the airport considers them to be a best practice. FCM does not have a family assistance program. Typical aircraft at this airport hold approximately one to 19 people (both passengers and crew members).



Diversion Tracking, Notification, and Communication: FCM, like all the MAC reliever airports, does not have an operations center; however, the airport receives notifications or information from the MSP Operations Center. MSP uses Flight Aware technology to track potential diverted aircraft. Formal notification of an incoming diversion comes from FAA air traffic control as soon as they get notified, typically 10 miles prior to arrival. FCM is typically notified via phone or email or a web-based system with flight information and intentions. FCM uses radios to communicate with the airport and emergency response personnel as well as with its FBOs and other network partners.

FCM has a social media presence, but they do not use it for emergency situations, but rather for marketing purposes as they are limited in how much effort they can put into this as they have a small staff.



Mutual Aid and Network Partners: FCM has agreements with firefighting, law enforcement, and medical assistance. The airport is located 2 miles from the closest fire department that provides fire, ambulance, and medical services. The airport is 5 miles away from the nearest health center. Events are fairly quick from start to finish.

If needed, a contract could be drafted or a credit card could be used for purchases of food and water and catered food.



Diversion Information: FCM receives diverted aircraft because of maintenance or weather concerns. The airport offers parking for small commuter aircraft on the airport terminal apron. FCM generally receives gas and go, mechanical, and medical emergencies. For most diversions that occur at FCM, the pilot contacts the local FBO for services and support. When the airport receives an emergency diversion, the Eden Prairie Police Department dispatch center communicates with all partners and mutual aid alongside the MSP operations center. When this happens, staff need to perform two different roles at the same time: emergency coordination and operations.

On July 15, 2021, FCM was in the process of conducting a fire department tabletop training exercise when a single-engine aircraft (Piper PA46) diverted to the airport. An emergency was declared by the aircraft, which had lost engine power. The aircraft had departed Alexandria, Minnesota, with the intent of flying to an out-of-state airport for maintenance. The aircraft developed engine power loss sometime after take-off but had enough altitude to reach FCM. The pilot was the only occupant onboard.

The airport contacted 911 and the dispatch call arrived on all fire department personnel's pagers. Police, fire, and ambulance were called and staged for aircraft arrival. Fire crews reported to their fire trucks and staged themselves at the base of the tower, per plan doctrine.

The MAC airport manager closed the airport down for the emergency response when the aircraft reported 5 miles out. Prior to this, the airport manager did open one runway to allow instrument flight rules and visual flight rules training traffic to land and vacate the airspace.

The pilot landed the diverted aircraft with limited power without incident and had enough power and speed remaining to taxi off of the runway onto a little-used taxiway. The local FBO was called to hook a tow bar up and tow the aircraft to the FBO ramp for maintenance.

The Airport Manager led emergency personnel from the staging area to another staging area near the approach end of the runway to be ready to take any corrective actions needed for

Copyright National Academy of Sciences. All rights reserved.

78 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

the aircraft emergency. Emergency crews led response efforts while MAC Airport staff trailed watching for foreign object debris and conducted runway safety checks to expedite the opening of the runway and movement surfaces. Conducting recovery efforts during the emergency response effort improved the airport's resiliency and enabled the airport to resume normal operations faster.

A short after-action session was held and a comment was made that in future events of non-powered emergency landings, to stage well away from the approach end in case of overshoots. Currently, the MAC does not pay for mutual aid at Flying Cloud Airport.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 7

Benefits of Collaborative Planning



The many benefits of collaborative planning have been studied over the past decade. Several concepts, such as airport collaborative decision-making, are being advocated through industry guidance as well. The support of collaborative concepts like these demonstrates the increased awareness and value of working together. As related to diverted flight response, when airports, mutual aid, and network partners work together, it results in the following:

- **Better overall mitigation** of routine, incident-related, and emergency-related response efforts. This can be demonstrated through the following:
  - Improved safety and security, as mutual aid partner agreements enable airports to access the most capable and experienced life safety personnel and specialized equipment during a diversion event, reducing risk for an airport.
  - Improved passenger experience, as airports and airlines can clarify roles and responsibilities related to passenger and family assistance and work together to fill in any gaps in service before they become problematic.
  - Enhanced collective reputation, as passengers often view their experience, whether at an airport or on an airplane, as one aviation experience. Therefore, when their experience with one mutual aid or network partner is positive, the rest of the industry partners get credit for that experience.
  - Improved regional connections and trust.
- **Improved relationships** among airport, mutual aid, and network partners, as follows:
  - When airports and other outside organizations and agencies work together, trust is built and respect is developed, potentially leading to new ways to cooperate.
  - A broader regional approach may expand to form a national network, as follows:
    - As mutual aid and network partners work together and discover ways to improve performance, this typically results in involving an expanded list of partners in the planning process.
    - Collaborative regions are forming around larger hub airports so that the various partners can collectively plan for and pool more resources in a region as well as incorporate more airlines operating at a variety of airports in a region.
- **Economic sustainability** via partnerships to mitigate diversions, which can support the local and regional economy (e.g., local stores and restaurants) and airport business partners (e.g., ground transportation).
- **Cost reduction**, as airports and their partners can avoid duplicate equipment, supplies, or even personnel. Training costs can also be spread among partners.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 8

Diverted Flight Challenges

There are many challenges associated with diverted flights. Some of these include uncertainties related to various future industry trends, the impact of small airports' staffing sizes, challenges with diversion planning and training, diversion costs, diversion coordination, and compounding issues associated with diverted flights.

Uncertainty Related to Increased Diversions Versus Smarter Planning

Accounting for the decrease in flight diversions in 2020 and 2021 because of the COVID-19 pandemic's impact on overall flight operations, using flight diversion data from 2011 through 2021, the number of overall flight diversions is still trending upward as demonstrated in Figure 10.

Additional data from the FAA's Core 30 airports also supports this trend, as demonstrated in Figure 11. The FAA tracks Core 30 airport data, as there are 30 commercial U.S. airports that operate around major metropolitan areas, serve as hubs for airline operations, and move 70% of all passengers in the United States From 2018 to 2019, ASPM data shows the number of diverted flights either to or from these Core 30 airports rose by 7% (FAA 2021b). This trend is expected to continue, as noted by the FAA in Figure 11.

Research also supports this trend. One industry article notes that aircraft diversions are gradually becoming more significant as the growing volume of air traffic reaches the limits of the current infrastructure (Spak, M., and P. Olexa 2020). In addition, in-flight medical emergencies average one out of every 604 flights and are expected to increase as the population ages and air travel increases (Lewis, B. A., et al. 2021). In addition, global weather trends may continue to exacerbate weather conditions as well, which, as the top causal factor for airport diversions, may contribute to their rise as well. Things like political unrest or future technology upgrades, such as 5G, may create diversion situations as well.

It is not known whether current levels of capital improvement at airports will be enough to increase overall airport capacity to take on the volume of air traffic. Nor is it known whether DOT airport fines related to the 3-hour and 4-hour rules are causing airlines to take a conservative approach and cancel more flights when bad weather can be predicted, potentially lowering the number of diverted flights.

However, there are indicators that smaller airports may experience a decline in diversions. For instance, emerging robust flight-tracking technologies are just beginning to enable air traffic controllers and pilots to see real-time capacities and capabilities of airports, enabling them to choose larger airports with more capacity or capabilities for diversions. Also, more regional airport planning groups have formed around major hubs, enabling airports, airlines, and government agencies

Copyright National Academy of Sciences. All rights reserved.

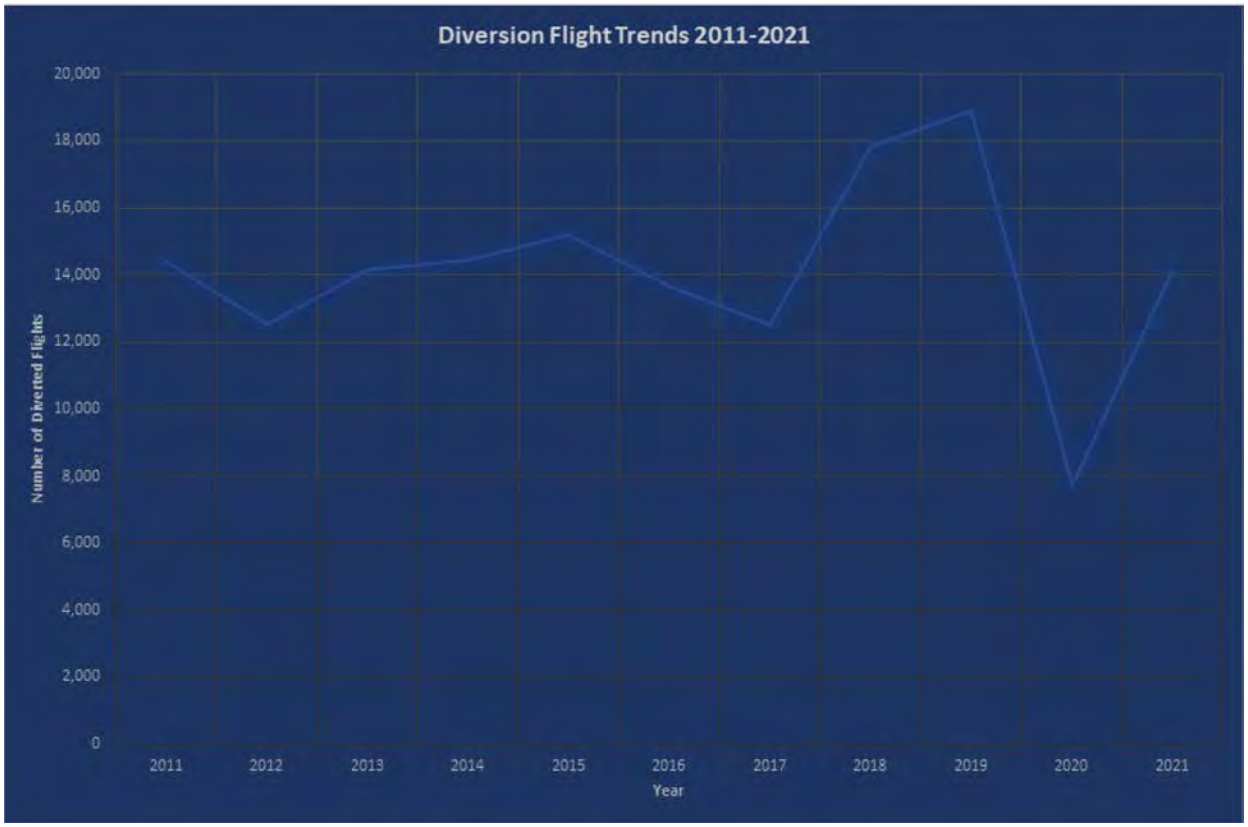


Figure 10. Diversion flight trends 2011–2021 (Source: Bureau of Transportation Statistics 2022).

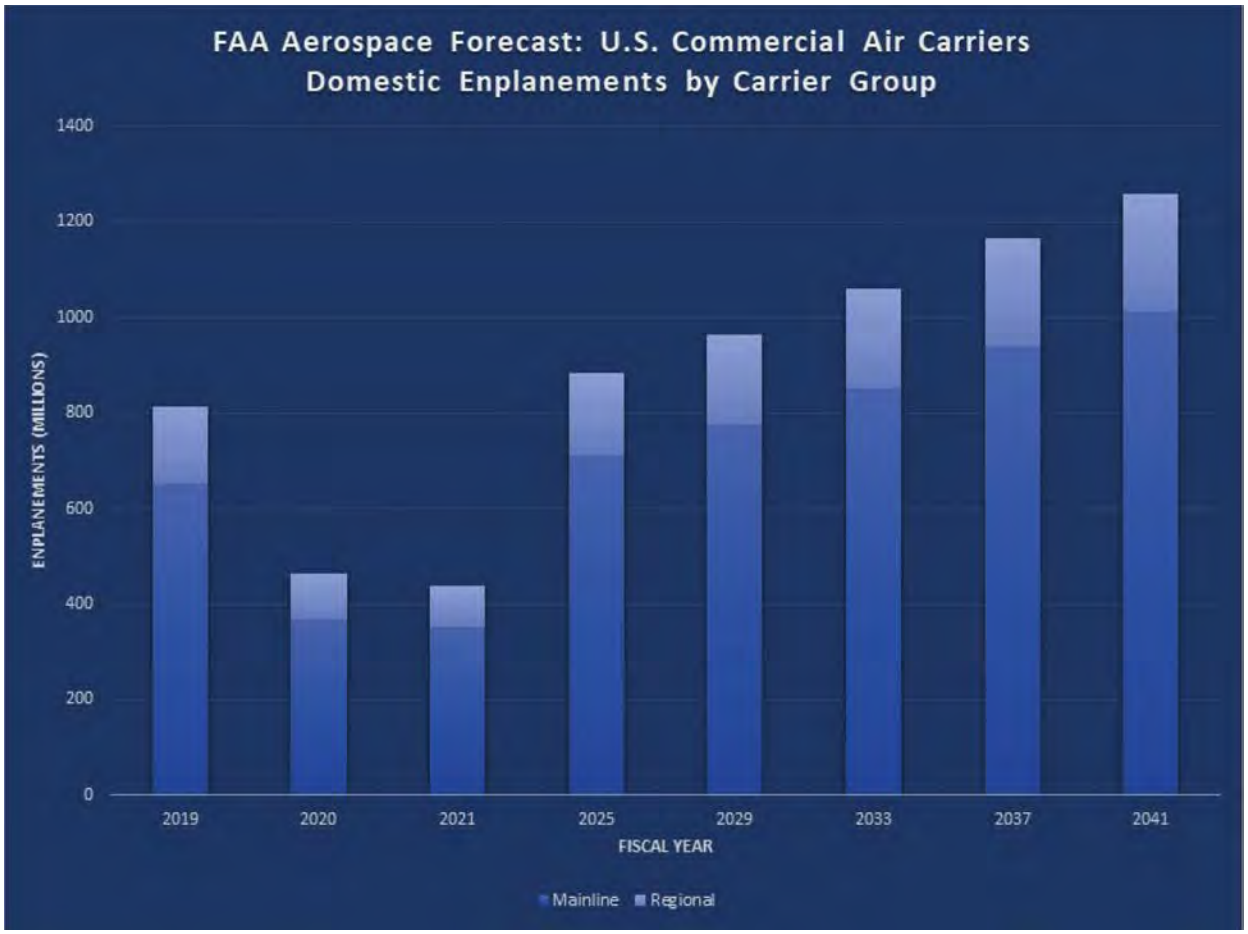


Figure 11. Anticipated rise in enplanements from 2025–2041 (Source: FAA 2021b).

Copyright National Academy of Sciences. All rights reserved.

to work together regularly to ensure that diverted flights are only routed to airports with the capacity and capability to manage them. In addition, the ramifications of the global pandemic and its associated workforce reduction combined with worldwide political uncertainties causing supply chain disruptions and steeply rising fuel costs have caused airlines to operate as efficiently as possible as they have minimal economical margins for error. This has translated into airlines reducing the number of overall flight routes and increasing pre-flight planning efforts to save fuel by avoiding holding patterns and only diverting to airports with available or certain capacity to handle them whenever possible. One airline's objective is to avoid diversions to smaller airports with limited resources and remote airports away from major hubs. All these factors point to potentially fewer diversions at the smallest of airports.



Small Staffing Size of Smaller Airports

When diversions occur at smaller airports, airport personnel often wear many hats at once—as a first responder, a media spokesperson, and sometimes as a caretaker for loved ones and families—while ensuring continued airport operations occur to the greatest extent possible. How the airport responds to these types of events, in all aspects of a response, will be judged critically and immediately by all observers, especially in this age of social media.

Playing many of these roles at one time can be particularly hard for smaller airports. Smaller airports do not have the advantage of larger airports with multiple aviation departments to call upon in major diversion event situations. As shown in Table 24, to illustrate this disparity in resources, the average number of full- and part-time employees at small hub, non-hub, and general aviation airports was compared to medium and large hubs.

The understanding of the disparity of resources is noteworthy when considering the lack of bench strength that smaller airports have available to draw upon when there is an incident or emergency associated with a flight diversion, especially a large commercial or international flight with hundreds of passengers, compared to larger airports.

To alleviate this problem, many smaller airports train a large percentage of their staff, typically maintenance personnel, to perform ARFF duties in addition to their regular ones. However, airports are now operating with minimal staff, so with fewer staff members, the ability to manage diversions is currently taxing.



Planning- and Training- and Exercise-Related Challenges

While many SH and NH have and use established plans to manage diversion events, the smallest airports, primarily GA airports, do not have plans in place to manage diversion events. They do not train or conduct exercises for diversions with mutual aid or network partners, either. Therefore, they figure out how to manage diversions as they happen. This approach can create challenges, particularly related to roles and responsibilities. It can challenge mutual aid and network partners, especially first responders as they may arrive on site not knowing the

Table 24. Numbers of employees at U.S. airports of varying size.

Airport Name	NPIAS Size Designation	Number of Airport Employees
Atlanta Hartsfield-Jackson International Airport	Large Hub	63,000
San Antonio International Airport	Medium Hub	5,000
Long Beach Airport	Small Hub	100
Fort Wayne International Airport	Non-hub	99
Lancaster Airport	General Aviation	23

Source: Baskas, H. 2016.

Copyright National Academy of Sciences. All rights reserved.

airport layout and may waste valuable time trying to figure out where to go and establish a staging location, for example.

For the smaller airports that do create diversion-related plans, it can be a challenge to keep information current, particularly regarding things like contact information and maps. It can also be challenging to find the time to update mutual aid and network partners on these changes as well. This can be compounded by high levels of turnover and attrition at the airports and in mutual aid and network partner organizations. Keeping up with and synchronizing multiple plans at smaller airports with limited staff is its own challenge. It can also be challenging to determine what to put in stand-alone diversion plans or checklists.

Diversion Costs



Diverted flights are costly to airlines, but not necessarily to airports. Airlines shoulder these expenses as the cost of doing business but have started passing the costs of diversions on to passengers if they are the cause of the diversion (e.g., unruly behavior), but this does not pertain to medical situations. On average, a diverted flight can cost airlines approximately $25,000 for a domestic flight on a narrow-body aircraft (single aisle, typically 120–150 seats) to over $100,000 for an international flight on a widebody aircraft (double aisle, typically over 180 seats) (Ayra, et al. 2014).

For airports, diversions can bring in revenue from landing fees from each aircraft that lands on an airfield. However, airports can incur expenses outside of typical operating costs if additional workers need to be called in, especially if they are called in after hours and receive overtime pay. There may be costs involved with maintaining equipment used or replenishing supplies, such as food and water, used in family assistance.

Coordination Issues



Smaller airports outside of regional networks tend to have limited coordination with airlines that very rarely use them for diverted flights. To that end, these same airlines tend to give smaller airports minimal notice, if any, of diversion aircraft headed to their respective airports. In these cases, these airports may not have adequate staffing available at the time of the diversions to accommodate the aircraft or its passengers.

Compounding Issues



Several factors can escalate diversion events, including the following:

- **After hours:** When diversions occur beyond an airport's regular hours of operation, the lack of coordination among airlines and airports is revealed. This has resulted in the issuance of DOT aviation enforcement orders that demonstrate how air carrier contingency plans have not been fully coordinated with all scheduled and diversion airports or with other agencies, CBP, and organizations such as FBOs.
- **Extended delay:** Long-term weather events can cause a diverted aircraft to remain at a diversion airport longer than expected. An unexpected number of stranded passengers at a smaller airport can exceed occupancy permit requirements and cause issues that can strain security staff.
- **Additional issues (power outage, flooding, etc.):** Long-term or severe issues as well as system overloads and network outages can all exacerbate a diversion. These can cause an incident to become an emergency or an emergency to become more severe.
- **Pandemic precautions:** As COVID-19 demonstrated at airports, additional health precautions like social distancing can make emergency response efforts more challenging.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 9

Conclusion and Summary of Findings



Conclusion and Summary

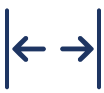
Interviews with smaller airports, regional groups, airline pilots, the FAA, and technology companies combined with recent literature and survey findings reveal that smaller airports receive and manage numerous gas and go diversions regularly using on-airport resources including FBOs and ground handlers. They may also receive several technical stop diversions at once, creating incident-level situations that they also regularly manage without exceeding their capacity and capabilities—some using formalized plans and/or checklists, and others not.

However, research indicates that smaller airports do not receive many emergency-level diversions annually. Smaller airports can typically expect to encounter one or two of these types of events on an annual basis. And, whether they can manage emergency diversions without mutual aid or network partner assistance primarily depends on how the authority over first responders is structured. And the success of managing these diversion events overall depends mainly on collaborative relationships—both internally at the airport and externally with mutual aid and network partners.

Questions asked in the survey and interviews focused on incident- and emergency-related diversion experience specifically from the past couple of years to ensure that staff involved with the event would still be working at the airports and would be familiar with diversion details. However, this short timeframe may have prohibited some airports surveyed from providing some additional relevant experience that may have been beneficial to this study. Additional airport interviews did uncover more diversion experiences from the past 3 to 6 years.

Also, focusing some survey questions on diversion events that "exceeded airport capacity" may have limited survey responses because of the various ways airports are governed, and the many ways they operate, often contracting out a variety of services to third parties. The timeframe requested for diversion examples was also impacted by pandemic conditions which experienced an overall reduction in enplanements and diversions, which may have skewed survey results as well.

A positive outcome of the survey is that some interviewed said that the survey questions gave them the idea to create a diversion checklist at their airports. Other survey respondents asked for copies of the questions so that they could evaluate some diversion response areas on the various criteria presented in the questions.



Gaps in Knowledge

The following are items that were unknown so no conclusions could be drawn.

- No peer-reviewed or airport industry research exists on how effective airports, especially smaller ones, are at communicating with internal staff as well as with on-airport partners and mutual aid and network partners during aircraft diversions.

Copyright National Academy of Sciences. All rights reserved.

- It was not clear whether many smaller airports have implemented newer communication technology, such as web-based mass notification and management software.
- Very little research is available specifically related to flight diversion events that assess how small airports can recover quickly and successfully from situations like these.
- The true cost of incident and emergency flight diversions for smaller airports is not understood.
- It was not clear why smaller airports are not sharing and coordinating diversion-related plans with mutual aid and network partners.
- Diverted cargo flights were not considered for this synthesis. It is unknown if cargo operators divert to smaller airports.

Further Areas of Study



The following are identified as potential future areas of study.

- **Impacts of Extended Delays**
  - The effects of flight diversions on smaller airports during prolonged weather events. Long-term weather events can wreak havoc on FBO and ground handler machinery and equipment needed for aircraft maintenance and deplaning, which can escalate a diversion situation.
- **How to Remain Ready and Resilient**
  - Ways for small airports, mutual aid, and network partners to remain prepared during long periods of "steady state" with no incident-level or emergency-level diversion events. Most smaller airports reported that they only receive diversions that cause them to call on assistance from partners about once per year.
  - How smaller airports maintain continuity of operations and resiliency during multiple or emergency-related diversion events. A look at how Business Continuity Plans can be used to maintain airport resiliency during diversion events might refine good practices needed by smaller staff sizes to be resilient.
  - Means for managing emergency diversions and related events with the number of senior airport leaders retiring and the upskilling of new hires needed on a small airport budget. What are the ways smaller airports can future-proof their organizations to build resiliency into their operations without seasoned leadership present?
- **Keeping Information Current**
  - Methods for collecting, maintaining, and updating contact lists and making them consistent across all airport plans and standard operating procedures. Many smaller airports do not have the staffing needed to regularly maintain a variety of plans and ensure all contacts are current. This is particularly a problem with the high turnover experienced in the aviation field.
  - Means for providing airport capability updates to various web-based sources for use in diversion-related decisions. There is no singular nationwide database of information that contains a clearinghouse of airport capabilities across the country.
- **Leveraging Existing Plans, Training, and Leadership**
  - Raising awareness of how DOT emergency contingency plans can be leveraged to include more diversion-related planning elements. These plans typically include basic provisions for deplaning and designating sterile areas, but they could be expanded upon and kept current since it is a requirement that they are regularly updated and submitted to the DOT and posted online.
  - Development of templates for emergency-level diversion exercises for small airports. A tabletop exercise is provided in Appendix C of this document, but more focused exercises can be developed.

Copyright National Academy of Sciences. All rights reserved.

86 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

  - Determine the need for and responsibilities of a diversion champion at smaller airports. This type of position may be an appropriate member of an IROPS Committee, an IROPS regional group, or an Emergency Working Group.
- **New Ways to Collaborate**
  - Focus on the benefits of regional groups (e.g., regional IROPS networks) and their successes with sharing resources and managing passenger care during diversion events. There are a few regions that discuss their challenges and successes regularly, and they may have good practice insights to share.
- **Using Social Media**
  - How smaller airports can utilize social media in emergency management situations. Build from ACRP guidance that advocates how smaller airports can use cost-effective industry solutions with even one part-time employee to manage social media during emergencies.
- **Financial Impacts of Diversions Airports**
  - Determine the return on investment from diverted flights. According to many smaller airports, they enjoy receiving diversions because of the financial benefits they bring from landing fees and other services.

Copyright National Academy of Sciences. All rights reserved.

Glossary

Agency: A division of government with a specific function offering a particular kind of assistance.

Aircraft rescue and firefighting (ARFF): Specialized firefighters, rescuers, procedures, and equipment to deal with aircraft accidents at an airport.

Airport emergency plan (AEP): A concise planning document developed by the airport operator that establishes airport operational procedures and responsibilities during various contingencies.

Airport security program: A security program approved by TSA. It describes requirements for security programs, including establishing secured areas, air operations areas, security identification display areas, and access to control systems.

ARFF index: The minimum aircraft rescue and firefighting personnel and equipment required for an airport as determined by the length of the longest air carrier aircraft for which there is an average of five or more daily departures. When there are fewer than five average daily departures of the longest air carrier aircraft serving the airport, the Index required by the airport will be the next lower Index group than the Index group prescribed for the longest aircraft.

Commercial service airport: *See* nonprimary commercial service airport.

Community: A political entity with the authority to adopt and enforce laws and ordinances for the area under its jurisdiction. In most cases, the community is an incorporated town, city, township, village, or unincorporated area of a county. However, each state defines its own political subdivisions and forms of government.

Continuity of business: Practices that provide the focus and guidance for the decisions and actions necessary for a business to prevent, mitigate, prepare for, respond to, resume, recover, restore, and transition from a disruptive (crisis) event in a manner consistent with its strategic objectives.

Continuity of operations: An effort within an organization to ensure that its primary mission essential functions continue during a wide range of emergencies, including localized acts of nature, accidents, and technological or attack-related emergencies.

Covered air transportation: Scheduled or public charter passenger air transportation provided by an air carrier that operates an aircraft that as originally designed has a passenger capacity of 30 or more seats.

Disaster: An occurrence of a natural catastrophe, technological accident, or human-caused event that has resulted in severe property damage, multiple injuries, and/or deaths.

Diversion airport: A U.S. DOT term that describes an airport that airlines have defined in their internal plans as those with which they must coordinate their tarmac delay contingency plans.

Copyright National Academy of Sciences. All rights reserved.

88 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Diverted flight: A flight operated from the scheduled origin point to a point other than the scheduled destination point in the carrier's published schedule. For example, a carrier has a published schedule for a flight from A to B to C. If the carrier were to actually fly an A to C operation, the A to B segment is a diverted flight, and the B to C segment is a canceled flight. (United States DOT definition; 14 CFR § 234.2)

Drill: A coordinated, supervised activity usually used to test a single specific operation or function in a single agency.

Emergency: Any occasion or instance that warrants action to save lives and to protect property, public health, and safety.

Emergency management (EM): The coordination and integration of all activities necessary to build, sustain, and improve the capabilities to prepare for, respond to, recover from, or mitigate against threatened or actual disasters or emergencies, regardless of cause.

Emergency operations center (EOC): A protected site from which emergency officials coordinate, monitor, and direct response activities during an emergency.

Enplanement: The boarding of a revenue passenger at an airport.

Exercise: A planned, staged implementation of the critical incident plan to evaluate processes that work and identify those needing improvement.

Fixed-base operator: A commercial entity providing aeronautical services such as fueling, maintenance, storage, ground, and flight instruction, etc., to the public.

Four-hour tarmac delay rule: A rule issued by the United States DOT that improves the air travel environment for consumers. One key provision of this rule is the requirement of air carriers (both domestic and international) to adopt and adhere to tarmac delay contingency plans. This rule establishes the "4-Hour Rule," which requires international air carriers to allow passengers to deplane after being out on the tarmac for 4 hours (also see the 3-hour tarmac delay rule).

General aviation (GA) airport: An airport that does not meet the criteria for classification as a primary service commercial service airport may be included in the NPIAS as a general aviation airport if they account for enough activity (having usually at least 10 locally based aircraft) and are at least 20 miles from the nearest NPIAS airport.

Hub: A commercial service airport that services one or more airlines.

Incident: An occurrence other than an accident with the operation of an aircraft, which affects or could affect the safety of operations.

Incident command system (ICS): A standardized organizational structure used to command, control, and coordinate the use of resources and personnel that have responded to the scene of an emergency.

Incident commander: The individual responsible for all incident activities, including the development of strategies and tactics and the ordering and release of resources.

Incident management team (IMT): An Incident Commander and the appropriate Command and General Staff personnel assigned to an incident; the level of training and experience of the IMT members, coupled with the identified formal response requirements and responsibilities of the IMT, are factors in determining "type," or level, of IMT.

Interlocal agreement: A written contract between local government agencies such as a city or county. Any time a public service involves the joint operations and budgets of two or more local government agencies, an interlocal agreement must be drawn up and approved by all parties.

Copyright National Academy of Sciences. All rights reserved.

Interoperability: The ability of systems, personnel, and equipment to provide and receive functionality, data, information, and/or services to and from other systems, personnel, and equipment, between both public and private agencies, departments, and other organizations, in a manner enabling them to operate effectively together.

Irregular operations (IROPS): Those actions taken to adjust for and recover from the impacts of disrupted airline schedules such as aircraft accidents, security incidents, crew absences, mechanical failures, and bad weather.

Large hub airport: An airport that accounts for at least 1% of total U.S. passenger enplanements.

Mass care: The actions taken to protect evacuees and other disaster victims from the effects of a disaster.

Medium hub airport: An airport that accounts for between 0.25% and 1% of total U.S. passenger enplanements.

Mutual aid: First-response entities (e.g., fire, medical, and law enforcement) that are usually contracted with an airport through interlocal or joint-use agreements or are managed through another department under municipal control.

Mutual aid agreement: A written agreement between agencies, organizations, or jurisdictions to lend assistance across jurisdictional boundaries. It agrees to assist by furnishing personnel, equipment, and expertise in a specified manner at the requisite time.

National incident management system (NIMS): A systematic, proactive approach guiding government agencies at all levels, the private sector, and nongovernmental organizations to work seamlessly to prepare for, prevent, respond to, recover from, and mitigate the effects of incidents, regardless of cause, size, location, or complexity, to reduce the loss of life and property and reduce harm to the environment.

National plan of integrated airport systems (NPIAS): A national airport plan prepared by the FAA in accordance with Section 47103 of Title 49 of the United States Code; the plan includes primary and commercial service airports, all general aviation airports designated as reliever airports by the FAA and selected general aviation airports.

Network partners: Agencies and organizations that provide personnel, equipment, and expertise under a predetermined plan.

Non-hub (NH) primary airport: An airport that enplanes less than 0.05% of all commercial passenger enplanements but has more than 10,000 annual enplanements.

Nonprimary commercial service airport: A public airport receiving scheduled passenger service and having 2,500 or more enplaned passengers per year but fewer than 10,000 annual passenger enplanements.

Non-station airline: An airline that does not operate from an airport.

Part 139 airport: An airport that serves scheduled and unscheduled air carrier aircraft with more than 30 seats, serves scheduled air carrier operations in aircraft with more than 9 seats but less than 31 seats, and that the FAA Administrator requires to have a certificate for operation.

Primary airport: Public airports that receive scheduled passenger service and have more than 10,000 annual passenger enplanements.

Reliever airport: A high-capacity general aviation airport in a major metropolitan area; such airports must have 100 or more based aircraft or 25,000 annual itinerant operations; the FAA officially designates reliever airports.

Copyright National Academy of Sciences. All rights reserved.

90 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Small hub (SH) airport: An airport that enplanes 0.05% to 0.25% of total U.S. passenger enplanements.

Tabletop exercise: An activity that involves key personnel discussing simulated scenarios in an informal setting. This type of exercise can be used to assess plans, policies, and procedures; or to assess the systems needed to guide the prevention of, response to, and recovery from a defined incident. Tabletop exercises typically are aimed at facilitating understanding of concepts, identifying strengths and shortfalls, and achieving changes in attitude. Participants are encouraged to discuss issues in depth and develop decisions through slow-paced problem solving, rather than the rapid, spontaneous decision-making that occurs under actual or simulated emergency conditions.

Technical stop: A stop commonly used to refuel the aircraft or to make unexpected essential repairs. No traffic is unloaded or loaded during a technical stop.

Three-hour tarmac delay rule: A rule issued by the United States DOT that improves the air travel environment for consumers. One key provision of this rule is the requirement of air carriers (both domestic and international) to adopt and adhere to tarmac delay contingency plans. This rule establishes the "3-Hour Rule," which requires domestic air carriers to allow passengers to deplane after being out on the tarmac for 3 hours (also see 4-hour tarmac delay rule).

Unified command: The unified command organization operating within NIMS consists of the Incident Commanders from various jurisdictions or organizations operating together to form a single command structure.

Copyright National Academy of Sciences. All rights reserved.

Acronyms

The following list also includes abbreviations that are found within the text.

A-CERT	Airport Community Emergency Response Team
A4A	Airlines for America
AAAE	American Association of Airport Executives
AAR	After-Action Report
AC	Advisory Circular
ACI-NA	Airports Council International–North America
A-CERT	Airport Community Emergency Response Team
ADA	Americans with Disabilities Act
AEP	Airport Emergency Plan
ANG	Air National Guard
AOC	Airport Operations Center
ARFF	Aircraft Rescue and Firefighting
ASPM	Aviation System Performance Metrics
ATC	Air Traffic Control
ATCT	Air Traffic Control Tower
BCP	Business Continuity Plan
CATSA	Canadian Air Transport Security Authority
CBP	Customs and Border Patrol
CBSA	Canada Border Services Agency
CCP	Crisis Communication Plan
CDC	Centers for Disease Control and Prevention
CDM	Collaborative Decision-Making
COOP	Continuity of Operations Plan
CS	Commercial Service Airport
DOD	Department of Defense
DOT	Department of Transportation
ECP	Emergency Communications Plan
EM	Emergency Management
EMS	Emergency Medical Services
EOC	Emergency Operations Center
ERP	Emergency Response Plan
EWG	Emergency Working Group
FBI	Federal Bureau of Investigation
FBO	Fixed-Base Operator
FEMA	Federal Emergency Management Agency
GA	General Aviation Airport

Copyright National Academy of Sciences. All rights reserved.

92 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

HSPD	Homeland Security Presidential Directive
IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
ICS	Incident Command System
IROPS	Irregular Operations
JIC	Joint Information Center
LEO	Law Enforcement Officer
LH	Large Hub Airport
MAC	Metropolitan Airport Commission
MH	Medium Hub Airport
MOU	Memorandum of Understanding
NH	Non-Hub Airport
NIMS	National Incident Management System
NPIAS	National Plan of Integrated Air Systems
NTSB	National Transportation Safety Board
R	Reliever Airport
RCAF	Royal Canadian Air Force
SH	Small Hub Airport
SOP	Standard Operating Procedures
TSA	Transportation Security Administration
UC	Unified Command
USAF	United States Air Force

Copyright National Academy of Sciences. All rights reserved.

References

Atlason, G., et al. (2014). *Airports Council International, Emergency Preparedness and Contingency Planning Handbook, First Edition*. ACI World, Montreal, Quebec, Canada.

Ayra, E. S., D. R. Insua, and J. Cano. (2014). To Fuel or Not to Fuel? Is That the Question? *Journal of the American Statistical Association*, Vol. 109.

Ball, J. (2008). The Impact of Training on General Aviation Pilots' Ability to Make Strategic Weather-Related Decisions. Office of Aerospace Medicine, Washington, D.C.

Barich, F., et al. (2020). *ACRP Web Resource 8: Integrating Social Media with Emergency Management at Airports.* Transportation Research Board, Washington, D.C. https://www.trb.org/ACRP/Blurbs/181207.aspx.

Baskas, H. (2016). How Many People Does It Take to Run an Airport? *USA Today*. https://www.usatoday.com/story/travel/flights/2016/03/30/airport-workers-employees/82385558/.

Bureau of Transportation Statistics. (2022). TranStats. U.S. Department of Transportation. https://transtats.bts.gov.

Cogliandro, B., et al. (2016). *ACRP Research Report 153: Guidebook for Stakeholder Communication and Coordination*. Transportation Research Board of the National Academies, Washington, D.C.

Epstein, C. R., J. M. Forbes, C. L. Futter, I. M. Hosegood, R. G. Brown, and A. A. Zundert. (2019). Frequency and Clinical Spectrum of In-Flight Medical Incidents During Domestic and International Flights. *Anaesthesia and Intensive Care*, Vol. 47, No. 1.

Federal Aviation Administration (FAA). (2009). Advisory Circular 150/5200-31C.

Federal Aviation Administration (FAA). (2012). General Aviation Airports: A National Asset. https://www.faa.gov/airports/planning_capacity/ga_study/.

Federal Aviation Administration (FAA). (2020a). Air Traffic By the Numbers. https://www.faa.gov/air_traffic/by_the_numbers/.

Federal Aviation Administration (FAA). (2020b). National Plan of Integrated Airport Systems (NPIAS)—Airports 2021–2025. www.faa.gov/airports/planning_capacity/npias/.

Federal Aviation Administration (FAA). (2021a). Regional Airports Offices. https://www.faa.gov/airports/regions/.

Federal Aviation Administration (FAA). (2021b). FAA Aerospace Forecast: Fiscal Years 2021–2041. Washington, D.C.

Federal Aviation Administration (FAA). (2022a). 5G and Aviation Safety. https://www.faa.gov/newsroom/faa-statements-5g.

Federal Aviation Administration (FAA). (2022b). Unruly Passengers. www.faa.gov/data_research/passengers_cargo/unruly_passengers/.

Federal Aviation Administration (FAA). (n.d.). Aviation System Performance Metrics (ASPM). https://aspm.faa.gov/aspmhelp/index/Aviation_System_Performance_Metrics_(ASPM).html.

Federal Emergency Management Agency (FEMA). (n.d.). *Managing Spontaneous Volunteers in Times of Disaster: The Synergy of Structure and Good Intentions.*

Griffith, D., et al. (2014). *ACRP Report 112: Airport Terminal Incident Response Planning*. Transportation Research Board of the National Academies, Washington, D.C.

International Air Transport Association (IATA). (2013). *Ground Operations Manual*.

International Civil Aviation Organization (ICAO). (2022). Emergency Response Planning. https://www.icao.int/sustainability/ERP/Pages/default.aspx.

IEM, Inc., et al. (2012). *ACRP Report 73: Airport-to-Airport Mutual Aid Programs Guidebook*. Transportation Research Board of the National Academies, Washington, D.C.

IEM, Inc., et al. (2013). *ACRP Report 95: Integrating Community Emergency Response Teams (A-CERTs) at Airports*. Transportation Research Board of the National Academies, Washington, D.C.

Lewis, B. A., et al. (2021). Data-Driven Estimation of the Impact of Diversions Due to In-Flight Medical Emergencies on Flight Delay and Aircraft Operating Costs. *Aerospace Medicine and Human Performance*, Vol. 92, No. 2.

Copyright National Academy of Sciences. All rights reserved.

94 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Malandri, C., L. Mantecchini, F. Paganelli, and M. N. Postorino. (2020). Impacts of Unplanned Aircraft Diversions on Airport Ground Operations. *Transportation Research Procedia*, Vol. 47.

Murphy, S., et al. (2021). A*CRP Synthesis 115: Practices in Airport Emergency Plans*. Transportation Research Board, Washington, D.C.

Nash, J. M., et al. (2012). *ACRP Report 65: Guidebook for Airport Irregular Operations (IROPS) Contingency Planning*. Transportation Research Board of the National Academies, Washington, D.C.

Price, M. (2014). *ACRP Report 116: Guidebook for Successfully Assessing and Managing Risks for Airport Capital and Maintenance Projects*. Transportation Research Board of the National Academies, Washington, D.C.

Ryerson, M. S. (2018). Diversion Ahead: Modeling the Factors Driving Diversion Airport Choice. *Journal of Infrastructure Systems,* Vol. 24, No. 1.

Shreveport Airport Authority Operations Division. (n.d.). Shreveport Regional Airport irregular operations plan (IROPS). http://flyshreveport.com/227/Shreveport-Regional-Airport-IROPS-2022.

Smith, J. F. (2014). *ACRP Synthesis 50: Effective Cooperation Among Airports and Local and Regional Emergency Management Agencies for Disaster Preparedness and Response*. Transportation Research Board of the National Academies, Washington, D.C.

Smith, J. F. (2019). *ACRP Synthesis 99: Emergency Working Groups at Airports*. Transportation Research Board, Washington, D.C.

Smith, J. F., and K. Kenville. (2013). *ACRP Synthesis 45: Model Mutual Aid Agreements for Airports*. Transportation Research Board of the National Academies, Washington, D.C.

Smith, J. F., et al. (2016). *ACRP Synthesis 73: Emergency Communications Planning for Airports*. Transportation Research Board, Washington, D.C.

Smith, J. F., K. Kenville, and J. M. Sawyer. (2015). *ACRP Synthesis 60: Airport Emergency Post-Event Recovery Practices*. Transportation Research Board of the National Academies, Washington, D.C.

Spak, M., and P. Olexa. (2020). Enhancement of the Diversion Airport Selection Methodology, *Transportation Research Procedia*, Vol. 51.

Stambaugh, H., M. Argabright, H. Benaman, and M. Cheston. (2014). *ACRP Report 103: A Guidebook for Integrating NIMS for Personnel and Resources at Airports*. Transportation Research Board of the National Academies, Washington, D.C.

U.S. Customs and Border Protection (CBP). (2019). Carrier Diversion Memo. U.S. Department of Homeland Security.

U.S. Department of Homeland Security. (2003a). Presidential Directive: 5 (HSPD-5)—National Preparedness.

U.S. Department of Homeland Security. (2003b). Presidential Directive: 8 (HSPD-8)—National Preparedness.

USLegal.com. (2022). Definition of Mutual Aid Agreement. uslegal.com.

Copyright National Academy of Sciences. All rights reserved.

Bibliography

ACRP Impacts on Practice. (2013). *Contingency Planning for Unexpected Passenger*. Transportation Research Board of the National Academies, Washington, D.C.

Barich, F., et al. (2013). *ACRP Report 88: Guidebook on Integrating GIS in Emergency Management at Airports*. Transportation Research Board of the National Academies, Washington, D.C.

Brause, L. M., A. Popa, T. Koch, A. Deutschmann, and M. Hellmann. (2020). Optimization of Resource Demand for Passenger Services at Airports During System Failures Such as Blackouts. *European Transport Research Review*, Vol. 12, No. 1: 54.

Corzine, S. (2013). *ACRP Report 93: Operational and Business Continuity Planning for Prolonged Airport Disruptions*. Transportation Research Board of the National Academies, Washington, D.C.

Federal Aviation Administration (FAA). (2021c). FAQ: Weather Delay. Washington, D.C. www.faa.gov/nextgen/programs/weather/faq/#faq4.

Federal Aviation Administration (FAA). (2022c). Aviation Forecasts: Aerospace Forecast. https://www.faa.gov/data_research/aviation/.

Federal Aviation Administration (FAA). (2022d). FAA Statements on 5G. https://www.faa.gov/newsroom/faa-statements-5g.

Federal Emergency Management Agency. (2017). *National Incident Management System Guideline for Mutual Aid*.

Hollis, B. F. (2016). Medical Incidents on Commercial Airlines, Doctor of Nursing Practice, An Analysis of Onboard Medical Incidents, Treatment versus Prevention. Ph.D. dissertation, Brandman University, Irvine, California.

ICF International, et al. (2016). *ACRP Report 160: Addressing Significant Weather Impacts on Airports: Quick Start Guide and Toolkit*. Transportation Research Board, Washington, D.C.

IEM. (2019). *ACRP Research Report 201: Airport Emergency Communications for People with Disabilities and Others with Access and Functional Needs*. Transportation Research Board of the National Academies, Washington, D.C.

Kaim, G. D., and N. A. Morrow. (2014). Examining the Standard of Care for Failure to Divert During In-Flight Emergencies. *Issues in Aviation Law and Policy*, Vol. 13, No. 2.

Karlsson, J., et al. (2016). *ACRP Report 106: Being Prepared for IROPS: A Business-Planning and Decision-Making Approach*. Transportation Research Board of the National Academies, Washington, D.C.

Kenville, K. A., et al. (2009). *ACRP Report 22: Helping Airport and Air Carrier Employees Cope with Traumatic Events*. Transportation Research Board of the National Academies, Washington, D.C.

Kreckie, J. (2020). Aircraft Rescue and Firefighting Strategies and Tactical Considerations for New Large Aircraft: Update. Federal Aviation Administration, U.S. Department of Transportation, Washington, D.C.

National Weather Service. (2022). Aviation Weather Center. Washington, D.C. http://aviationweather.gov.

Naz, Y., and M. S. Akturk. (2021). Aircraft and Passenger Recovery During an Aircraft's Unexpected Unavailability. *Journal of Air Transport Management*, Vol. 91.

Newmeyer, D. A., M. C. Modjeski, J. K. Voges, and J. T. Bryant. (2007). The Importance of Reliever Airports in the USA. *Collegiate Aviation Review,* Vol. 25, No. 1.

Paxinos, O., et al. (2021). In-Flight Medical Events and Cabin Crew First Aid Response. *Aerospace Medicine and Human Performance*, Vol. 92, No. 2.

Ruskin, K. J., E. M. Ricaurte, and P. M. Alves. (2018). Medical Guidelines for Airline Travel: Management of In-Flight Cardiac Arrest. *Aerospace Medicine and Human Performance*, Vol. 89, No. 8.

Smith, J. F. (2010). The Status of General Aviation Airports in Disaster Response Planning: A Reconnaissance Study. American Public University System.

Copyright National Academy of Sciences. All rights reserved.

Smith, J. F., R. E. Garcia, J. M. Sawyer, and K. Kenville. (2016). *ACRP Synthesis 72: Tabletop and Full-Scale Emergency Exercises for General Aviation, Non-Hub, and Small Hub Airports*. Transportation Research Board, Washington, D.C.

Sonmez, K., and S. Uslu. (2020). The Effect of Management Practices on Aircraft Incidents. *Journal of Air Transport Management*, Vol. 84.

Stambaugh, H., et al. (2009). *ACRP Report 12: An Airport Guide for Regional Emergency Planning for CBRNE Events*. Transportation Research Board of the National Academies, Washington, D.C.

Varma, A., S. Germolus, and D. Beaver. (2016). *ACRP Synthesis 78: Continuity of Operations Planning for Small Airports*. Transportation Research Board, Washington, D.C.

Warner-Bean, S., K. Jenkins, J. S. Miller, C. Parkins, and R. Hoaglund. (2017). *ACRP Research Report 171: Establishing a Coordinated Local Family Assistance Program for Airports*. Transportation Research Board, Washington, D.C.

Zellers, S. J., et al. (2019). *ACRP Research Report 16: Guidebook for Managing Small Airports, Second Edition*. Transportation Research Board, Washington, D.C.

Copyright National Academy of Sciences. All rights reserved.

APPENDIX A

Survey Questionnaire and Interview Guide

AIRPORT BACKGROUND INFORMATION

1. Please provide your airport's name. *Please note that this information will remain confidential. We only need this information for statistical purposes, ensuring we calculate the total number of airports that respond to the survey correctly.*

2. Please indicate your airport's NPIAS designation:
   - ☐ Small airport
   - ☐ Non-hub primary airport
   - ☐ Nonprimary commercial service airport
   - ☐ Reliever airport
   - ☐ General aviation airport

3. Which incident- and emergency-related services does your airport manage under its own authority (part of the airport organization). Please choose all that apply:
   - ☐ ARFF
   - ☐ Medical
   - ☐ Law Enforcement
   - ☐ None
   - ☐ Other (please specify)

DIVERSION-RELATED AIRPORT PLANS

4. Where do your airport's diversion-related plans reside? Please choose all that apply:
   - ☐ Airport Diversion Plan
   - ☐ Aircraft Emergency Response Plans (e.g., Standard Operating Procedures)
   - ☐ Airport Emergency Plan (AEP)
   - ☐ AEP Annex
   - ☐ Airport Training and Exercise Plan
   - ☐ Airport EOC Plan
   - ☐ City/County/Local Emergency Plan

Copyright National Academy of Sciences. All rights reserved.

98 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

- ☐ Business Continuity Plan (BCP)/Continuity of Operations Plan (COOP)
- ☐ Crisis Communication Plan/Emergency Communications Plan
- ☐ Family Assistance Plan
- ☐ Hazard-Specific Plans (weather, pandemic etc.)
- ☐ Irregular Operations (IROPS) Plan
- ☐ Mutual Aid Plan
- ☐ My airport does not have any provisions for diversion planning
- ☐ Unsure
- ☐ Other (please specify)

DIVERSION-RELATED AIRPORT PLANS

5. Does your airport have a **practical** and **actionable** plan, SOP, or checklist to use during flight diversions?
   - ☐ Yes
   - ☐ No

6. What **elements** does your airport's plan, SOP, or flight diversion checklist include?
   - ☐ Notification triggers (e.g., when to call upon mutual aid network partners)
   - ☐ Notification procedures (e.g., how to contact mutual aid network partners)
   - ☐ Communication procedures to maintain common situational awareness among mutual aid network partners during an event (e.g., email, text, virtual dashboard)
   - ☐ Escalation triggers (e.g., defines when an incident becomes an emergency)
   - ☐ Coordination procedures to access resources, staff and equipment from mutual aid network partners (e.g., sharing agreements in place)
   - ☐ Special conditions procedures (e.g., after-hours staffing procedures, aircraft/passenger surge management, prolonged event considerations)
   - ☐ Monitoring procedures (e.g., weather, flight status, and regional/national airspace status)
   - ☐ Other (please specify)

7. What makes your flight diversion checklist **easy to use**?
   - ☐ Digital plan (laptop/smartphone accessible)
   - ☐ Available on a smartphone app
   - ☐ Quick reference (1-2 pages)
   - ☐ Separate stand-alone document
   - ☐ Easy to find (e.g., posted on a bulletin board)
   - ☐ Easy to access (e.g., tear out sheet in binder)
   - ☐ Specific, clear and precise instructions
   - ☐ Bullet point format
   - ☐ Divides complicated tasks into simple ones
   - ☐ Grouped by role/responsibility
   - ☐ Grouped by task
   - ☐ Tasks ordered sequentially or by escalating factors
   - ☐ Verified to be current/up-to-date (e.g., includes time stamp)
   - ☐ Other (please specify)

Copyright National Academy of Sciences. All rights reserved.

MUTUAL AID NETWORK PARTNERS

DEFINITION: Mutual aid network partners are not just the agencies that your airport has formal agreements with to assist you during both emergency and non-emergency events but also are other local organizations that support your airport when its capabilities have been exceeded.

8. With the above definition in mind, who do you consider to be your **public safety** and **government agency** mutual aid network partners? With which do you share/coordinate your diversion-related plans? With which do you collaboratively train (at least annually) With which do you train specifically for diverted flights? Please choose as many as applicable.

	Mutual Aid Network Partner	Share/Coordinate Diversion-Related Plans	Regular Collaborative Training	Diverted Flight Training
Law Enforcement	☐	☐	☐	☐
ARFF	☐	☐	☐	☐
Fire Department	☐	☐	☐	☐
EMS	☐	☐	☐	☐
Local EM Agency/Office	☐	☐	☐	☐
Local Healthcare/ Hospitals	☐	☐	☐	☐
FAA	☐	☐	☐	☐
FBI	☐	☐	☐	☐
CBP	☐	☐	☐	☐
CDC	☐	☐	☐	☐
NTSB	☐	☐	☐	☐
TSA	☐	☐	☐	☐
ANG, USAF, or other military joint use on airport	☐	☐	☐	☐
U.S. Marshals	☐	☐	☐	☐
Air Marshals	☐	☐	☐	☐

Copyright National Academy of Sciences. All rights reserved.

100 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

9. Again, with the above definition in mind, which **airline, airport and local organizations** do you consider to be your airport's mutual aid network partners? With which do you share/coordinate your diversion-related plans? With which do you collaboratively train (at least annually)? With which do you train specifically for diverted flights? Please choose as many as applicable.

	Mutual Aid Network Partner	Share/Coordinate Diversion-Related Plans	Regular Collaborative Training	Diverted Flight Training
Airlines	☐	☐	☐	☐
FBO/Fuelers	☐	☐	☐	☐
Hangar/Aircraft Owners	☐	☐	☐	☐
Concessionaires	☐	☐	☐	☐
Other Airport Tenants	☐	☐	☐	☐
Ground Transportation	☐	☐	☐	☐
Red Cross	☐	☐	☐	☐
Counselors	☐	☐	☐	☐
Local Stores	☐	☐	☐	☐
Local Restaurants	☐	☐	☐	☐
Local Hotels	☐	☐	☐	☐
Airport or Other Volunteer Group	☐	☐	☐	☐

10. Who is your **airport's main point of contact** to coordinate with your mutual aid network partners during incidents and emergencies when your capabilities have been exceeded?
    - ☐ The airport's director or general manager acting 1:1 with all mutual aid network partners
    - ☐ The airport's operations manager
    - ☐ The airport's emergency manager
    - ☐ The local (city, county, or local multi-county regional) EM agency
    - ☐ The state EM agency
    - ☐ A federal agency
    - ☐ Don't know
    - ☐ Other (please specify)

Copyright National Academy of Sciences. All rights reserved.

11. In the past 12-18 months, how often has your airport asked your mutual aid network partners to assist with incident- or emergency-related flight diversions?
    - ☐ 1-5 times
    - ☐ 5-10 times
    - ☐ 10-15 times
    - ☐ 15-20 times
    - ☐ More than 20 times per year
    - ☐ Unsure
    - ☐ Other (please specify)

INDUSTRY PRACTICES RELATED TO DIVERSION EVENTS

12. Does your airport implement any of these industry-recommended **notification practices** with your mutual aid network? Indicate all that you implement:
    - ☐ Update and test your 24/7 mutual aid network partner contact list regularly (at least annually)
    - ☐ Identify both primary and secondary contacts at all network partners with various communication methods identified for each (e.g., office phone number, cell phone number and email address)
    - ☐ Establish communication protocols (e.g., email, text, dashboard, etc.) that all partners can access for sharing notifications and real-time, accurate and consistent updates throughout the event
    - ☐ Use common definitions and acronyms following National Incident Management System (NIMS) and Incident Command Structure (ICS) during events, using common/plain language and avoiding unique codes/phrases whenever possible
    - ☐ Test your airport's notification system (technology) via "live" tests throughout the year
    - ☐ Develop a backup plan for technology during power failures, cyber-attacks, and network overloads
    - ☐ Unsure
    - ☐ None of the above

13. Does your airport plan for any of these industry-recommended **contingency practices** that can help prevent flight diversion events from **escalating**?
    - ☐ Specific aircraft-related practices (e.g., plans for wide body, international flights, non-station airline needs)
    - ☐ Surge of passengers (Plans that delineate overflow areas) or aircraft (Plans that delineate hard stand areas, parking areas)
    - ☐ After hours conditions (e.g., plans/agreements for on-call or late shift staffing)
    - ☐ Additional impacts to airport facilities unrelated to the diversion, including landside impacts (e.g., plans for power outages, weather-related damages) and airside impacts (e.g., alternate plans for damaged runway)
    - ☐ Compounded communication issues (e.g., plans for network overload or power outages)
    - ☐ Extended delay situations (e.g., plans for preventing equipment overuse, plans)
    - ☐ Long-term weather event
    - ☐ National Airspace System conditions (e.g., natural disaster in one part of the country causing flight diversions)
    - ☐ Unsure
    - ☐ None of the above

Copyright National Academy of Sciences. All rights reserved.

102 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

14. Does your airport implement any of these industry-recommended **coordination practices** with your mutual aid network partners? Indicate all that you implement:
    - ☐ Meet regularly with your mutual aid network partners via meetings, workshops or training sessions
    - ☐ Ensure the airport's diversion-related plans are coordinated and aligned with mutual aid partner plans (e.g., regional/municipal contingency plans, mutual aid operational plans, emergency response plans, etc.)
    - ☐ Establish, update and share a centralized list of local equipment and resources for use in flight diversion events
    - ☐ Establish and clarify funding sources for incident- and emergency-related costs associated with diversions (e.g., food and water for stranded passengers)
    - ☐ Identify and establish alternative or multiple service or supply vendors
    - ☐ Conduct an after-action review with mutual aid network partners, assigning personnel with follow-up responsibilities to ensure accountability
    - ☐ Update diversion-related plans and training based upon findings from after-action review
    - ☐ Disseminate updated diversion-related plans both internally at your airport and to mutual aid network partners
    - ☐ Unsure
    - ☐ None of the above

INDUSTRY PRACTICES RELATED TO SPECIFIC PARTNERS

15. Does your airport implement any of these industry-recommended coordination practices with **airlines**? Indicate all that you implement:
    - ☐ Understand each airline's priority airport list for flight diversions
    - ☐ Understand each airline's aircraft towing/gate and deplaning requirements (e.g., wide body aircraft) and available equipment on site at your airport
    - ☐ Understand typical daily staffing availability as well as whether they use any third-party staffing vendors and how they communicate with them
    - ☐ Establish agreements to clearly define the assistance airlines can provide to flight diversions associated with non-station airlines at your airport
    - ☐ Establish agreements to clearly define who (airport or airline) manages passenger assistance / victims' assistance and what resources, supplies and staffing they will provide in various incident- and emergency-situations
    - ☐ Establish equipment sharing agreements with airlines (e.g., air stairs for deplaning, etc.)
    - ☐ Unsure
    - ☐ None of the above

16. Does your airport implement any of these industry-recommended coordination practices with the following **agencies** and **airport tenants**?
    - ☐ FAA - Establish and regularly review capacity constraints policy
    - ☐ TSA - Establish and regularly review security procedures for stranded passengers
    - ☐ CBP - Establish and review deplaning plans for diverted international flights
    - ☐ VARIOUS - Establish and regularly review after-hour staffing plans with TSA, CBP and concessions
    - ☐ Unsure
    - ☐ None of the above

Copyright National Academy of Sciences. All rights reserved.

DIVERTED FLIGHT EXPERIENCE

17. Has your airport recently (past 12-24 months) received a diverted flight due to either an incident or emergency that exceeded your capacity and required the use of your mutual aid network partners (e.g., emergency responders, agencies, healthcare, community groups, Red Cross, etc.) to respond?
    - ☐ Yes
    - ☐ No
18. This diverted flight event involved the following circumstances. Please choose as many as applicable:
    - ☐ Poor weather conditions
    - ☐ Multiple aircraft diversions
    - ☐ Minor mechanical aircraft issue (e.g., oil leak, warning light)
    - ☐ Aircraft emergency (major mechanical failure, fire)
    - ☐ Medical emergency (e.g., heart attack)
    - ☐ Passenger disruption on flight (e.g., security issue)
    - ☐ Passenger deplanement into the airport (e.g., flight cancellation, extended flight delay)
    - ☐ Safety / security issue
    - ☐ Victim assistance
    - ☐ Family assistance
    - ☐ International flight
    - ☐ After airport normal business hours
    - ☐ Other (please specify)

19. This diverted flight event involved the following type of aircraft:
    - ☐ Large commercial aircraft
    - ☐ Small commuter aircraft (single engine/multi-engine piston)
    - ☐ Corporate / private jet (turboprop)
    - ☐ Experimental aircraft
    - ☐ Rotorcraft
    - ☐ Military aircraft
    - ☐ Other (please specify)

Copyright National Academy of Sciences. All rights reserved.

104 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

20. If you were to conduct an after-action report related to your recent diverted flight situation, what would you say **worked well** and **did not work well**, specifically related to coordinating with your mutual aid network partners (e.g., emergency responders, agencies, healthcare, community groups, Red Cross, etc.)? Please choose as many as applicable:

	Mutual Aid Network Partner	**Share / Coordinate Diversion-Related Plans**
COMMUNICATION: Early notification (advanced warning)	☐	☐
COMMUNICATION: Contacting mutual aid network partners (correct contact information on file)	☐	☐
COMMUNICATION: Information-sharing among mutual aid network partners (specifics / needs)	☐	☐
COMMUNICATION: Crisis communication (if applicable)	☐	☐
COMMUNICATION: Social media used for emergency management	☐	☐
COMMUNICATION: Use of technology (e.g., dashboard)	☐	☐
PLANNING: Following a practical checklist or plan	☐	☐
PLANNING: Preparation (e.g., diversion-related training)	☐	☐
PLANNING: Understanding the airport's capability and capacity (e.g., staffing, facilities, resources/supplies, and equipment available for the diverted flight)	☐	☐
INCIDENT MANAGEMENT: Understanding roles and responsibilities of each network partner (e.g., chain of command)	☐	☐
INCIDENT MANAGEMENT: Response time from mutual aid resources	☐	☐
OUTCOME: Adequate passenger care (e.g., facilities, equipment, resources)	☐	☐
OUTCOME: Adequate victim's assistance from mutual aid network partners (if applicable)	☐	☐

Other (please specify)

Copyright National Academy of Sciences. All rights reserved.

21. Would you be open to sharing your good practices and lessons learned associated with diverted flights with our research team?
    - ☐ Yes
    - ☐ No

If yes, please provide your name, airport and email address so one of our team members can contact you.

Thank you!

Thank you for your time. Your participation in this survey is greatly appreciated!

Copyright National Academy of Sciences. All rights reserved.

106 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

AIRPORT INTERVIEW QUESTIONS

DIVERSION PLANNING

Diversion-related Plans

- **Stand-alone vs. comprehensive**
  - Do you use a detailed plan that includes SOPs for diverted flights?
  - Does your plan stand alone or is it part of other plans (e.g., AEP, IROPS, BCP, etc.)?
  - Are your diversion-related plans easy to use? Are they practical?
  - What works and what doesn't?
  - Do you conduct after-action reports for diversions? How is this knowledge gained integrated back into your plans? Have you updated your plans recently?
- **Checklists**
  - Do you have a diversion checklist or procedures? Are you willing to share these with the larger airport/stakeholder industry?
- **Customer Care/Victims Assistance**
  - What kind of customer care do you plan for diverted passengers?
  - What kind of victim/family assistance service do you plan for (e.g., Red Cross, counselors and other medical resources, local stores, hotels, etc.)?
- **Capability and Capacity**
  - How many people do you have to respond to diversions? (For GA – Do you operate 24/7?)
  - What facilities and equipment do you have to accommodate diverted flights and passengers? (Particularly deplaning and hold room capacity)
  - What technology do you use for diversions?
  - Do you keep a list of your capability and capacity (e.g., personnel, equipment, technology) needed for diversions?
  - Have you planned for non-station diverted flights, particularly international flights? Have you coordinated with the CBP?
  - How do you handle crisis communication?
  - Does social media play a role in managing emergency response?
- **Mutual Aid Network Partners**
  - Do you participate in a network (Regional IROPS Network? Emergency Working Group - EWR)?
  - Do you have a 24/7 email/contact list of airport and mutual aid network? How often are your notification lists updated? Do you test the numbers for accuracy? Do you have a backup contact for each major mutual aid partner? Do you publish these numbers? Are these contacts stored and shared in an easy-to-find location?
  - Do you have formal notification and communication procedures for diversions that work well with your network?
  - Do you use any LOAs or MOUs with your mutual aid network to assist with diversions (emergency or incident) that you can recommend or share?
  - Have you created a common terminology, acronyms and definitions related to diversions with your mutual aid network?
  - Do you have an escalation process (both formal and informal) from incident to emergency? Do you have specific triggers and thresholds for activating plans and measure for this escalation process?
  - Do you have good collaboration/partnering with emergency management / mutual aid network? What is going well? What would you improve?

Copyright National Academy of Sciences. All rights reserved.

- **DOT Tarmac Delay Plans**
  - Have you updated your DOT Tarmac Delay Plan from 2017 due to the impacts of the Pandemic and/or diversion events?
  - Do you have a detailed IROPS Contingency Plan (reference Report 65 Template) that includes plans for diverted flights?
  - Are there specific activities you plan before, during and after major disruptive/diversion events?
  - Have your airline partners provided an update of their DOT Airline Tarmac Delay Plans from 2021?
- **Training**
  - Do you train with your mutual aid network?
  - How often does your (stakeholder specific) practice/train for irregular operations or emergency events? Have you conducted diversion training as part of these?
  - Do you have training and/or exercises for diversions? Can you share these with the larger airport/stakeholder industry?
- **Risk**
  - Does your airport have a Risk Tolerance Policy?
  - Do you have a list of essential services for business continuity?
  - Do you track the cost of diversions – specifically medical diversions?
  - Any performance measurements?

DIVERSION EXPERIENCE

- What types of diversions have you received in the last 12-18 months (gas and go, severe weather, medical emergency, flight disruption, international/CBP assistance or unknown)?
- Are there media accounts of the diversion (with links)?
- Is there other background information or reports available about the diversion?
- Walk us through diversions you have received (a variety of emergency, medical or incident level if possible)

Notification. Discuss if or how you were notified of the diverted aircraft. How and where did you receive the notification? Do you use any specific flight tracking tools to better prepare for a diversion? Do you have a diversion tracking log?

Technology. Did you use specific technology to manage diversions such as Flight Aware?

Mitigation. Describe how you mitigated the diversion event.

Mutual Aid Network. Who did you contact/how? Did you have any challenges reaching them? What was their response to your request?

Passenger Care. How many passengers were on board? What were the needs of the passengers? (emergency, medical, basic). Did they have to deplane? Did you have the deplaning equipment needed? Did you have the room for them in your airport facilities? Did they require additional resources?

Preparation. If you were not expecting this type of diversion, what would have helped you be better prepared?

Commercial vs. Corporate. What are the differences between commercial and business/corporate jet diversions? Notification? Communication? Passenger needs? Mutual aid assistance?

Copyright National Academy of Sciences. All rights reserved.

108 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Pros and cons.
What do you like about diversions at your airport?
What are the biggest challenges you face in relation to diversions at your airport?

OTHER

- **Tools**. Do you use any ACRP tools related to diversions?
- **Gaps**. Have you identified any current gaps in service related to the diverted flights (e.g., deplaning equipment, international passenger processing services, after hours/extended delay considerations, basic passenger needs, etc.)?
- **Good Practices**. Do you have any recommended good practices associated with managing diversions?
- **Lessons Learned**. Do you have any lessons learned associated with managing diversions?

Copyright National Academy of Sciences. All rights reserved.

PILOT/AIRLINE INTERVIEW QUESTIONS

DIVERTED FLIGHT EXPERIENCE

1. Have you recently (past 12-24 months) piloted a diverted flight due to either an incident or emergency that required the use of your Airlines mutual aid network partners (e.g., emergency responders, agencies, healthcare, community groups, Red Cross, etc.) to respond?

2. This diverted flight event involved the following type of aircraft:
   - ☐ Large commercial aircraft
   - ☐ Regional aircraft

3. This diverted flight event involved the following circumstances. Please choose as many as applicable:

Poor weather conditions	Medical emergency	Safety or other concerns
Mechanical issues/failures	Passenger disruption	OTHER

4. If you were to conduct an after-action report related to your recent diverted flight situation, what would you say worked well, specifically related to coordinating with the airport and your mutual aid network partners (e.g., emergency responders, agencies, healthcare, community groups, Red Cross, etc.)? Please choose as many as applicable:

Pre-Event Planning with Airline System Operations Center (SOC or OCC)

- Early notification (advanced warning)
- Technology Tools such as Flight Aware or Custom Diversion Planning Tools
- Airport Capability and Capacity
- Gate capacity and capability
- Hours of operations
- Handler Staffing
- Servicing (fuel)
- Servicing (catering)
- Servicing (Lavs)
- Rescue Flight Options
- Hotels
- Customs
- Passenger rebook through airline website
- Gate slot for arrival to Airport X
- Recovery plan for arrival into Airport X
- Update local systems to reflect passenger volume and times update
- OTHER

Diversion Event: Aircraft on the Ground and Coordination with Local Airport Officials

- Gate or Hard Stand
- Fuel Service
- Crew Times
- Maintenance on Site
- Airline Rep for customer interaction
- Passenger Connections
- Customer amenities (Water, food, lavs. Social media)
- Rescue Flight Support – Red Cross - other
- Resolve the Diversion Issue for continuation as quickly as possible or enable recovery plan

Copyright National Academy of Sciences. All rights reserved.

110 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

5. Diversion Flight Decision Points: In flight aircraft or to which airport to divert.
   - ☐ Discuss impacts
     - Aircraft Fleet Schedule
     - Crew Schedule
     - Maintenance Schedule
     - Passenger Schedules
     - Other
   - ☐ How does Airline X/Pilot see the associated affects of diversion decisions?

6. Would you be open to sharing your good practices and lessons learned associated with diverted flights with our research team?

Copyright National Academy of Sciences. All rights reserved.

TECHNOLOGY FIRM INTERVIEW QUESTIONS

Overview: What Are IROPS Events? Events which disrupt optimized flight schedules and negatively impact the normal flow of passengers through the air transportation system (i.e. severe weather, air traffic control (ATC) programs, airport service issues, passenger emergencies, and airline operations difficulties). The outcome includes crowded terminal spaces, distressed passengers, swamped airline passenger service agents, and other situations which will demand the attention of the airport and airline.

Before Flight Diversion Events: What Is Diversion Response Planning? A deliberate and well planned /coordinated process for addressing unique passenger needs in recognition of the scale of incident/passenger requiring attention during an unplanned diversion. Planning takes place before the diversion event and is carried out on the spot through the delivery of special services during an event and after a diversion event by finding ways to continually improve the passenger recovery process.

Elements of Airline/Airport Technology:

Aircraft Autodocking System	Automated Boarding Gates (including biometric-exit through CBP Traveler Verification Service)		Baggage Handling Systems (inbound bags, dynamic carousel asgmt., first/last bag automation/data, flexibility on make-up pier/carousel)	
Baggage Tracking Systems	Billing System	Content Management System for Information Displays and enhanced content including full-motion HD video		Data Exchange and Analytics
Facility Infrastructure (power & network connectivity)	Jet Bridge	Lightning Detection / Warning System	Off-site Passenger and/or Baggage Processing	Ramp Control
RMS	Self-Bag-Drop (attended & unattended with biometric identification)	Video Surveillance System (including ramp views available to airlines)	Wi-Fi & Distributed Antennae System (public & non-public roadway, terminal, ramp)	Wayfinding

Pre-Event Questions: How does your diversion tracking tool address the following:

- Airlines need to know where all aircraft is diverting (not just their own flights) and selected diversion airport capability status (gates, fuel, deicing fluid, hardstands, local support infrastructure.). How does your tool manage this?
- Airports, CBP, and Ground Handlers need to know how many diversions are headed to them, what type of diversion, what type of aircraft, which airlines, and whether crews are likely to time-out. How does your tracking tool achieve this?

Copyright National Academy of Sciences. All rights reserved.

112 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

During Diversion Event Questions:

- Please describe your decision-support information tool for diversions.
- How do multiple users (Airline, Airports, FAA, CBP, FBO, other) see real-time information which enables them to address their specific diversion event(s)?
- What organizations do you collaborate with to implement this diversion communication tool?
- How do you address the following in your tool?



Diversion Recovery Planning:

- Where do airports/airlines use your diversion tool in After-Action Reporting and obtaining lessons learned?

FBO [Fixed-Base Operator] INTERVIEW QUESTIONS

Before

- What plans do you use to manage diversions?
- Does your airport share their diversion plans with you?
- How do you plan with your airport for diversions? Any training involved?

During

- What equipment do you typically use to manage diversion events?
- How do communicate with other organizations during diversions?
- Do you share resources? How do you share (e.g. formal agreements, verbal, etc.)?
- What are challenges you experience related to diversions?
- How often do you (or your airport) need to ask for mutual aid to manage emergency situations?
- What things typically go well during diversions?
- What could be improved?

After

- Are you involved in the after action reporting at your airport?
- What are the typical costs you incur during a diversion?

Copyright National Academy of Sciences. All rights reserved.

APPENDIX B

Survey Responses

Q1: Please provide your airport's name. Please note that this information will remain confidential. We only need this information for statistical purposes, ensuring we calculate the total number of airports that respond to the survey correctly:
Answered: 71 Skipped: 0

- ☐ Small Hub Airports
    1. Blue Grass Airport, Lexington, KY
    2. Colorado Springs Airport, Colorado Springs, CO
    3. Dane County Regional Airport, Madison, WI
    4. Des Moines International Airport, Des Moines, IA
    5. Duluth International Airport, Duluth, MN
    6. El Paso International Airport, El Paso, TX
    7. Eugene Airport, Eugene, OR
    8. Gerald R. Ford International Airport, Grand Rapids, MI
    9. Northwest Florida Beaches International Airport, Panama City, FL
    10. Orlando Sanford International Airport, Orlando, FL
    11. Portland International Jetport, Portland, ME
    12. Punta Gorda Airport, Punta Gorda, FL
    13. Savannah Hilton Head International, Hilton Head, SC
    14. Spokane International Airport, Spokane, WA
    15. Springfield-Branson National Airport, Springfield, MO
    16. Syracuse Hancock International Airport, Syracuse, NY
    17. The Eastern Iowa Airport, Cedar Rapids, IA
    18. Tucson International Airport, Tucson, AZ
    19. Tulsa International Airport, Tulsa, OK

Copyright National Academy of Sciences. All rights reserved.

- ☐ Non-hub Airports
    1. Asheville Regional Airport, Asheville, NC
    2. Canyonlands Regional Airport, Moab, UT
    3. Chippewa Valley Regional Airport, Eau Claire, WI
    4. Columbia Metropolitan Airport, West Columbia, SC
    5. Corpus Christi International Airport, Corpus Christi, TX
    6. Flagstaff Pulliam Airport, Flagstaff, AZ
    7. Gainesville Regional Airport, Gainesville, FL
    8. General Wayne A. Downing Peoria International Airport, Peoria, IL
    9. Grand Forks International Airport, Grand Forks, ND
    10. Green Bay Austin Straubel International Airport, Green Bay, WI
    11. La Crosse Regional Airport, La Crosse, WI
    12. Laughlin/Bullhead International Airport, Bullhead City, AZ
    13. Lebanon Municipal Airport, West Lebanon, NH
    14. Lincoln Airport, Lincoln, NE
    15. Norfolk International Airport, Norfolk, VA
    16. Phoenix-Mesa Gateway Airport, Mesa, AZ
    17. Richmond International Airport, Richmond, VA
    18. Rickenbacker International Airport, Columbus, OH
    19. Roanoke-Blacksburg Regional Airport, Roanoke, VA
    20. San Antonio International Airport, San Antonio, TX
    21. San Luis Obispo County Regional, San Luis Obispo, CA
    22. Sarasota Bradenton International Airport, Sarasota, FL
    23. Shreveport Regional Airport, Shreveport, LA
    24. State of Alaska Airports, AK
    25. Watertown International Airport, Watertown, SD
- ☐ General Aviation Airports
    1. Arlington Municipal Airport, Arlington, WA
    2. Atlanta Regional Airport - Falcon Field, Peachtree City, GA
    3. Sebring Regional Airport, Sebring, FL
    4. Livermore Municipal Airport, Livermore, CA
    5. Madisonville Regional Airport, Madisonville, KY
    6. Moore County Airport, Moore County, NC
    7. Mount Carmel Municipal Airport, St. Francisville, IL
    8. Naples Airport, Naples, FL
    9. Northeast Ohio Regional Airport, Jefferson, OH
    10. Ocala International Airport, Ocala, FL
    11. Pompano Beach Airpark, Pompano Beach, FL
    12. Sheboygan County Memorial Airport, Sheboygan Falls, WI
    13. Shreveport Downtown Airport, Shreveport, LA
    14. Springfield-Beckley Municipal Airport, Springfield, OH
    15. Suwannee County Airport, Live Oak, FL

Copyright National Academy of Sciences. All rights reserved.

- ☐ Reliever Airports
    1. Albert Whitted Airport, St. Petersburg, FL
    2. Boca Raton Airport, Boca Raton, FL
    3. Kissimmee Gateway Airport, Kissimmee, FL
    4. McKinney National Airport, McKinney, TX
    5. Orlando Executive Airport, Orlando, FL
    6. Scottsdale Airport, Scottsdale, AZ
    7. St. Charles County Smartt Airport, Portage Des Sioux, MO
- ☐ Nonprimary Commercial Service Airports
    1. Concord-Padgett Regional Airport, Concord, NC
    2. Hector International Airport, Fargo, ND
    3. Arnold Palmer Regional Airport, Latrobe, PA
    4. Mason City Municipal Airport, Clear Lake, IA
    5. Friedman Memorial Airport, Sun Valley, ID

Q2: Please indicate your airport's NPIAS designation:
Answered: 71 Skipped: 0

Answer Choices	Responses	
Small hub airport	26.76%	19
Non-hub primary airport	33.80%	24
Nonprimary commercial service airport	7.04%	5
Reliever airport	9.86%	7
General aviation airport	22.54%	16
	Total Respondents	**71**

Q3: Which incident- and emergency-related services does your airport manage under its own authority (part of the airport organization)? Please choose all that apply:
Answered: 71 Skipped: 0

Answer Choices	Responses	
ARFF	49.30%	35
Medical	22.54%	16
Law Enforcement	38.03%	27
None	40.85%	29
Other (please specify)	9.86%	7
	Total Respondents	**71**

Copyright National Academy of Sciences. All rights reserved.

Q4: Where do your airport's diversion-related plans reside? Please choose all that apply
Answered: 69 Skipped: 2

Answer Choices	Responses	
Airport Diversion Plan	14.49%	10
Aircraft Emergency Response Plans (e.g., Standard Operating Procedures)	28.99%	20
Airport Emergency Plan (AEP)	36.23%	25
AEP Annex	1.45%	1
Airport Training and Exercise Plan	5.80%	4
Airport EOC Plan	5.80%	4
City/County/Local Emergency Plan	7.25%	5
Business Continuity Plan (BCP) / Continuity of Operations Plan (COOP)	4.35%	3
Crisis Communication Plan / Emergency Communications Plan	4.35%	3
Family Assistance Plan	4.35%	3
Hazard-Specific Plans (weather, pandemic etc.)	7.25%	5
Irregular Operations (IROPS) Plan	40.58%	28
Mutual Aid Plan	4.35%	3
My airport does not have any provisions for diversion planning	23.19%	16
Unsure	2.90%	2
Other Plan (please specify)		3
Total Respondents		**69**

Q5: Does your airport have a practical and actionable plan, SOP, or checklist to use during flight diversions?
Answered: 51 Skipped: 20

Answer Choices	Responses	
Yes	50.98%	26
No	49.02%	25
Total Respondents		**51**

Copyright National Academy of Sciences. All rights reserved.

Q6: What elements does your airport's plan, SOP, or flight diversion checklist include?
Answered: 25 Skipped: 46

Answer Choices	Responses	
Notification triggers (e.g., when to call upon mutual aid network partners)	80.00%	20
Notification procedures (e.g., how to contact mutual aid network partners)	92.00%	23
Communication procedures to maintain common situational awareness among mutual aid network partners during an event (e.g., email, text, virtual dashboard)	72.00%	18
Escalation triggers (e.g., defines when an incident becomes an emergency)	68.00%	17
Coordination procedures to access resources, staff and equipment from mutual aid network partners (e.g., sharing agreements in place)	72.00%	18
Special conditions procedures (e.g., after-hours staffing procedures, aircraft/passenger surge management, prolonged event considerations)	72.00%	18
Monitoring procedures (e.g., weather, flight status, and regional/national airspace status)	68.00%	17
Other (please specify)	8.00%	2
Total Respondents		25

Q7: What makes your flight diversion checklist easy to use?
Answered: 25 Skipped: 46

Answer Choices	Responses	
Digital plan (laptop/smartphone accessible)	24.00%	6
Available on a smartphone app	4.00%	1
Quick reference (1-2 pages)	72.00%	18
Separate stand-alone document	56.00%	14
Easy to find (e.g., posted on a bulletin board)	32.00%	8
Easy to access (e.g., tear out sheet in binder)	28.00%	7
Specific, clear and precise instructions	36.00%	9
Bullet point format	36.00%	9
Divides complicated tasks into simple ones	8.00%	2
Grouped by role/responsibility	44.00%	11
Grouped by task	20.00%	5
Tasks ordered sequentially or by escalating factors	28.00%	7
Verified to be current/up-to-date (e.g., includes time stamp)	28.00%	7
Other (please specify)	8.00%	2
Total Respondents		25

Copyright National Academy of Sciences. All rights reserved.

Q8: With the above definition in mind, who do you consider to be your public safety and government agency mutual aid network partners? With which do you share/coordinate your diversion-related plans? With which do you collaboratively train (at least annually) With which do you train specifically for diverted flights? Please choose as many as applicable.

Answered: 55 Skipped: 16

	Mutual Aid Network Partner		Share / Coordinate Diversion-Related Plans		Regular Collaborative Training		Diverted Flight Training		Total
Law Enforcement	87.27%	48	36.36%	20	41.82%	23	10.91%	6	55
ARFF	67.74%	21	54.84%	17	48.39%	15	32.26%	10	31
Fire Department	92.31%	48	28.85%	15	46.15%	24	3.85%	2	52
EMS	92.00%	46	28.00%	14	38.00%	19	4.00%	2	50
Local EM Agency/Office	87.23%	41	23.40%	11	31.91%	15	4.26%	2	47
Local Healthcare / Hospitals	89.74%	35	25.64%	10	28.21%	11	0.00%	0	39
FAA	86.11%	31	41.67%	15	16.67%	6	8.33%	3	36
FBI	84.85%	28	24.24%	8	27.27%	9	3.03%	1	33
CBP	82.61%	19	34.78%	8	30.43%	7	21.74%	5	23
CDC	88.89%	8	11.11%	1	0.00%	0	0.00%	0	9
NTSB	92.86%	13	7.14%	1	7.14%	1	0.00%	0	14
TSA	84.62%	33	35.90%	14	33.33%	13	15.38%	6	39
ANG, USAF, or other military joint use on airport	87.50%	14	31.25%	5	37.50%	6	18.75%	3	16
U.S. Marshals	75.00%	3	25.00%	1	0.00%	0	0.00%	0	4
Air Marshals	66.67%	4	16.67%	1	16.67%	1	0.00%	0	6
								Total Respondents	**55**

Copyright National Academy of Sciences. All rights reserved.

Q9: Again, with the above definition in mind, which airline, airport and local organizations do you consider to be your airport's mutual aid network partners? With which do you share/coordinate your diversion-related plans? With which do you collaboratively train (at least annually) With which do you train specifically for diverted flights? Please choose as many as applicable.

Answered: 54 Skipped: 17

	Mutual Aid Network Partner		Share/ Coordinate Diversion-Related Plans		Regular Collaborative Training		Diverted Flight Training		Total
Airlines	77.50%	31	65.00%	26	40.00%	16	20.00%	8	40
FBOs/Fuelers	74.51%	38	56.86%	29	31.37%	16	15.69%	8	51
Hangar / Aircraft Owners	78.95%	15	26.32%	5	0.00%	0	0.00%	0	19
Concessionaires	80.95%	17	57.14%	12	23.81%	5	23.81%	5	21
Other Airport Tenants	76.19%	16	33.33%	7	19.05%	4	14.29%	3	21
Ground Transportation	78.95%	15	47.37%	9	15.79%	3	15.79%	3	19
Red Cross	88.46%	23	38.46%	10	34.62%	9	3.85%	1	26
Counselors	80.00%	4	40.00%	2	20.00%	1	0.00%	0	5
Local Stores	100.00%	2	0.00%	0	0.00%	0	0.00%	0	2
Local Restaurants	75.00%	3	25.00%	1	0.00%	0	0.00%	0	4
Local Hotels	78.57%	11	28.57%	4	7.14%	1	7.14%	1	14
Airport or Other Volunteer Group	85.71%	6	14.29%	1	0.00%	0	0.00%	0	7
							Total Respondents		**54**

Q10: Who is your airport's main point of contact to coordinate with your mutual aid network partners during incidents and emergencies when your capabilities have been exceeded?

Answered: 58 Skipped: 13

Answer Choices	Responses	
The airport's director or general manager acting 1:1 with all mutual aid network partners	43.10%	25
The airport's operations manager	50.00%	29
The airport's emergency manager	5.17%	3
The local (city, county, or local multi-county regional) EM agency	17.24%	10
The state EM agency	1.72%	1
A federal agency	0.00%	0
Don't know	0.00%	0
Other (please specify)	13.79%	8
Total Respondents		**58**

Copyright National Academy of Sciences. All rights reserved.

120 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Q11: In the past 12-18 months, how often has your airport asked your mutual aid network partners to assist with incident- or emergency-related flight diversions?
Answered: 57 Skipped: 14

Answer Choices	Responses	
1-5 times	50.88%	29
5-10 times	5.26%	3
10-15 times	1.75%	1
15-20 times	1.75%	1
More than 20 times per year	3.51%	2
Unsure	7.02%	4
Other (please specify)	31.58%	18
Total Respondents		57

Q12: Does your airport implement any of these industry-recommended notification practices with your mutual aid network? Indicate all that you implement:
Answered: 55 Skipped: 16

Answer Choices	Responses	
Update and test your 24/7 mutual aid network partner contact list regularly (at least annually)	63.64%	35
Identify both primary and secondary contacts at all network partners with various communication methods identified for each (e.g., office phone number, cell phone number and email address)	50.91%	28
Establish communication protocols (e.g., email, text, dashboard, etc.) that all partners can access for sharing notifications and real-time, accurate and consistent updates throughout the event	32.73%	18
Use common definitions and acronyms following National Incident Management System (NIMS) and Incident Command Structure (ICS) during events, using common/plain language and avoiding unique codes/phrases whenever possible	58.18%	32
Test your airport's notification system (technology) via "live" tests throughout the year	47.27%	26
Develop a backup plan for technology during power failures, cyber-attacks, and network overloads	30.91%	17
Unsure	5.45%	3
None of the above	14.55%	8
Total Respondents		55

Copyright National Academy of Sciences. All rights reserved.

Q13: Does your airport plan for any of these industry-recommended contingency practices that can help prevent flight diversion events from escalating?

Answered: 55 Skipped: 16

Answer Choices	Responses	
Specific aircraft-related practices (e.g., plans for wide body, international flights, non-station airline needs)	36.36%	20
Surge of passengers (Plans that delineate overflow areas) or aircraft (Plans that delineate hard stand areas, parking areas)	32.73%	18
After hours conditions (e.g., plans/agreements for on-call or late shift staffing)	45.45%	25
Additional impacts to airport facilities unrelated to the diversion, including landside impacts (e.g., plans for power outages, weather-related damages) and airside impacts (e.g., alternate plans for damaged runway)	41.82%	23
Compounded communication issues (e.g., plans for network overload or power outages)	16.36%	9
Extended delay situations (e.g., plans for preventing equipment overuse, plans)	34.55%	19
Long-term weather event	29.09%	16
National Airspace System conditions (e.g., natural disaster in one part of the country causing flight diversions)	16.36%	9
Unsure	1.82%	1
None of the above	27.27%	15
Total Respondents		**55**

Copyright National Academy of Sciences. All rights reserved.

122 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Q14: Does your airport implement any of these industry-recommended coordination practices with your mutual aid network partners? Indicate all that you implement:
Answered: 55 Skipped: 16

Answer Choices	Responses	
Meet regularly with your mutual aid network partners via meetings, workshops or training sessions	54.55%	30
Ensure the airport's diversion-related plans are coordinated and aligned with mutual aid partner plans (e.g., regional/municipal contingency plans, mutual aid operational plans, emergency response plans, etc.)	27.27%	15
Establish, update and share a centralized list of local equipment and resources for use in flight diversion events	36.36%	20
Establish and clarify funding sources for incident- and emergency-related costs associated with diversions (e.g., food and water for stranded passengers)	10.91%	6
Identify and establish alternative or multiple service or supply vendors	14.55%	8
Conduct an after-action review with mutual aid network partners, assigning personnel with follow-up responsibilities to ensure accountability	47.27%	26
Update diversion-related plans and training based upon findings from after-action review	27.27%	15
Disseminate updated diversion-related plans both internally at your airport and to mutual aid network partners	23.64%	13
Unsure	1.82%	1
None of the above	25.45%	14
	Total Respondents	55

Q15: Does your airport implement any of these industry-recommended coordination practices with airlines? Indicate all that you implement:
Answered: 55 Skipped: 16

Answer Choices	Responses	
Understand each airline's priority airport list for flight diversions	27.27%	15
Understand each airline's aircraft towing/gate and deplaning requirements (e.g., wide body aircraft) and available equipment on site at your airport	43.64%	24
Understand typical daily staffing availability as well as whether they use any third-party staffing vendors and how they communicate with them	34.55%	19
Establish agreements to clearly define the assistance airlines can provide to flight diversions associated with non-station airlines at your airport	14.55%	8

Copyright National Academy of Sciences. All rights reserved.

Establish agreements to clearly define who (airport or airline) manages passenger assistance / victims' assistance and what resources, supplies and staffing they will provide in various incident- and emergency-situations	25.45%	14
Establish equipment sharing agreements with airlines (e.g., air stairs for deplaning, etc.)	21.82%	12
Unsure	1.82%	1
None of the above	40.00%	22
	Total Respondents	**55**

Q16: Does your airport implement any of these industry-recommended coordination practices with the following agencies and airport tenants?

Answered: 53 Skipped: 18

Answer Choices	Responses	
FAA - Establish and regularly review capacity constraints policy	15.09%	8
TSA - Establish and regularly review security procedures for stranded passengers	37.74%	20
CBP - Establish and review deplaning plans for diverted international flights	35.85%	19
VARIOUS - Establish and regularly review after-hour staffing plans with TSA, CBP and concessions	16.98%	9
Unsure	13.21%	7
None of the above	37.74%	20
	Total Respondents	**53**

Q17: Has your airport recently (past 12-24 months) received a diverted flight due to either an incident or emergency that exceeded your capacity and required the use of your mutual aid network partners (e.g., emergency responders, agencies, healthcare, community groups, Red Cross, etc.) to respond?

Answered: 55 Skipped: 16

Answer Choices	Responses	
Yes	10.91%	6
No	89.09%	49
	Total Respondents	**55**

Copyright National Academy of Sciences. All rights reserved.

124 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Q18: This diverted flight event involved the following circumstances. Please choose as many as applicable:

Answered: 6 Skipped: 65

Answer Choices	Responses	
Poor weather conditions	66.67%	4
Multiple aircraft diversions	33.33%	2
Minor mechanical aircraft issue (e.g., oil leak, warning light)	50.00%	3
Aircraft emergency (major mechanical failure, fire)	50.00%	3
Medical emergency (e.g., heart attack)	33.33%	2
Passenger disruption on flight (e.g., security issue)	0.00%	0
Passenger deplanement into the airport (e.g., flight cancellation, extended flight delay)	50.00%	3
Safety / security issue	0.00%	0
Victim assistance	0.00%	0
Family assistance	0.00%	0
International flight	0.00%	0
After airport normal business hours	33.33%	2
Other (please specify)	0.00%	0
Total Respondents		**6**

Q19: This diverted flight event involved the following type of aircraft:

Answered: 6 Skipped: 65

Answer Choices	Responses	
Large commercial aircraft	66.67%	4
Small commuter aircraft (single engine/multi-engine piston)	16.67%	1
Corporate / private jet (turboprop)	16.67%	1
Experimental aircraft	0.00%	0
Rotorcraft	0.00%	0
Military aircraft	16.67%	1
Other (please specify)	16.67%	1
Total Respondents		**6**

Copyright National Academy of Sciences. All rights reserved.

Q20: If you were to conduct an after-action report related to your recent diverted flight situation, what would you say worked well and did not work well, specifically related to coordinating with your mutual aid network partners (e.g., emergency responders, agencies, healthcare, community groups, Red Cross, etc.)? Please choose as many as applicable:
Answered: 6 Skipped: 65

	Went Well		Did Not Go Well		Total
COMMUNICATION: Early notification (advanced warning)	83.33%	5	16.67%	1	6
COMMUNICATION: Contacting mutual aid network partners (correct contact information on file)	100.00%	5	0.00%	0	5
COMMUNICATION: Information-sharing among mutual aid network partners (specifics / needs)	100.00%	6	0.00%	0	6
COMMUNICATION: Crisis communication (if applicable)	100.00%	3	0.00%	0	3
COMMUNICATION: Social media used for emergency management	100.00%	4	0.00%	0	4
COMMUNICATION: Use of technology (e.g., dashboard)	66.67%	2	33.33%	1	3
PLANNING: Following a practical checklist or plan	100.00%	6	0.00%	0	6
PLANNING: Preparation (e.g., diversion-related training)	80.00%	4	20.00%	1	5
PLANNING: Understanding the airport's capability and capacity (e.g., staffing, facilities, resources/supplies, and equipment available for the diverted flight)	100.00%	5	0.00%	0	5
INCIDENT MANAGEMENT: Understanding roles and responsibilities of each network partner (e.g., chain of command)	100.00%	6	0.00%	0	6
INCIDENT MANAGEMENT: Response time from mutual aid resources	100.00%	5	0.00%	0	5
OUTCOME: Adequate passenger care (e.g., facilities, equipment, resources)	100.00%	6	16.67%	1	6
OUTCOME: Adequate victim's assistance from mutual aid network partners (if applicable)	100.00%	5	0.00%	0	5
Other (please specify)					1
				Total Respondents	**6**

Copyright National Academy of Sciences. All rights reserved.

Q21: Would you be open to sharing your good practices and lessons learned associated with diverted flights with our research team?

Answered: 6 Skipped: 65

Answer Choices	Responses	
Yes	66.67%	4
No	33.33%	2
If yes, please provide your name, airport and email address so one of our team members can contact you.		3
Total Respondents		**6**

Copyright National Academy of Sciences. All rights reserved.

APPENDIX C

Sample Diversion Plans, Checklists, and Training Exercises

SAMPLE GRAB AND GO DIVERSION CHECKLIST (FOR AIRPORT SUPERVISORS) COURTESY OF AVIATION INNOVATION, LLC

☐ **RECORD DIVERTED FLIGHT INFORMATION:**

- Notified by: _____
- Time of notification: _____
  - Flight information: _____
  - Aircraft Operator: _____
  - Ground Handler: _____
  - Flight number: _____Origin: _____ Destination: _____
  - Aircraft type: _____
    - ☐ Domestic ☐ International (Pre-cleared status: _____)
  - Number of passengers: _____ Number of crew: ____ Crew time left: _____
  - Special needs passengers: _____ Pets/animals on board: _____

☐ **RECORD DIVERSION TYPE AND TIMING**

- ☐ Gas and Go (See gas and go section)
- ☐ Medical Emergency (See medical emergency section)
- ☐ Multiple Diversions / Deplaning (See multiple diversions / deplaning section)
- ☐ International Diversion (see international section)
- ☐ Extended Delay (see extended delay section)
- ☐ Estimated time of arrival _____
- ☐ Time landed _____
- ☐ Gate or location _____
- ☐ Estimated departure time _____

☐ **GAS AND GO DIVERSION**

- ☐ Contact airline station manager, ATC and FBO/ground handler – Coordinate gate or parking area
- ☐ Contact FBO/ground handler – Request airstairs and equipment for refueling
- ☐ Monitor timing (See extended delays section)
- ☐ Record departure time _____

☐ **MEDICAL EMERGENCY DIVERSION**

- ☐ Receive notification via crash phone / alarm system
- ☐ Contact airline station manager, ATC and FBO/ground handler
  - ☐ Determine aircraft location and coordinate gate or parking area
  - ☐ Secure airstairs for deplaning; communicate special needs (e.g., ADA equipment) requests
- ☐ Contact law enforcement to standby aircraft and ensure no one other than a patient and family deplane; assist with crowd control (curious bystanders, volunteers)
- ☐ If international flight – designate law enforcement to escort patient and family to hospital until cleared by CBP; provide CBP with patient and family names
- ☐ Record departure time _____

Copyright National Academy of Sciences. All rights reserved.

- ☐ **MULTIPLE DIVERSIONS / DEPLANING**
  - ☐ Contact airline station manager, ATC and FBO/ground handler (Coordinate gates/parking areas)
  - ☐ Contact FBO/ground handler/ARFF to secure airstairs for deplaning; communicate special needs (e.g., ADA equipment) requests
  - ☐ Notify law enforcement to assist with escorting passengers from remote parking to the terminal; assist with crowd control; animal/pet relief assistance
  - ☐ Conduct recovery procedures
    - ☐ Coordinate with airline station managers for passenger care provisions or see extended delay section for passenger care provisions
    - ☐ Notify maintenance of additional terminal cleaning duties; check on supplies status
    - ☐ Notify TSA of after-hours or extra staffing needed
    - ☐ Monitor timing (See extended delays section)
  - ☐ Record departure time(s) _____
- ☐ **INTERNATIONAL DIVERSION**
  - ☐ Contact airline station manager, ATC and FBO/ground handler – Coordinate gate or parking area
  - ☐ Contact CBP – Request gas and go procedures (or deplaning procedures – see below)
  - ☐ Contact FBO/ground handler or ARFF – Request standby or secure airstairs for refueling
  - ☐ Notify law enforcement to standby aircraft
  - ☐ Monitor timing (See extended delays section)
  - ☐ Implement procedures for deplaning international passengers:
    - ☐ Contact CBP to allow deplaning into an isolated / sterile / secure area
    - ☐ Contact FBO/ground handler/ARFF to secure airstairs for deplaning; communicate special needs (e.g., ADA equipment) requests
    - ☐ Contact law enforcement to:
      - ☐ Set up sterile area (place barriers at selected location)
      - ☐ Restrict access to doors at selected location
      - ☐ Ensure minimum number of approved security are available
    - ☐ Notify CBP and law enforcement of all employees working in the sterile area
  - ☐ Conduct recovery procedures
    - ☐ Coordinate with airline station manager for passenger care items or see extended delay section for passenger care provisions
    - ☐ Receive notification from CBP to release passenger and crew
    - ☐ Notify TSA if rescreening involved
    - ☐ Ensure trash is collected from the sterile area and give to operating airline for placement on international aircraft
    - ☐ Collect names of all local employees involved in an international diverted flight
  - ☐ Record departure time _____

- ☐ **EXTENDED DELAY**
  - ☐ Passenger Care Provisions
    - ☐ Notify operations/maintenance/ARFF to assist with passenger care items (water, blankets, etc.) and animal/pet relief assistance
    - ☐ Notify concessions (for supplies/staffing) and/or local stores
    - ☐ Contact Convention and Visitors Bureau for status update on hotel room availability
    - ☐ Notify ground transportation (buses) and local hotels
    - ☐ Notify TSA of after-hours or increased staffing needed
    - ☐ Notify law enforcement to assist with crowd control
  - ☐ Timing:
    - ☐ 60 min – Talk with station manager; coordinate parking; brief operations department
    - ☐ 90 min – Request updated departure/deplaning timeline
    - ☐ 120 min – Ensure passengers have access to snack/beverage and lavatory; request update from airline – if longer than 30 minutes, request deplaning services
    - ☐ 150 min (threshold) – Require deplaning to commence if no safety issues exist; contact concessions and maintenance for additional services/staffing
  - ☐ Record departure time _____

Copyright National Academy of Sciences. All rights reserved.

SAMPLE AIRPORT DIVERSION CHECKLIST

Courtesy of Dane County Regional-Truax Field

IROPS – INTERNATIONAL DIVERT (1 of 3)

Airport Operations Supervisor Checklist

NOTIFIED BY:_____ TIME: _____
AIRCRAFT OPERATOR: _____
GROUND HANDLER: _____

FLIGHT #: _____

ORIGIN:_____DESTINATION: _____

AIRCRAFT TYPE: _____

PRE-CLEARED STATUS: _____

OF PASSENGERS: _____ # OF CREW: _____

ESTIMATE TIME OF ARRIVAL: _____

EXTENDED STAY: ☐ **FUEL & GO:** ☐ **MEDICAL EMERGENCY:** ☐

TIME LANDED: _____

GATE # OR LOCATION: _____

ANTICIPATED DEPATURE TIME: _____

Airport Operations will serve as point of contact and coordination with airlines, CBP, DCSO, and TSA.

	To be completed by one of the following:							
TASK	**Ops 4**	**Ops 3**	**Sh 1**	**ATC**	**CBP**	Airline	**TSA**	**Co 16**
Airport Operations notified of International Divert								
Obtain pertinent flight information from airline (listed above in red)	☐							
Contact CBP	☐					☐		
Determine aircraft location	☐							
Notify DCSO	☐							
Notify TSA Coordination Center	☐							
Notify Ops 3	☐							

Copyright National Academy of Sciences. All rights reserved.

IROPS – INTERNATIONAL DIVERT (2 of 3)

TASK	Ops 4	Ops 3	Sh 1	ATC	CBP	Airline	TSA	Co 16
Extended Stay								
Place barriers at selected location according to plan	☐	☐	☐				☐	☐
Restrict access to doors at selected location	☐	☐	☐					
Ensure minimum number of approved guard are available	☐							
Notify Concessions for extended hours or kiosk service	☐							
CBP allows passengers into isolated area					☐			
Passengers deplane into isolated area								
Notify CBP of all employees working in sterile area	☐							
CBP on-site					☐			
CBP releases passengers / crew					☐			
Medical Emergency								
Notified via Crash Phone				☐				
Determine aircraft location	☐							
Ensure no one other than patient and family deplane	☐	☐	☐					
LEO will escort patient and family to hospital until cleared by CBP			☐					
Provide CBP with patient and family names	☐	☐				☐		
Aircraft Departs								

Copyright National Academy of Sciences. All rights reserved.

IROPS – INTERNATIONAL DIVERT (3 of 3)								
TASK	**Ops 4**	**Ops 3**	**Sh 1**	**ATC**	**CBP**	**Airline**	**TSA**	**Co 16**
Fuel & Go								
CBP allows refueling								
DCSO stand-by aircraft								
Notify ARFF stand-by or airstairs for refueling								
Aircraft departs								
Recovery								
CBP releases passengers and crew								
Determine if passengers need TSA screening								
Collect all trash from isolated area and give to operating airline for placement on international aircraft								
Collect names of all local employees involved in international divert								

Copyright National Academy of Sciences. All rights reserved.

SAMPLE DIVERSION CHECKLIST FROM ACRP GUIDANCE

From ACRP Report 65: Guidebook for Airport Irregular Operations (IROPS) Contingency Planning (Tool 16)

TOOL 16 – DIVERSION CHECKLIST

(Edit as necessary to meet your airport's needs)

Purpose: When aircraft are diverted, passengers end up in unexpected places at unexpected times, and service providers need to coordinate efforts to accommodate their needs. In the event an aircraft is diverted to a diversion airport, this tool provides the steps that should be taken by airlines and the diversion airport (including their public safety department [airport police]) before, during, and after a diversion event. Roles of airlines, airport operation (specifically the operations manager and duty manager), and the public safety department are outlined in this tool.

AIRLINE

Before:

- Notify airport operations. Include:
  - Airline
  - Approximate arrival time
  - Approximate departure time – if available
  - Reason for potential diversion
  - Intentions (examples: gas and go, extended delay, or unknown)
  - Potential services needed
  - Number of passengers on board

During:

- Communicate plane's intentions to airport operations.
- Confirm airport operations and ensure that the duty manager will assist with communication.
- If necessary, ask for assistance. Determine who will coordinate passenger accommodations, including:
  - Food
  - Transportation
  - Lodging
  - Security
  - Special Needs
- Communicate status to necessary service providers at least every 30 minutes.

After:

- Supervisor – obtain feedback from employees about what went well, what did not, and what changes could be made.
- Manager and supervisor join post-diversion conference call with airport.

Copyright National Academy of Sciences. All rights reserved.

AIRPORT OPERATIONS

Before:

- Create a 24/7 email contact/distribution list of major airport stakeholders in your region, including diversion airports, to communicate status and track diverted flights. For hubs and large airports, establish a conference call with key stakeholders 24 to 48 hours prior to severe weather forecasts to facilitate communications and coordination (i.e., National Weather Service, FAA, airlines, CBP, TSA, and airport departments).
- When notified by airline of a diversion, communicate to airlines that airport operations will be the point of contact during the event.
- Determine whether this is a regular diversion (airline and aircraft that are regularly serviced at airport).
  - If regular aircraft/airlines, determine and communicate equipment available to help service (see attached sample equipment list)
  - If airline has no representation at airport, determine potential services needed and communicate what equipment/options are available to service particular aircraft (see included sample equipment list)
- International diversions: Have a plan in place ahead of time with CBP to handle and/or offload passengers from international diversions, especially if there are no CBP officers or facilities present at an airport. At a minimum, coordinate with the regional CBP official and local law enforcement to share important CBP contact information, such as 24/7 phone numbers.

During:

Operations Center

Notify:

- Airport duty manager in charge
- Law enforcement officer (LEO) in charge
- Federal security director (FSD)
- Senior duty manager (or deputy aviation director – airside operations)
- Concessions, if services are needed
- CBP (if international flight, need 24/7 contact information)
- Communicate with airlines frequently during event (at least every half hour).
- Remind airlines of available assistance, including:
  - Additional resources (If aircraft cannot taxi from its location, coordinate to use local FBOs and/or aircraft recovery service to have aircraft removed.)
  - Ability to contact resources for airlines if requested
  - Use of social media to inform passengers
  - Providing of flight information display systems (FIDS) updates

Communicate status to necessary service providers at least every 30 minutes.

Ascertain who is making the decisions about the status of an aircraft regarding loading and unloading of passengers, bags, and cargo. This is especially important if an airline is not represented at an airport; airport staff should find out from the flight crew some of the system operations centers (SOCs) or headquarters phone numbers so that they can contact someone in a position to make a decision at critical times (such as when the 3- and 4-hour rule is reached). This should be done as soon as the aircraft is grounded and parked.

Copyright National Academy of Sciences. All rights reserved.

134 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Airport Operations Manager

- When notified of a possible diversion, contact the applicable airline to determine the potential length of the delay.
- Record in diversion contact log:
  - Date/time
  - Air carrier name and contact information
  - Flight number
  - Aircraft type and tail number
  - Passenger count
  - Arriving from/original route
  - Parking location
  - Reason for diversion
  - ETA/ETD
  - Jet bridge use and departing flight number
  - Crew time left (international flights only)
  - Services needed
- Determine gate needs (coordinate a gate from which to deplane if delay exceeds 3 hours for domestic flights and 4 hours for international flights), whether airline will accommodate aircraft at their regularly assigned gate(s), and can or will they accommodate other airlines. Gate options must take into consideration:
  - Aircraft type/size
  - Access to restroom facilities and restroom service needs
    Access to vending machines
  - Access to drinking fountains
  - Food and beverage services through tenant restaurant vendor
  - Ability to restrict international passengers from mixing with domestic passengers*
  - Airline support to contain passengers isolated from domestic passengers*
  - No CBP processing available for international flights*

    *international flights only

- If no gates are available:
  - Coordinate with airlines and ATC services to direct aircraft to park at alternate parking location, escort marshaling/ground handling crew as necessary
  - Coordinate with airline or ground handlers to provide access to aircraft for air stairs, refueling, lavatory services, ground power units (GPUs), and other ground service equipment (GSE)
  - Coordinate deplaning of passengers via air stairs and buses or via loading bridge at terminal when delay exceeds 3 hours (4 hours for international flights) and/or when airline requests access to terminal
- If the aircraft delay is a departure and the passengers are deplaned at the terminal:
  - Screening for passengers who leave the concourses must be provided or passengers must remain in the sterile area and food, beverage, and restroom facilities must be provided until the passengers are reboarded for departure

Copyright National Academy of Sciences. All rights reserved.

- If the projected time at the gate is after the time that screening is closed:
  - Coordinate passenger screening operations to remain open or coordinate with the LEO to provide staffing of the checkpoint to prevent re-entry of unscreened passengers
- Coordinate provisions with the airport's concessions.
- Ensure that LEO is available to assist with disruptive passenger(s).
- Maintain contact with the airline representative to determine if the flight may be cancelled and, if so, the airline's intentions concerning its passengers.
- For international flights:
- Coordinate with CBP port director for any concerns for passenger boarding/containment
- Arrange for LEO to monitor passengers to prevent mixing with domestic passengers (must be local airline employee or air crew members when no local representative is available)
- Establish visual or physical perimeter stanchions, seating, and so forth to contain passengers (perimeter should allow restroom access without escort)
- Communicate status to necessary service providers at least every 30 minutes.

After:

- Initiate conference call:
  - Obtain feedback on what went well, what didn't go well, and any changes that need to be made
- Type up notes from conference call – disseminate to all entities as lessons learned/action items.
- Check that the following entities attended conference call: –
  - Airport operations
  - Airlines
  - FAA
  - TSA
  - CBP
  - LEO
  - Public safety
  - Concessions
  - Car rental
  - Parking
  - Military (if on-site)
  - FBO

Copyright National Academy of Sciences. All rights reserved.

PUBLIC SAFETY DEPARTMENT

Before (if notified prior to aircraft landing):

- Notify airport operations specialist on duty.
- Fill out diversion contact log.
- If warranted, notify additional personnel or entities such as concessions, FBO, and the like.
- For extended delays at the airport, determine the resources to accommodate the situation and call up resources as appropriate.

During:

- Obtain additional information about aircraft:
  - Tail number
  - Time landed
  - Any other pertinent information
  - Fill out diversion contact log
- Inform airlines of public safety assistance available.
- Communicate status to necessary service providers at least every 30 minutes.

After:

- Obtain feedback from officers regarding what went well, what didn't, and any changes that need to be made (similar to post-incident discussion).
- Join the post diversion conference call and provide input.

Copyright National Academy of Sciences. All rights reserved.

Sample Equipment Checklist

(Edit as necessary to meet your airport's needs)

Sample Equipment List	Airline	Airline	Airline	Airline	Airline	Airport	Other
737 Tow Bar (TB)							
757 TB							
767 TB							
A319/320 TB							
AIRBUS TB							
A320 TB							
MD80/90 TB							
CRJ200 TB							
CRJ700/900 TB							
E190 TB							
Q-400 TB							
Universal TB							
737 Pushup stairs							
737 Air stairs							
757 Cabin access stairs							
757 Passenger stairs non-motorized							
767 Stairs							
767 Pushup stairs							
747/777 Stairs							
A320 Diesel powered air stairs							
A320 Pushup stairs							
MD80 Galley access stairs							
Air start							
Air Start Wide body Capable							
Ground Power Unit (GPU)							
Bottle air start							
Lavatory service truck							
Lavatory service truck wide body capable							
Potable water cart							
Cabin service lift truck wide body							
Pushback tractor, wide body capable							
Pushback tug							

Copyright National Academy of Sciences. All rights reserved.

SAMPLE DIVERSION RECOVERY CHECKLIST FOR AIRPORT SUPERVISORS

Courtesy of Aviation Innovation, LLC

- ☐ **MONITOR AIR TRAFFIC STATUS**
  - Continue to communicate with the ATC and airlines for status updates
  - Continue to use tracking technologies to monitor weather and flights
- ☐ **MONITOR AIRSIDE STATUS**
  - Communicate with airlines, FBOs, ground handlers, and other hangar tenants:
    - Check for impacts to current physical infrastructure (e.g., limited gate/parking) and equipment capacity and/or operability (jet bridges)
    - Check aircraft maintenance/repair status (status of labor, equipment and supplies)
  - Recommend backup staffing if needed
  - Recommend calling alternate supply providers for additional supplies if needed
  - Understand shift changes (flight crews and ground crews) and impact to current situation
  - Understand equipment maintenance schedule and how it aligns with current situation
- ☐ **MONITOR LANDSIDE STATUS**
  - Check status: staffing levels/shifts, equipment use/maintenance schedule, and supply levels from:
    - Operations
    - Facilities Management/Maintenance
    - Concessions
    - Retailers
    - Other tenants
  - Contact backup staff or switch to alternate shift schedule
  - Contact alternate supply providers if needed
- ☐ **MONITOR STATUS OF ANY DEPLANED INTERNATIONAL DIVERTED FLIGHTS**
  - TSA: Notify of additional re-screening needed
  - CBP: Obtain status updates on release of passenger and crew
  - Law Enforcement: Restore/remove barriers from sterile area
  - Maintenance: Ensure trash has been collected from the sterile area and give to operating airline for placement on international aircraft
- ☐ **MONITOR STATUS OF MULTIPLE DIVERSIONS DURING EXTENDED DELAYS**
  - Delegate roles/responsibilities as needed:
    - Consider leveraging unused staff for surge situations
    - Contact list of A-CERTS and other community volunteers to backfill as needed
  - Community resources:
    - Ground Transportation – Status of taxis and ride share service or passenger buses
    - Convention and Visitors Bureau – Status update on hotel room availability
  - Maintenance: Collect all passenger care items (e.g., blankets etc.), arrange cleaning, and re-store; take inventory of supplies and order additional items to replenish stock
  - TSA: Notify of additional re-screening needed
  - Concessions/Retailers: Status of supplies and staff

Copyright National Academy of Sciences. All rights reserved.

SAMPLE CHECKLIST FOR CREATING AN EMERGENCY WORKING GROUP

Courtesy of Aviation Innovation, LLC

CREATE EWG BUY-IN

- ☐ Try different methods such as:
  - ☐ Give an EWG presentation at a station manager's meeting or at an airport tenants meeting
  - ☐ Discuss at a triennial event or along with a diversion tabletop training event
  - ☐ Discuss after an emergency or diversion event
  - ☐ Demonstrate EWG usefulness beyond aircraft crash assistance; helpful during multiple diversions at once and diversions with extended delays
- ☐ Talk 1:1 to key leaders from potential EWG organizations:
  - ☐ Airport Operations and Emergency Management
  - ☐ Airline Station Managers
  - ☐ FBOs/Fuelers/Ground Handlers
  - ☐ Airport Tenants: Concessionaires, Ground Transportation
  - ☐ City/County EM
  - ☐ Military on airport
  - ☐ Community – Busing companies, stores, restaurants, hotels
  - ☐ Volunteers
- ☐ Write EWGs into airport plans such as AEPs etc.

ASSIGN ROLES AND RESPONSIBILITIES

- ☐ Develop one champion to oversee the group (and identify one alternate)
- ☐ Create training and exercise programs
- ☐ Create and refine Family Assistance Plans, Passenger Care Plans and AEPs

CONDUCT MEETINGS AND TRAINING SESSIONS

- ☐ Meet regularly (e.g., monthly or quarterly)
- ☐ Conduct regular training for:
  - ☐ Victim and Family Assistance
  - ☐ Passenger Care
  - ☐ Emergency / International Diversion Situations

DEVELOP ACTIVATION

- ☐ Develop an alert notification process/group for the EWG (e.g., Alert 3 for crashes and develop one for specific diversion types)

Copyright National Academy of Sciences. All rights reserved.

SAMPLE INTERNATIONAL DIVERSION PLAN TEMPLATE

Courtesy of Aviation Innovation, LLC

Part 1: Airport Information

Airport Name
Address
Phone
Website
Social Media identification

Operations Contact
Name
Title
Address
Phone number

General Information

General airport information.

Airlines served.

Irregular Operations / Contingency Plans

Information about airports that regularly send diversions to this airport.

List plans used to manage diversions.

Copyright National Academy of Sciences. All rights reserved.

Runway Information

Discuss the runway configuration.



Runway Data	Runway (Information)	Runway (Information)
Runway Length/Width		
Marking		
Approach Category		
Slopes		
Navaids Approach Lighting		
Landing System		
Visual Aids		
Pavement Strength in lbs: Single (S)		
Dual (D)		
Double Tandem (DDT) / Dual Tandem (DT)		
Pavement Type		

Copyright National Academy of Sciences. All rights reserved.

Airport Resources

Airport's (insert name of plan) Plan encourages the sharing of resources among the airport, airlines and FBOs when needed. A summary of relevant resources available along with who owns these resources are listed below:

Inventory Item	Ownership
Air Conditioning Cart	
Bag Belts	
Deicing Units	
Fuel Truck	
Heater Cart	
Jacks (Hydraulic)	
Ice	
Lavatory Cart	
Lift Truck	
Oxygen/Nitrogen cart	
Push Stairs	
Truck Stairs	
Safety Cones	
Graders Front End Loaders	
Tow Bar	
Tug	
Wheel Chocks	
Passenger Buses	

Copyright National Academy of Sciences. All rights reserved.

Area Amenities

Arrangements for buses, overnight hotels and meals should be made by the airline directly. Feel free to use the business partners listed or use business partners of your own choice. Direct billing arrangements are to be made with the service provider.

Special Requests / Special Needs

Local Pizza Delivery	24/7 Big Box Stores	

Hotels

Transportation

Rental Cars

Luxury Sedan and Taxi Services Cars

Public Transit

Copyright National Academy of Sciences. All rights reserved.

Airport Contact Information for Diverted International Flights

AIRPORT: Airport requests notification of diversion as early as possible in advance of flight arrival; however, Airport does understand that many times this cannot be accomplished. Airport will activate its (Name) Plan. In the event of an emergency, Airport's (Name) Plan will be active and airport personnel will respond quickly.

KEY CONTACT NUMBERS:	
Airport Operations	Tel #
CBP	Tel #
CBP Service Desk 24/7	Tel #
TSA	Tel #
FAA Tower	Tel #
FBO	Tel #
Deicing	Tel #

CBP: Airlines should contact the CBP for a "Permit to Proceed from One Airport to Another" as soon as possible.

TSA: Airline should contact the TSA if deplaning is anticipated, especially after hours, to ensure that staff is available to re-screen passengers.

FBO: A Fixed Base Operator needs to be contacted by the airline prior to their arrival at Airport. Airlines should contact (Name) at (telephone #) to indicate FBO needs and so aircraft can be guided to the correct parking or gate location.

AIRPORT'S FBOs

Airport offers the full service of the Fixed Base Operator listed below, who can provide all ground handling services.

FBO: Name
Tel:
Fax:
Email:
Frequency:
SITA Address:

FBO: Name
Tel:
Fax:
Email:
Frequency:
SITA Address:

Comprehensive services include:

- *Marshalling and Parking*
- *Fueling*
- *Deicing*
- *Aircraft Pushback and Towing*
- *Flight Planning*
- *Lavatory Service*
- *Potable Water Services*

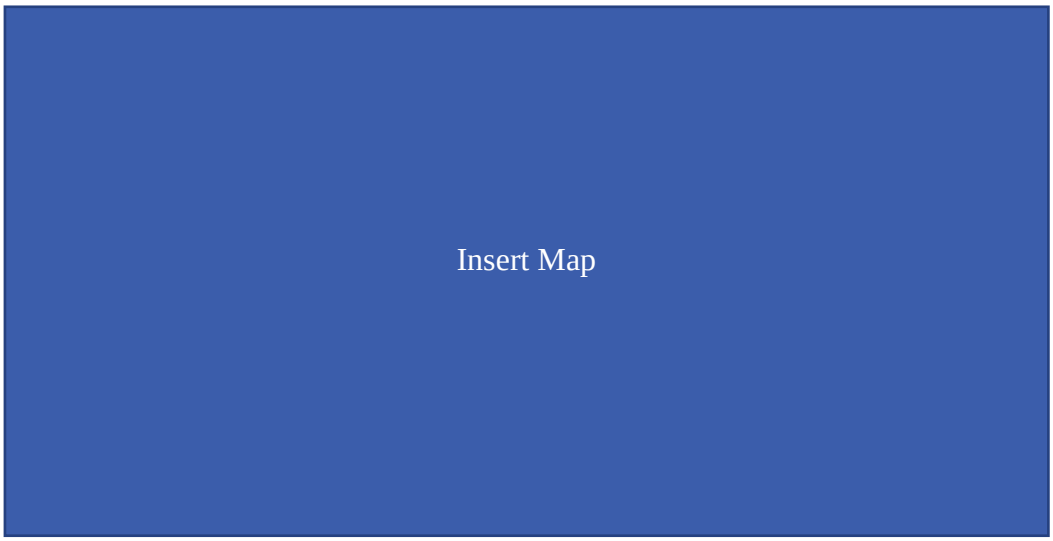
Copyright National Academy of Sciences. All rights reserved.

INTERNATIONAL FLIGHT INFORMATION

International Air Carrier Code Sharing Agreements with Other Air Carriers at Airport:

Air Carrier without Presence at Airport	Air Carrier with Presence at Airport (Code Sharing Partners are checked)						
	Name Tel #	Name Tel #	Name Tel #	Name Tel #	Name Tel #	Name Tel #	Name Tel #
Name							
Name							
Name							
Name							
Name							

Current Map of International Flights Typically Diverted to Airport



Copyright National Academy of Sciences. All rights reserved.

Part 2: Checklist for International Aircraft Diversions at Airport

This section provides a checklist for handling a diverted international aircraft that must be processed through a Federal Inspection Station (CBP). The checklist will be used in the event of a diverted passenger aircraft arriving at Airport.

Note: Every effort will be made to prevent any passenger from being subjected to a tarmac delay of three hours or more without being afforded the opportunity to get off the aircraft per this plan. Clock starts when aircraft touches down at Airport. Consideration will be given to the total number of hours already spent aboard the aircraft – passengers can be removed sooner if requested by the applicable airline station chief or their designated representative.

Checklist for International Diverted Flights

- ☐ Upon notification or unannounced arrival of any and all international aircraft diversions, the responsible Airline Station Manager will immediately contact:
  - Operations Tel # (24/7 number)
  - CBP (0600-1200) Tel #
  - CBP Service Desk 24/7 Tel #
  - TSA Tel #
  - Airline Tel #
  - Airline Tel #

- ☐ Airport Operations will contact ARFF or Fire Department
  - IMPORTANT: Fire Chief and Police Chief or their designated representatives have full authority from Customs and Border Protection to immediately conduct (without delay) whatever fire suppression and rescue actions deemed necessary to preserve safety and health, including immediate transport of any international passenger or crew to the hospital. Afterwards, they will inform the CBP Port Director at the earliest opportunity.

Checklist for Less Than 2 Hour Delays

If aircraft is expected to depart after a short delay (no mechanical or other reason to delay departure) – expected time on ground less than two hours:

- ☐ Parking location for international aircraft refueling operations:
  - List them here.
- ☐ Airline Station Manager will contact CBP before crew or cargo is allowed off the aircraft – Exception: Safety.
- ☐ Airline Station Manager will continually update CBP, up to and including the aircraft's departure.
- ☐ Monitor situation. If the aircraft is on the ground at the two hour mark, implement the next checklist (i.e. if aircraft is not expected to depart in less than two hours).

Copyright National Academy of Sciences. All rights reserved.

Checklist for More Than 2 Hour Delays

If aircraft is not expected to depart in less than two hours:

Note: Every effort will be made to prevent any passenger from being subjected to a tarmac delay of three hours or more without being afforded the opportunity to get off the aircraft per this plan. Clock starts when aircraft touches down at Airport. Consideration will be given to the total number of hours already spent aboard the aircraft – passengers can be removed sooner if requested by the applicable airline station chief or their designated representative.

- ☐ Airport Operations will contact the following for an immediate meeting inside the (room location):
  - Port Director, CBP
  - TSA
  - On-duty Airport Police Supervisor
  - On-duty Airport Authority Supervisor
  - Responsible Airline Station Manager
  - Airport Fire Department

Officials at this meeting will:

- ☐ Determine where to park the aircraft to deplane the passengers. Location must have a bathroom available. Designated location is as follows:
  - Areas (indicate) will be blocked off and guarded.

Copyright National Academy of Sciences. All rights reserved.

Checklist for Deplaning

If the decision is made to deplane the passengers (note three hour rule above):

- ☐ Before deplaning, CBP, TSA, and the Airport Police will sterilize the area chosen to secure the passengers.
- ☐ Airport Operations will provide busses as needed for transportation.
- ☐ CBP, TSA and the Airport Police will provide enough personnel to segregate and maintain security of the off-loaded passengers to prevent them from associating in any way with other passengers, domestic or foreign.
- ☐ The Airline Station Manager will make arrangements for food and water as well as other comfort items.
- ☐ Airline will coordinate aircraft repairs. If this is not feasible, the airline will attempt to provide another aircraft to transport the passengers to an International Airport with a Federal Inspection Station preferably the aircraft's original destination.
- ☐ If another aircraft is used, CBP and TSA will sterilize the aircraft upon arrival. All passengers and cargo will be loaded onto the new aircraft under the supervision of CBP, TSA and the Airport Police.
- ☐ The Airline Station Manager will notify CBP at the next port of arrival.

Copyright National Academy of Sciences. All rights reserved.

Appendix

International Air Carrier Contact Information for Airport

In order to provide you with the best possible service during a diversion to Airport, please provide the following information to Airport Operations at your earliest convenience.

Operations Contact
Name
Address
Telephone
Email

Please insert the information required below and provide a copy of this page to the above contact. Be sure to keep this page with this document as a record of all changes made.

Airline	Station Manager Contact Name	Telephone Email	Date Submitted/ Updated

Copyright National Academy of Sciences. All rights reserved.

150 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

SAMPLE TABLETOP EXERCISE FOR INTERNATIONAL DIVERSIONS

From ACRP Report 153

International Diversions Tabletop Scenario

<table>
<tr><th colspan="3">Tabletop Exercise Package – International Diversion</th></tr>
<tr><td>1. Date Issued/Revised:</td><td>2. Type: Tabletop

International Diversion</td><td>3. IROPS Impact:
✓ Airline Delays
✓ Airline Cancellations
✓ Airline Diversions
✓ Airline Deplanement
✓ Airline Gate/Facility Sharing
✓ Airport Surge
✓ Airport Capacity
✓ Airport Off Hours
✓ Airport Extended Stay</td></tr>
<tr><td>4. Date Scheduled:</td><td>5. Time:</td><td>6. Duration of the Tabletop Exercise:</td></tr>
<tr><td colspan="2">7. IROPS Committee Chairperson:</td><td>8. Tabletop Facilitator:</td></tr>
<tr><td colspan="3">9. Tabletop Participants:

IROPS Committee members

Additional staff from
• <Airport> Operations
• Airlines
• Concessions
• Ground Handlers
• FBOs
• TSA
• FAA - Tower
• CBP Required – Local / Regional
• Hotels
• Public Affairs</td></tr>
<tr><td colspan="3">10. Tabletop Purpose/Scope:

<u>Purpose:</u> To test the ability of <Airport> to develop the communication and coordination aspects of their IROPS plans and procedures to successfully address their contingency response to an international diversion event which results in disruptive airline and airport impacts (as listed above).</td></tr>
</table>

Copyright National Academy of Sciences. All rights reserved.

Tabletop Exercise Package – International Diversion
Scope: Scope of this exercise involves considerations of IROPS Planning, the initial notification, response and coordination with other aviation service providers and stakeholders (i.e., various airport departments, airlines, FAA, TSA, CBP, ground handlers, concessions, local hotels, the public, etc.) during an international diversion event that impacts continued service to passengers. Scope includes the communication protocols involving support personnel to provide initial response, decision-making, notifications, and recovery planning.
11. Tabletop Objectives: **Overall Objective:** Upon completion of this exercise, participants will be better prepared to respond to an international diverted flight event via IROPS plan procedures, direction and control by the IROPS Committee, interface with aviation service providers, the effectiveness of notification and response protocol, and timely status reporting to the traveling and general public, and other stakeholders. • Test the communication and coordination aspects of the <Airport> IROPS Plan to prepare their response to an international diversion resulting in airline and airport impacts (as listed above). • Validate procedures within the IROPS Plan. • Provide effective management and direction of response procedures to the event. • Test the IROPS planning and notification process for various entities including airport departments, airlines, FAA, TSA, CBP, ground handlers, concessions, local hotels, relief organizations, the public, etc. • Test the ability of personnel to recognize triggers and implement IROPS procedures. • Understand how social media can be used to communicate with various entities and the public. • Test the response effort.
12. Tabletop Scenario Narrative Summary
NOTE: ***"ABC" Airport is used throughout as the primary airport undergoing this tabletop exercise.***
Day One, 10:00 p.m.: **Current weather at ABC Airport**: Temp 48 degrees F, winds light from the south, skies partly cloudy, low pressure area to the south, slowly drawing a warm front up from the gulf. Expected high of 55 degrees F, today and into the high 60s by the end of the week. The Air Canada Flight had been delayed for about 2 hours in Cancun. When the plane finally pushed back and took off for Toronto, the passengers settled in for what they hoped would be a routine flight. However, in flight the aircraft developed fuel system related problems and was forced to divert to ABC Airport in the U.S.

Copyright National Academy of Sciences. All rights reserved.

152 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Tabletop Exercise Package – International Diversion

Once on the ground work to resolve the fueling issue caused the need to deplane passengers after 3 hours in a remote parking location. ABC Airport does not have Air Canada Service or a Code-Sharing Partner nor an FIS Sterile area to process or hold international passengers.

The diverted flight arrived at 10:00 p.m.

Air Canada Pilot contacted the local ground agent and the FBO contacted ABC Airport Operations to begin the following discussion:

Does ABC Airport have a pre-approved CBP procedure for deplaning passengers per DOT Regulation?

Upon notification that an aircraft diverted to ABC Airport, did the airline representatives contact nearest Large Hub with FIS Facilities?

Is ABC Airport prepared to give the CBP the following required information during the initial notification?

- Airline and flight number
- Passenger and crew count
- Departure airport/country
- Original port of arrival
- Estimated time of arrival
- Arrival terminal and gate
- Status of crew (time-out)
- Cause of diversion
- Anticipated action
- Airline point of contact and number

Has Air Canada requested permission from CBP to disembark their passengers and crew into a sterile and secure area?

Day Two, 12:00 a.m.:

Air Canada and ABC Airport began communications with the Regional CBP Office. CBP granted permission for passengers and crew to be held "in transit" providing the Air Canada:

- Communicate[s] with the Watch Commander (WC) as to the nature of the diversion and the anticipated length of delay.
- Update[s] the status of the diversion upon any changes as they occur.
- Control[s] the movement and secure the passengers from contact with any other domestic or foreign flight.
- Request[s] access to a securable area in advance in order to care for their passengers.

Air Canada and ABC Airport agree to the terms and work with the Regional CBP Office to establish a sterile/secure area. They determine:

Copyright National Academy of Sciences. All rights reserved.

Tabletop Exercise Package – International Diversion

- How to employ various types of security at the gate/airport to ensure that passengers remain on board, or if deplaned, that they remain in a secure area while awaiting CBP processing and clearance or a continuation of their flight.
- Location of the secure area.
- How security of the area will be maintained.
- How passengers will access restrooms since there are none in the secure area.
- How the Airport/Air Carrier will provide food and water to passengers while maintaining the sterility of the secure area.
- How CBP- regional HUB with FIS will validate passenger information through law enforcement databases as applicable.
- How CBP-Local / Regional Office will assist with any complex immigration issues that responding officers are not able to address.

Day Two, 2:00 a.m.:

A secure and sterile gate area was established with the assistance of TSA, Local Police and Airport Operations. This provided the opportunity for the passenger to disembark and wait in the restricted gate area.

ABC Airport contacted a local 24/7 service and ordered pizza and water for passengers. When the pizza arrived it was determined that it had to be processed due to special requirements for an international flight.

- Did ABC Airport have a 24/7 CBP stakeholder contact numbers local, regional and national?
- What if CBP had to drive 6 hours to ABC Airport and set-up a temporary FIS to screen passengers?
- Did ABC airport have a deplanement plan for handling passengers on a remote parking location?
- What would ABC Airport have done if they had passengers for 12 more hours in sterile area?
- Did ABC Airport contact local Red Cross and or local mutual aid to give them a 'heads-up' in the case of a lengthy delay due to CBP requirements?
- Did ABC Airport have procedures for handling International garbage and de-catering (Is this a real term?) How about using Fleet Service instead for the diverted flight?

Day Two, 4:00 a.m.:

The airline representatives coordinated with an approved flight/ground services company for refueling, ground handling, and catering (and de-catering) services, which takes additional time. CBP assists the airlines and flight service company in providing passengers any food or comfort items while delayed.

Airlines were reminded that all passengers/crewmembers, baggage and cargo need to remain onboard the aircraft until CBP gives them authorization to offload. No personnel, other than emergency personnel, are to board the aircraft without CBP authorization.

Copyright National Academy of Sciences. All rights reserved.

154 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

Tabletop Exercise Package – International Diversion

Day Two, 6:00 a.m.:

Air Canada had the fueling issue resolved and the service dog was cared for. Passenger re-board took place and breakfast was served on the flight back to Toronto.

END OF EXERCISE

Debrief /Hotwash

Recap event:

- Summarize the event.

Debrief Questions:

General:

- **What went well?**
- **What needs to be improved?**
- **What would you change?**

Specific:

- Did the airport clearly understand the DOT requirement for sterile area?
- Does the airport have a procedure for non-station airlines?
- Does the FBO have fueling agreements with International Carriers?
- Should ABC Airport have a pre-approved International Diversion Plan which has been reviewed with the Regional CBP Officer?
- What was the communication and social media strategy for this exceptional flight?

13. Tabletop Control, Simulations and Logistics Requirements

U-Shaped Table formation
Multimedia Projector with wireless remote
White Board – 3m sticky notes
Screen
Flip Chart(s) Number____
Easel(s) Number____
Marker(s) Number____
Extension Cord(s) Number____ Length_____
Power Strip(s) Number _____

For each Attendee, please provide:
_____Handout(s) Number _____
_____Notepad(s) Number _____
_____Pen(s) Number _____
_____Nametag(s) Number _____

Copyright National Academy of Sciences. All rights reserved.

Tabletop Exercise Package – International Diversion		
14. Number of Participants:	15. Signature of Manager:	16. Date:

Copyright National Academy of Sciences. All rights reserved.

156 Managing a Flight Diversion with an Emergency Response at Small, Non-Hub, or General Aviation Airports

SAMPLE MEMORANDUM OF UNDERSTANDING (MOU)

Memorandum of Understanding

By And Between

The Arkansas Air Coordination Group

And Memorial Field Airport

Purpose

This Memorandum of Understanding (MOU) is to promote coordination between the Arkansas Division of Emergency Management (hereafter referred to as ADEM), Arkansas Department of Transportation (hereafter referred to as ARDOT), Air Coordination Group (hereafter referred [to] as the ACG), and the Memorial Field Airport (hereafter referred to as the Airport) and effectively utilize the airport in response to emergencies and disasters. It also documents that the airport and the Air Coordination Group (ACG) recognize that the airport may be needed to facilitate the response to a disaster. The airport may be used in support of missions assigned by the State Emergency Operations Center (SEOC) and coordinated by the ACG. Missions may also originate from the U.S. Department of Homeland Security (DHS), Federal Emergency Management Agency (FEMA); U.S. Department of Defense (DOD), the private sector; and/or the non-profit sector.

Hazards

Arkansas has identified 15 natural and manmade disasters that can occur in the state. These are: tornadoes, severe winter storms, floods, earthquakes, wildfire, landslides, expansive soil/land subsidence, dam and levee failure, drought, severe thunderstorms (winds, hail, and lightning), hazardous materials, nuclear events, terrorism, major disease outbreak, and cyber incident[s].

Responsibilities

1. **ADEM** –Coordinates state resources in response to requests for support [in] an emergency, when local governments reach or anticipate exhaustion of their capabilities.
2. **SEOC** – Supports response and recovery activities of local jurisdictions. The SEOC will coordinate needed state, nongovernment, private sector, Emergency Management Assistance Compact (EMAC) member state, and federal resources.
3. **ARDOT** – Primary agency responsible for coordinating transportation activities. ARDOT is responsible for coordinating state resources needed to restore and maintain transportation routes

Copyright National Academy of Sciences. All rights reserved.

Memorial Field Airport MOU necessary to protect lives and property during an emergency/disaster. They will coordinate the available state air assets through the ACG according to the Arkansas Aviation Operations Plan annex to the Arkansas Comprehensive Emergency Management Plan.

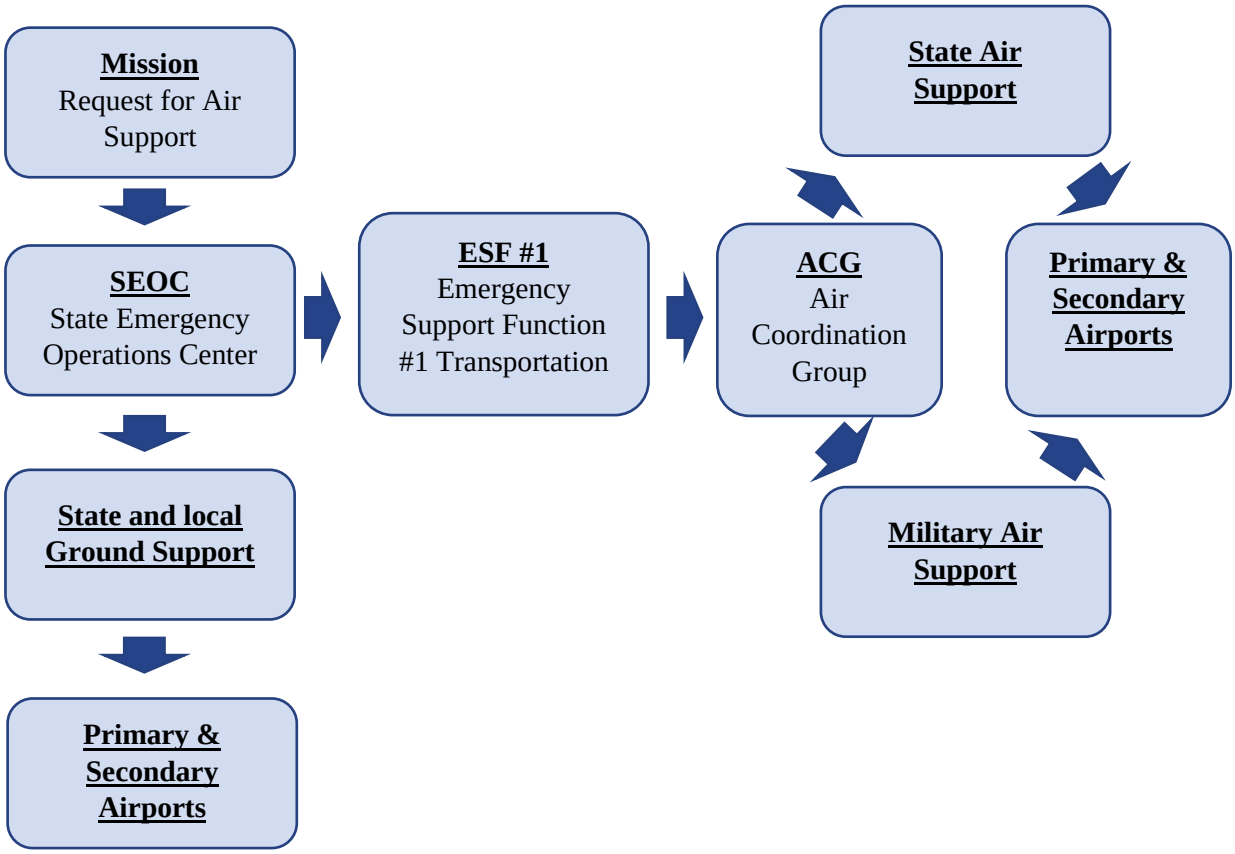
4. **ACG** – Coordinates between agencies while providing a point of contact for aviation coordination to and from the disaster area. The agency providing the aircraft for approved mission tasks by the ACG maintains the responsibility to operate the aircraft. The ACG will have a minimal role in the direct Command and Control (C2) of aviation resources.
5. **Airport** – The Memorial Field Airport is a city-owned, public-use airport located approximately three miles southwest of the central business district of Hot Springs in Garland County, Arkansas.

Area of Application

1. This MOU applies to the facilities, ground equipment, runway use, and personnel required for aircraft operation support at Memorial Field Airport.
2. Refer to the Arkansas Aviation Operation Plan for the following:
    a. ACG/State roles and responsibilities.
    b. Administration and resources.
    c. Coordinating instructions and communications.

Copyright National Academy of Sciences. All rights reserved.

ACG Mission Organization



Copyright National Academy of Sciences. All rights reserved.

Effective Date, Term, Modifications, and Termination

Effective Date: The terms of this MOU will become effective on the date of the last signature where all parties have signed below.

Term: This MOU will become effective when signed by all Parties. This MOU will remain in effect for six years from the date of the latest signature.

Modifications: This MOU may be modified at any time by written notification between each party.

Termination: Each party involved shall have the right to terminate this MOU by giving sixty (60) days written notice to the other party at any time.

_____	_____
Airport Manager	Date Signed
_____	_____
Mayor	Date Signed
_____	_____
Director, Arkansas Division of Emergency Management	Date Signed
_____	_____
Director, Arkansas Department of Transportation	Date Signed

Copyright National Academy of Sciences. All rights reserved.

Copyright National Academy of Sciences. All rights reserved.

Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

Copyright National Academy of Sciences. All rights reserved.

Transportation Research Board
500 Fifth Street, NW
Washington, DC 20001

ADDRESS SERVICE REQUESTED

NATIONAL ACADEMIES Sciences Engineering Medicine

The National Academies provide independent, trustworthy advice that advances solutions to society's most complex challenges.
www.nationalacademies.org

ISBN 978-0-309-68773-7



Copyright National Academy of Sciences. All rights reserved.